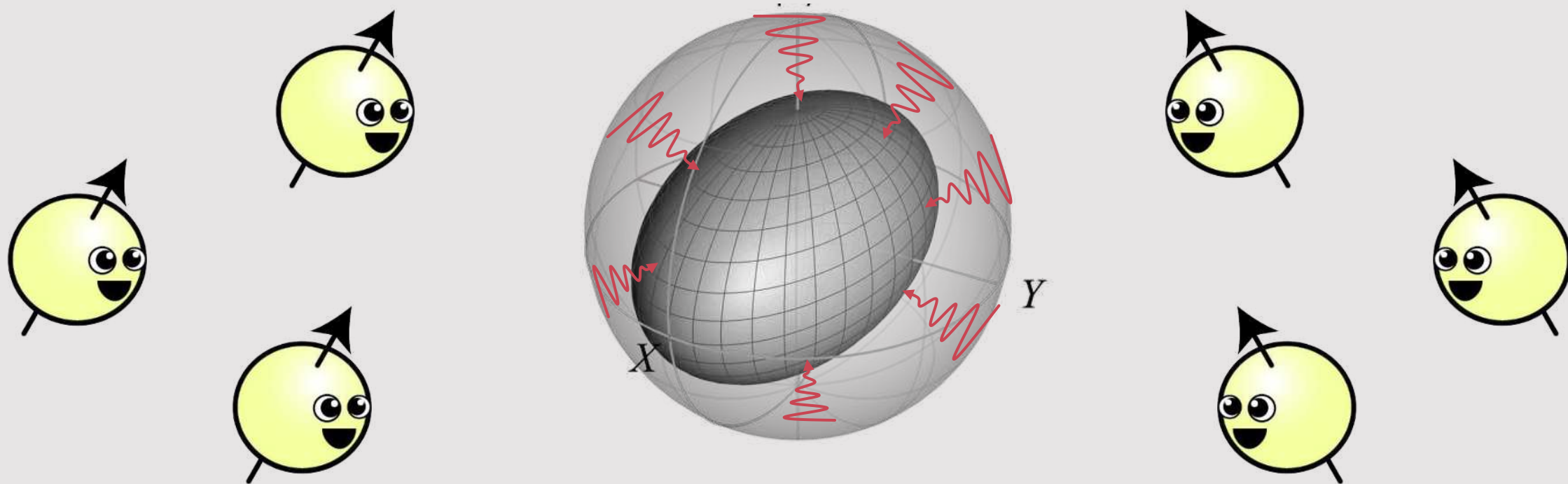


Introduction to Quantum Noise

Qiskit Global Summer School: Quantum Simulations



Zlatko K. Minev

IBM Quantum

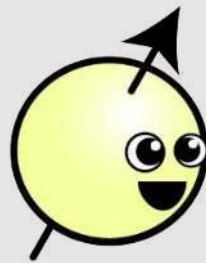


@zlatko_minev



zlatko-minev.com

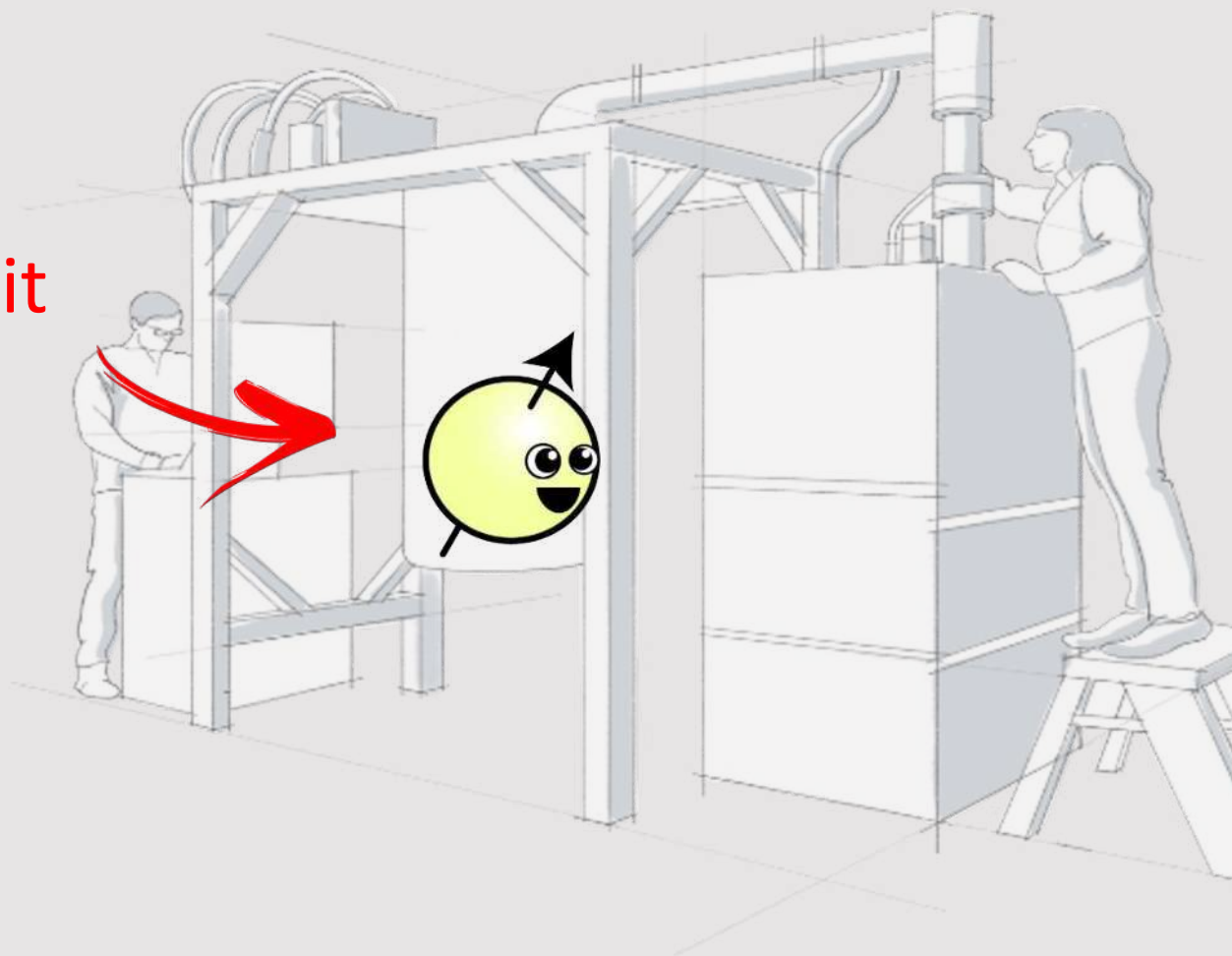
What do I need to know before
I run quantum simulation on a
real, noisy quantum processor?

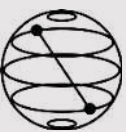




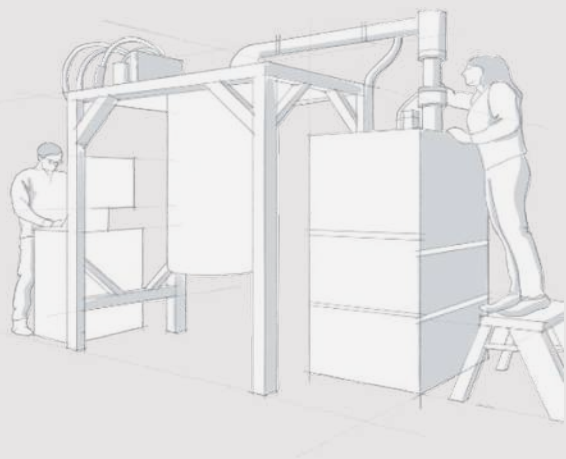
Chapter 1: Hello World!

A qubit

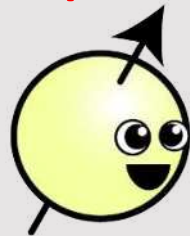




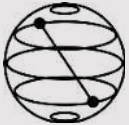
Hello World! building blocks



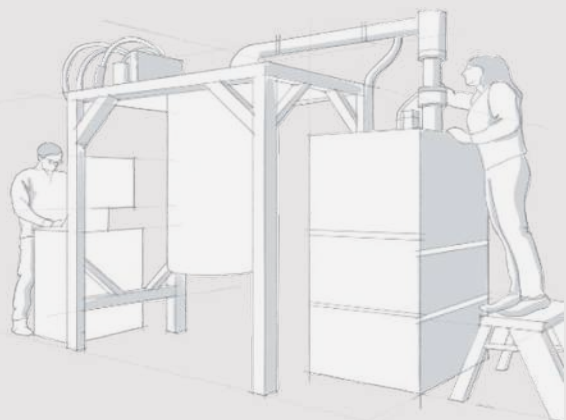
A qubit

 $|1\rangle$ $|0\rangle$

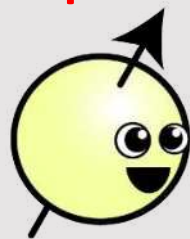
Computational
basis states



Hello World! building blocks

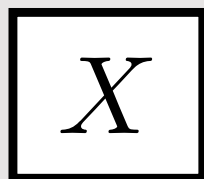


A qubit

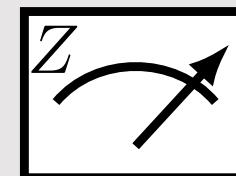
 $|1\rangle$ $|0\rangle$

Computational
basis states

Operations: qubit gate



Measurements: qubit observable



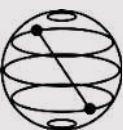
refresher:

$$X |0\rangle = |1\rangle$$

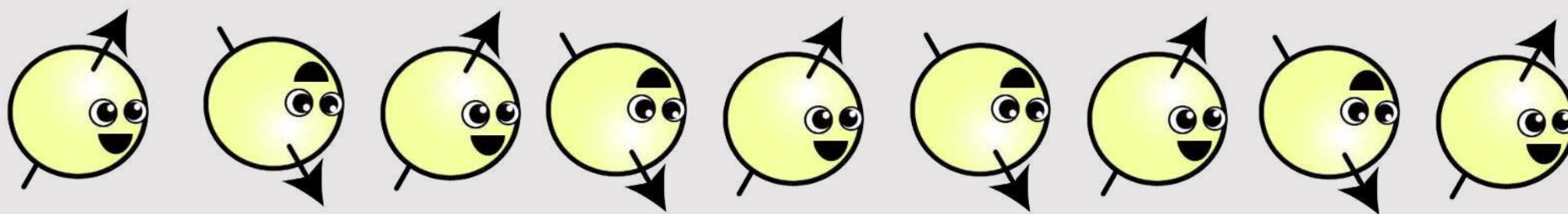
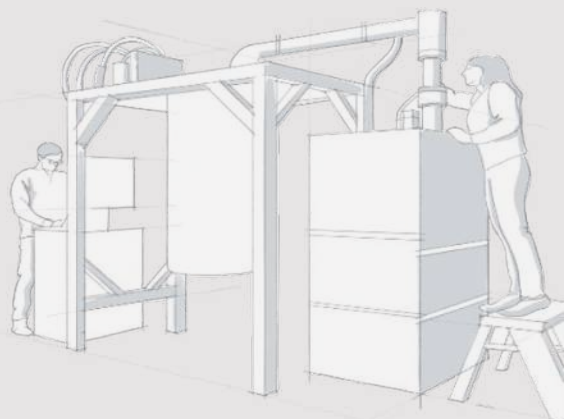
$$X |1\rangle = |0\rangle$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$



Hello World! Even-odd algo: qubit flipper



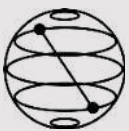
refresher:

$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

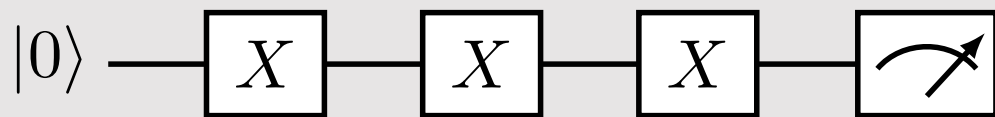
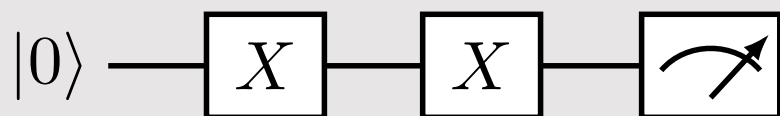
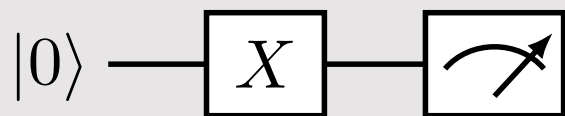
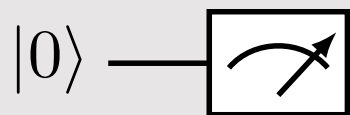
$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$



Hello World! qubit flipper quantum circuits

depth



⋮

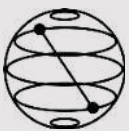
refresher:

$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

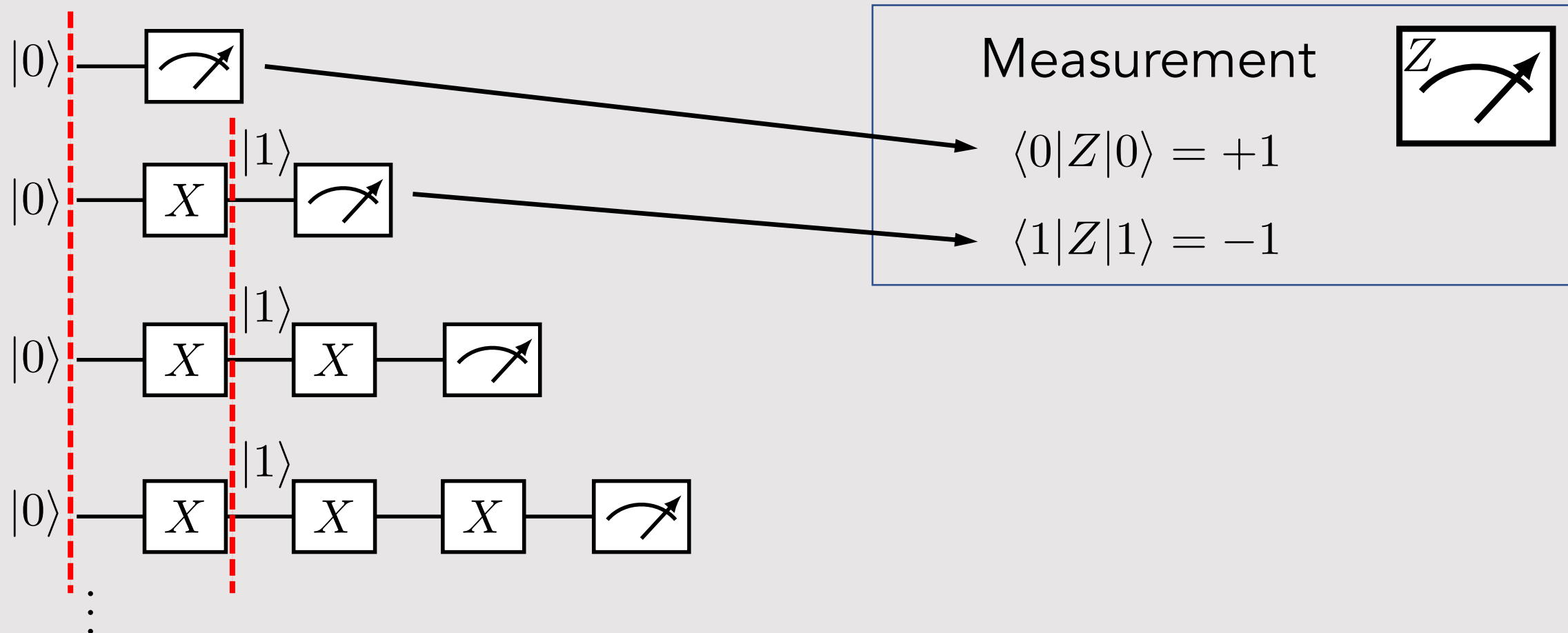
$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$



Hello World! “debugger” step through

depth



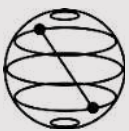
refresher:

$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

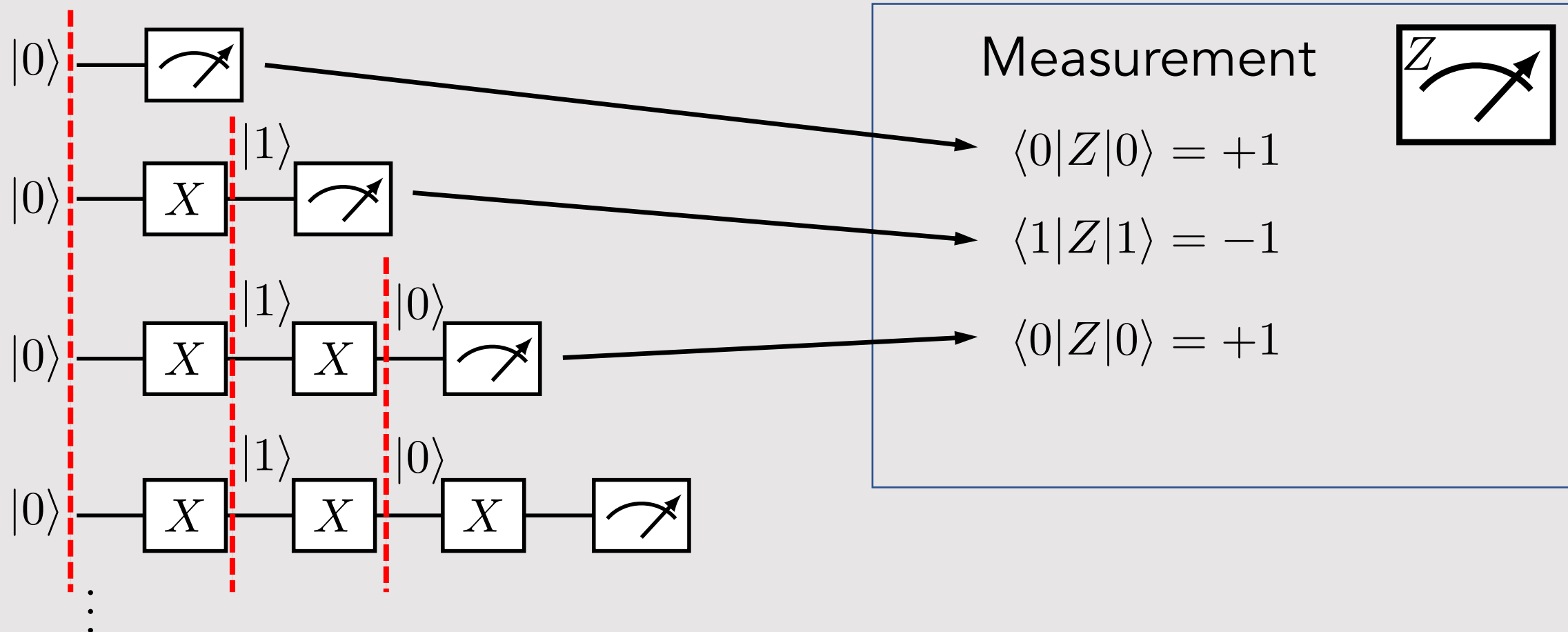
$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$



Hello World! “debugger” step through

depth



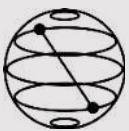
refresher:

$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

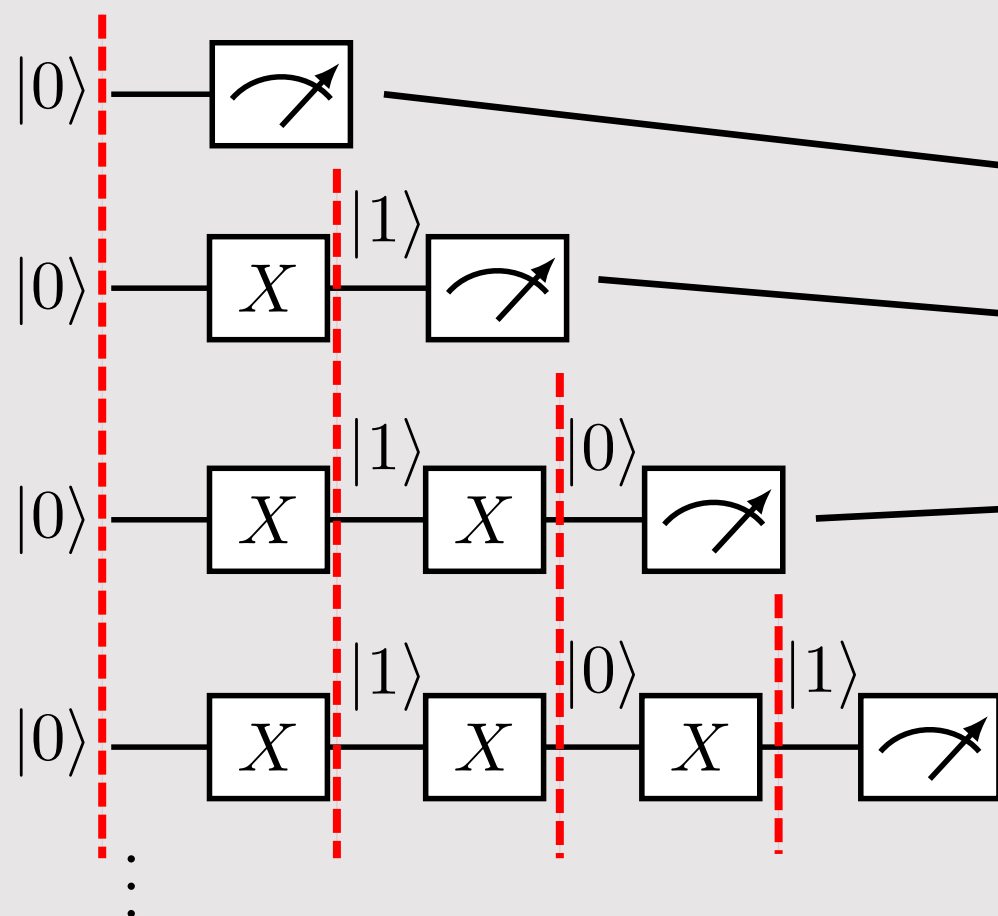
$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

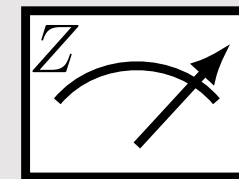


Hello World! “debugger” step through

depth



Measurement



$$\langle 0|Z|0\rangle = +1$$

$$\langle 1|Z|1\rangle = -1$$

$$\langle 0|Z|0\rangle = +1$$

$$\langle Z \rangle = (-1)^d$$

where d is the circuit depth

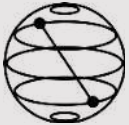
refresher:

$$X |0\rangle = |1\rangle$$

$$X |1\rangle = |0\rangle$$

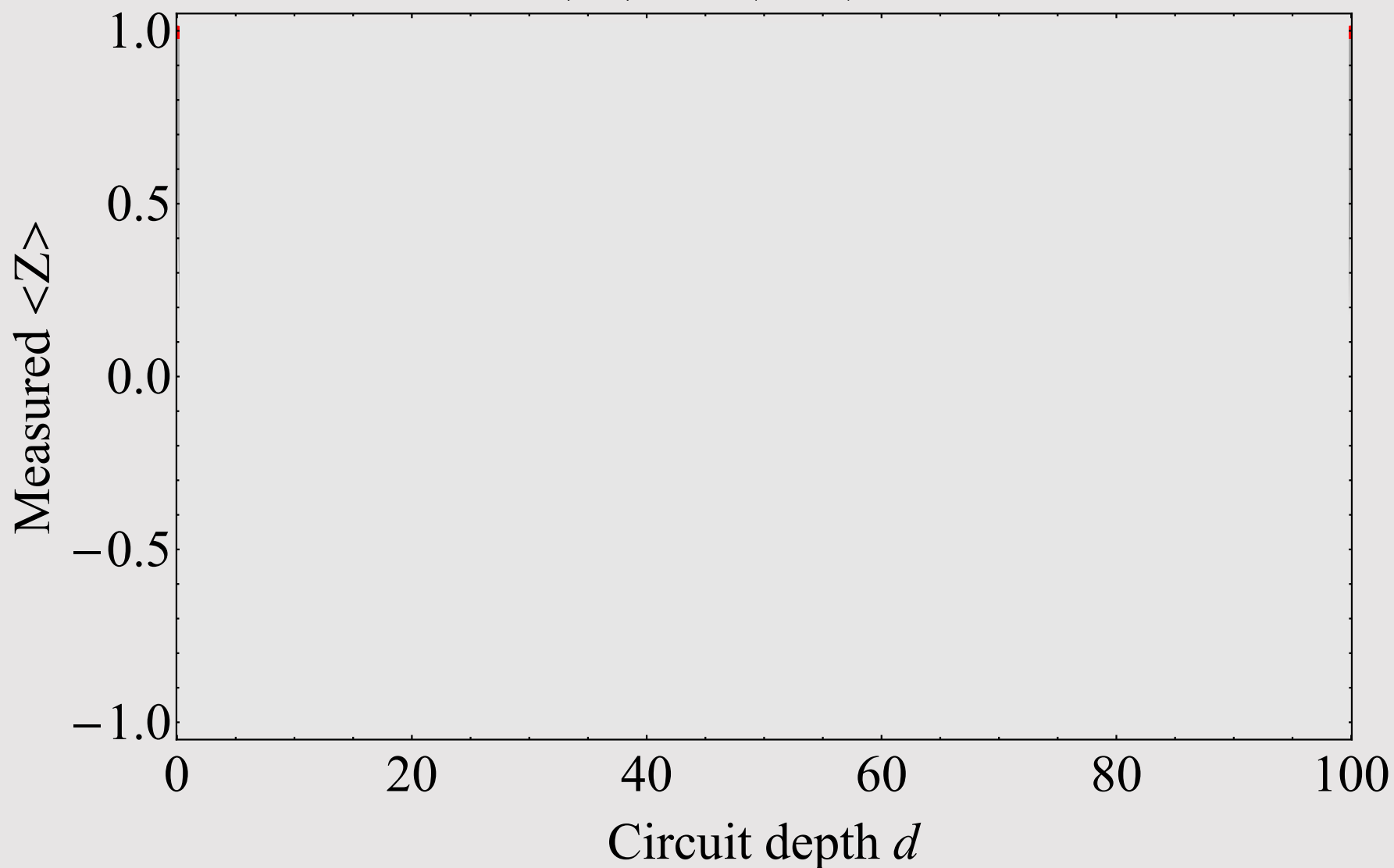
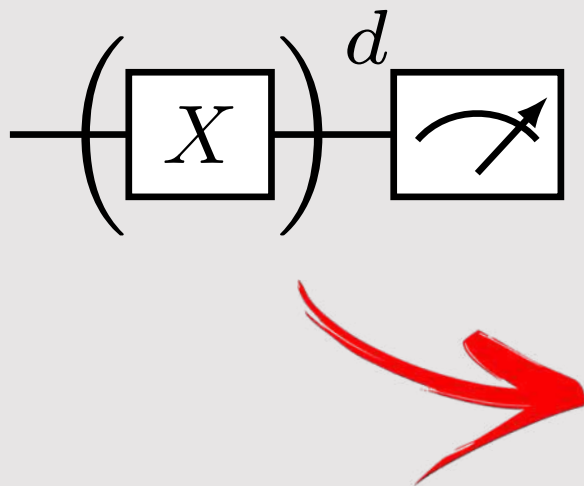
$$Z |0\rangle = +1 |0\rangle$$

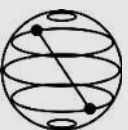
$$Z |1\rangle = -1 |1\rangle$$



Hello World! Ideal expectation results

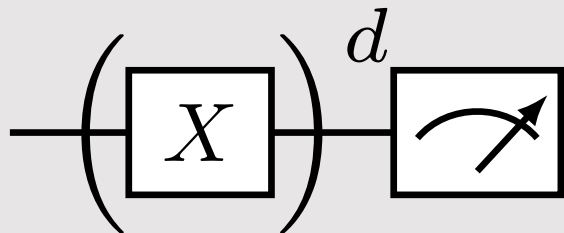
$$\langle Z \rangle = (-1)^d$$



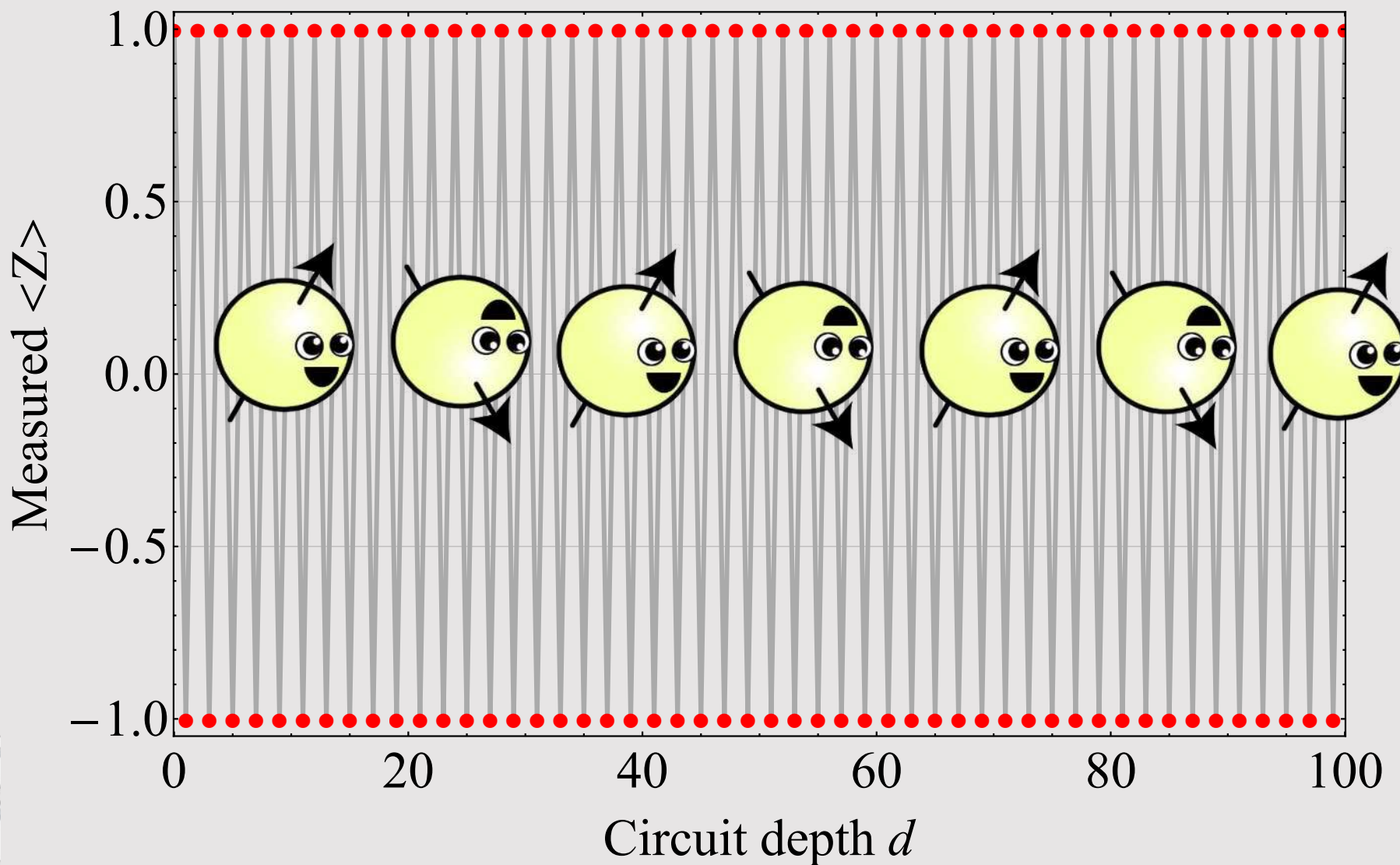
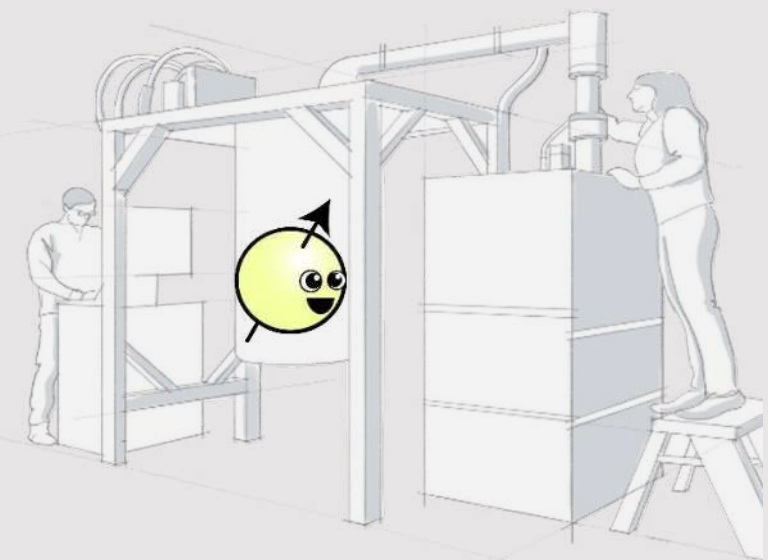


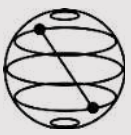
Hello World! Ideal expectation results

$$\langle Z \rangle = (-1)^d$$

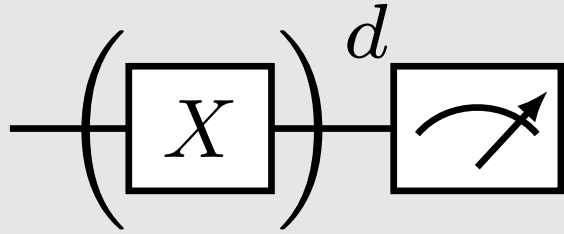


Let's run on a real device!

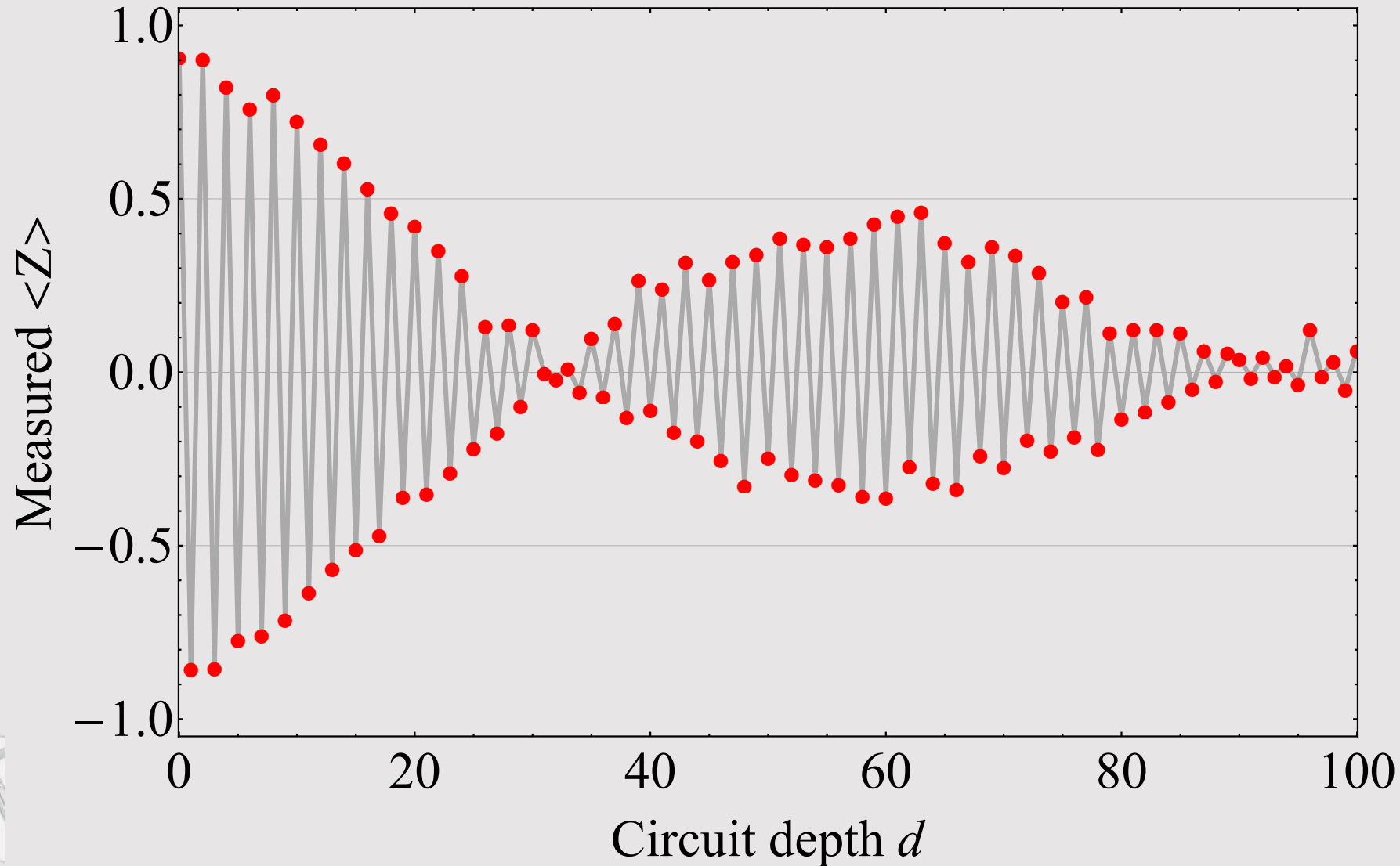
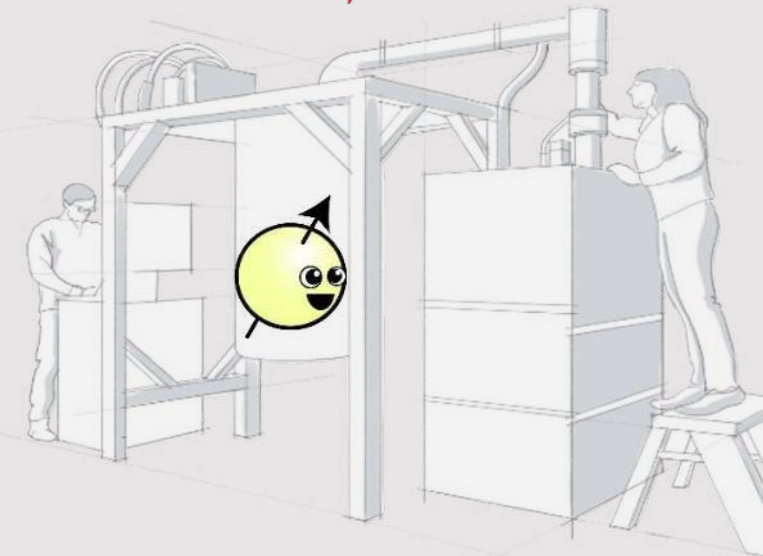


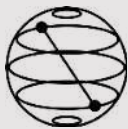


Hello World! Real expectation results



Real data!?!?



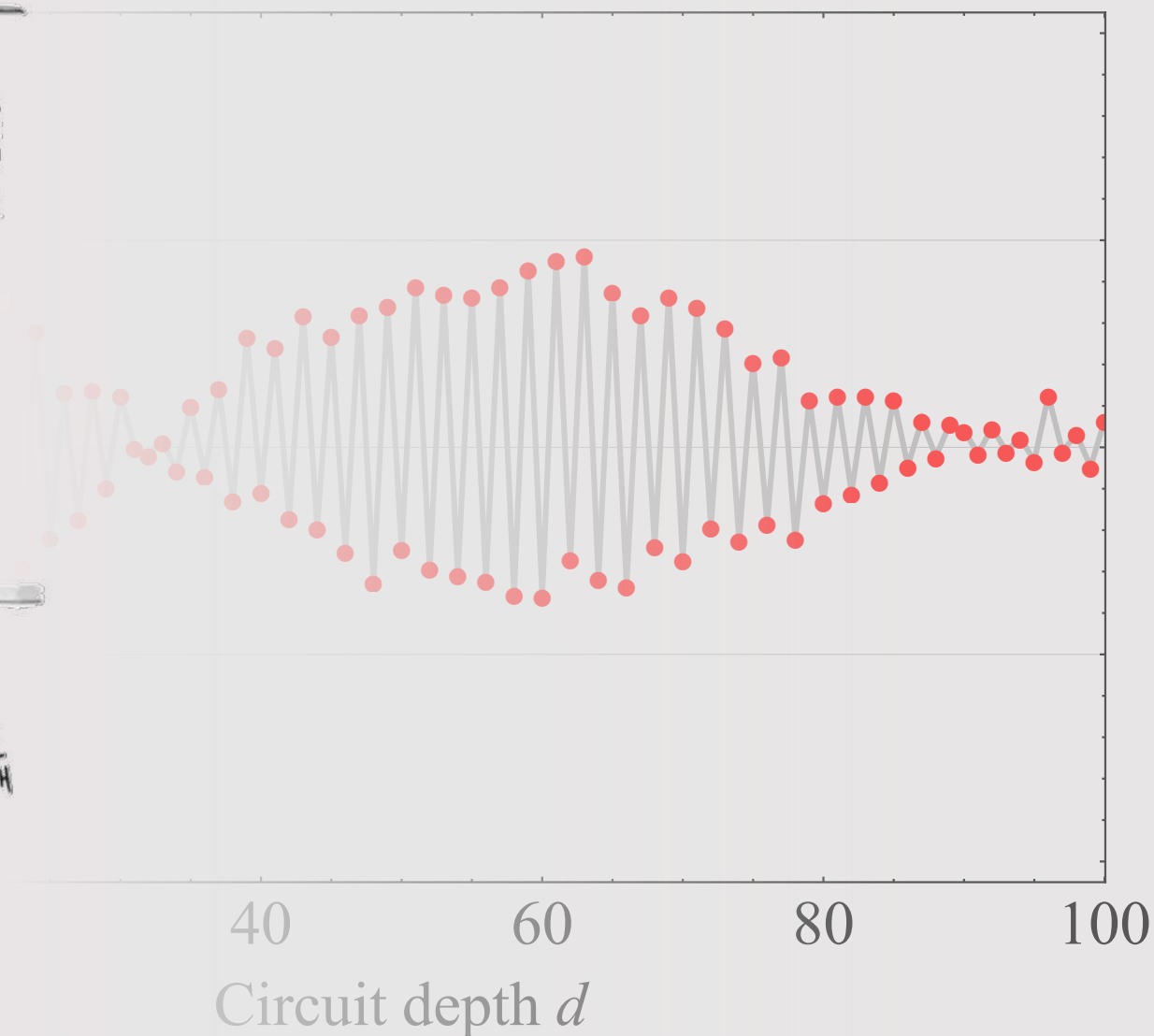


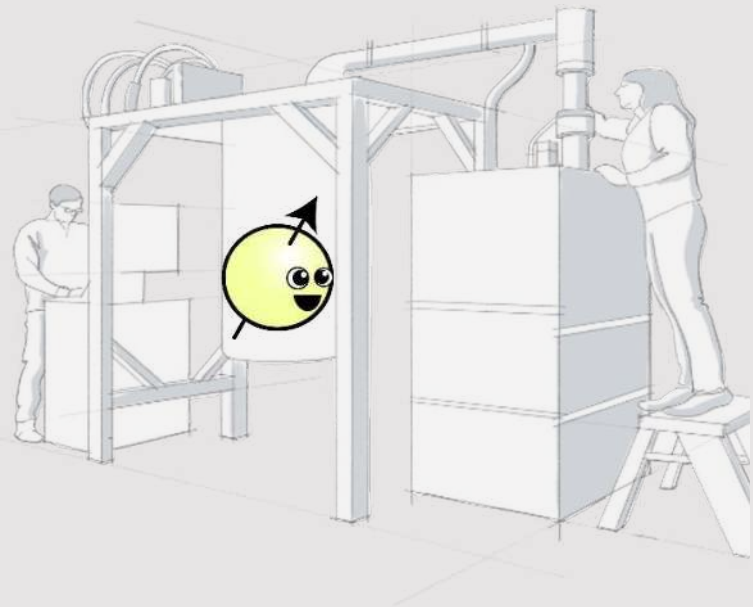
Real & noisy quantum processors: Why study noise?



“Well, your quantum computer is broken in every way possible simultaneously.”

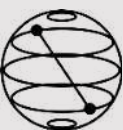
PAUL
NOTH



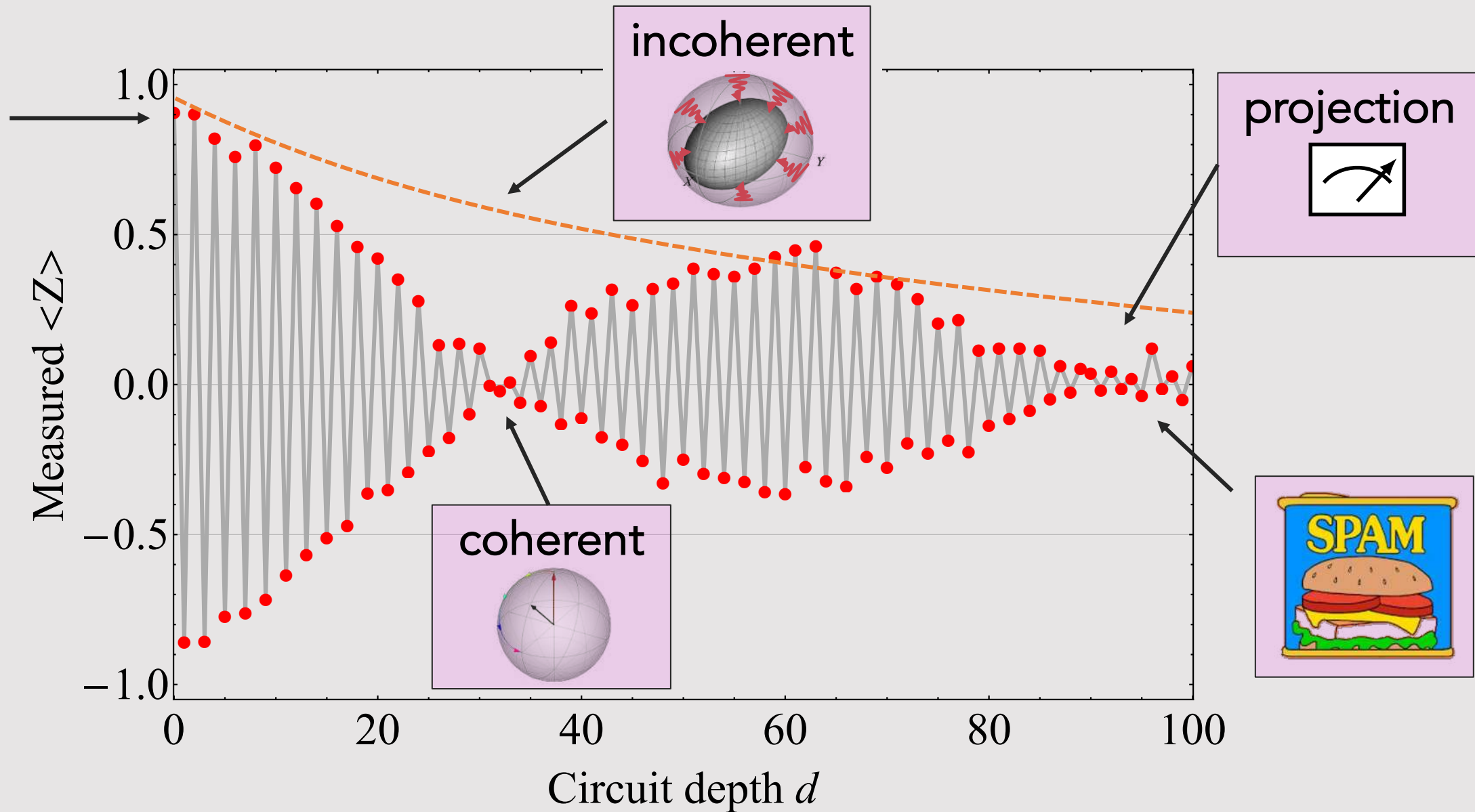


"Quantum phenomena
do *not* occur in a Hilbert space,
they occur in a laboratory."

Asher Peres

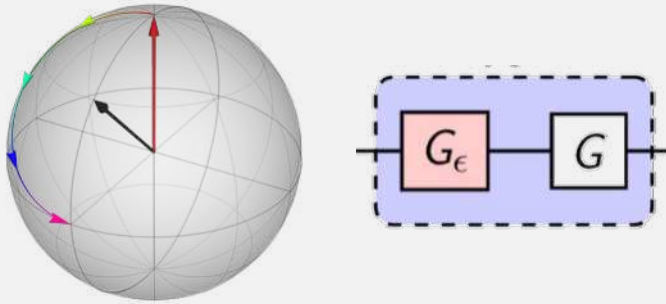


Elements of 🤯 noise

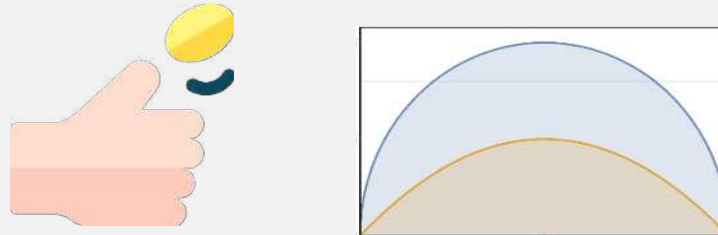


The road ahead

Coherent noise



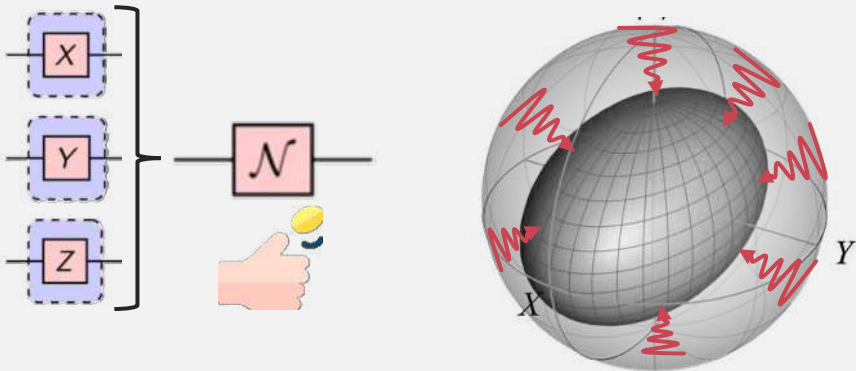
Measurement in a nutshell Projection noise



SPAM: Noisy meters



Incoherent noise

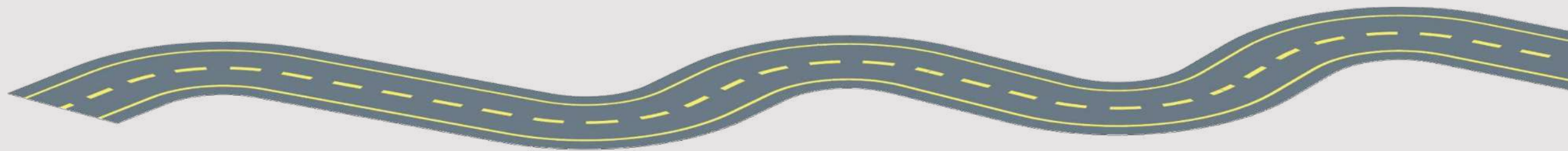


Bonus content Coherent ZZ State preparation



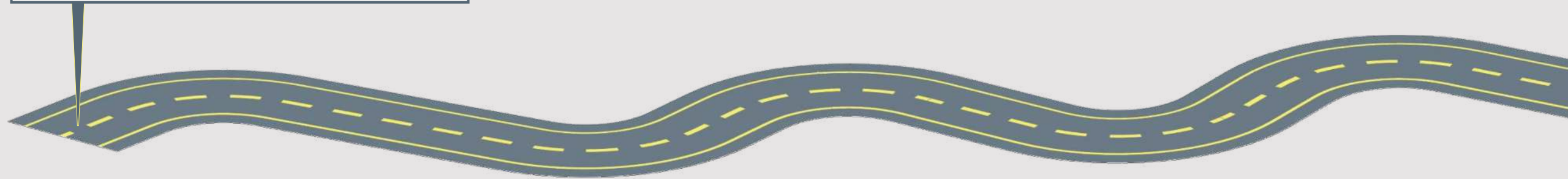
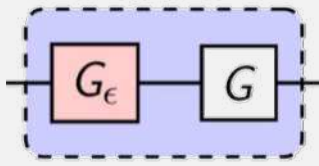
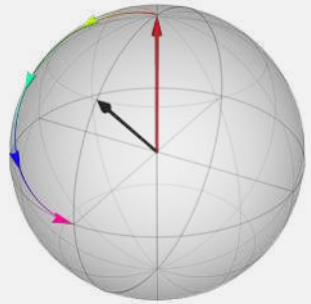
coin toss: flaticon; spam: make it move;
road based on: freepik

The destination is the journey

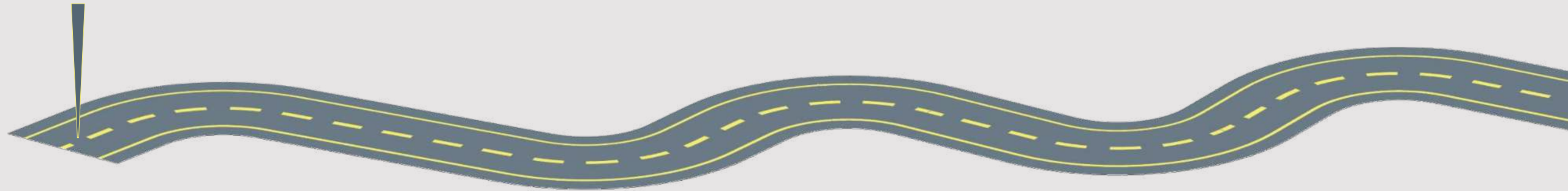
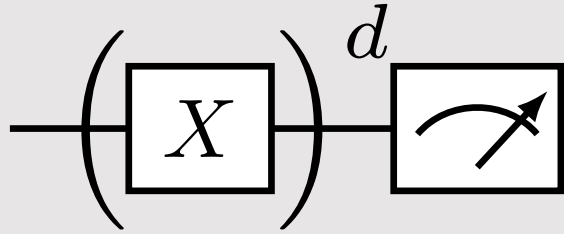


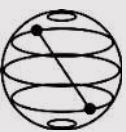
Chapter 2: Coherent noise

Coherent



Return to the Hello World example

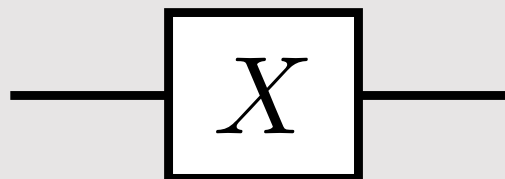




Origin of our X gate: time evolution

$$X = R_X(\pi)$$

(up to global phase)



Refresher:

$$\hat{H} = \frac{\hbar\omega}{2} X$$

* We will often drop hats on Paulis I, X, Y, Z

$$= \frac{\hbar\omega}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

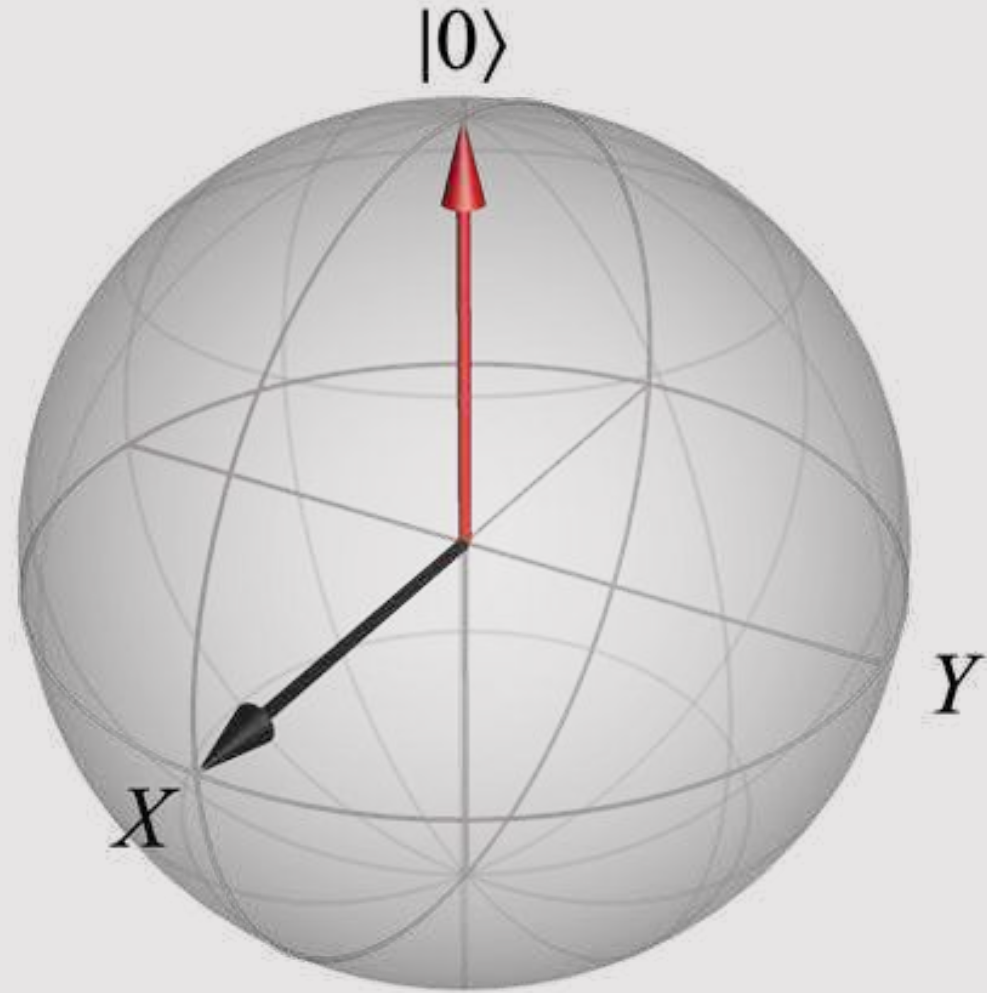
$$U(t) = \exp\left(-it\hat{H}/\hbar\right)$$

$$\theta := \omega t$$

$$R_X(\theta) = \exp\left(-\frac{i\theta}{2} X\right)$$

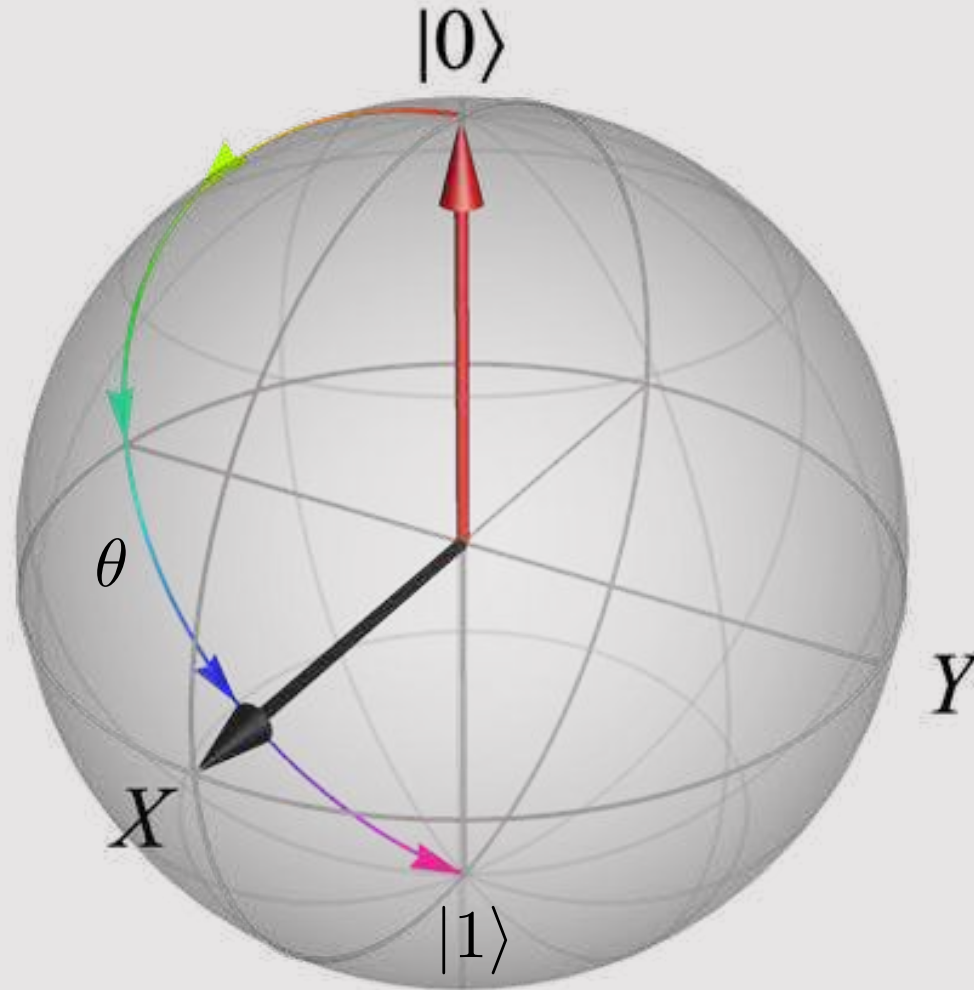
$$= \cos(\theta/2)I - i \sin(\theta/2)X$$

Visualize: Bloch sphere



Visualize: Evolution on the Bloch sphere

$$R_X(\theta) = \exp\left(-\frac{i\theta}{2}X\right)$$

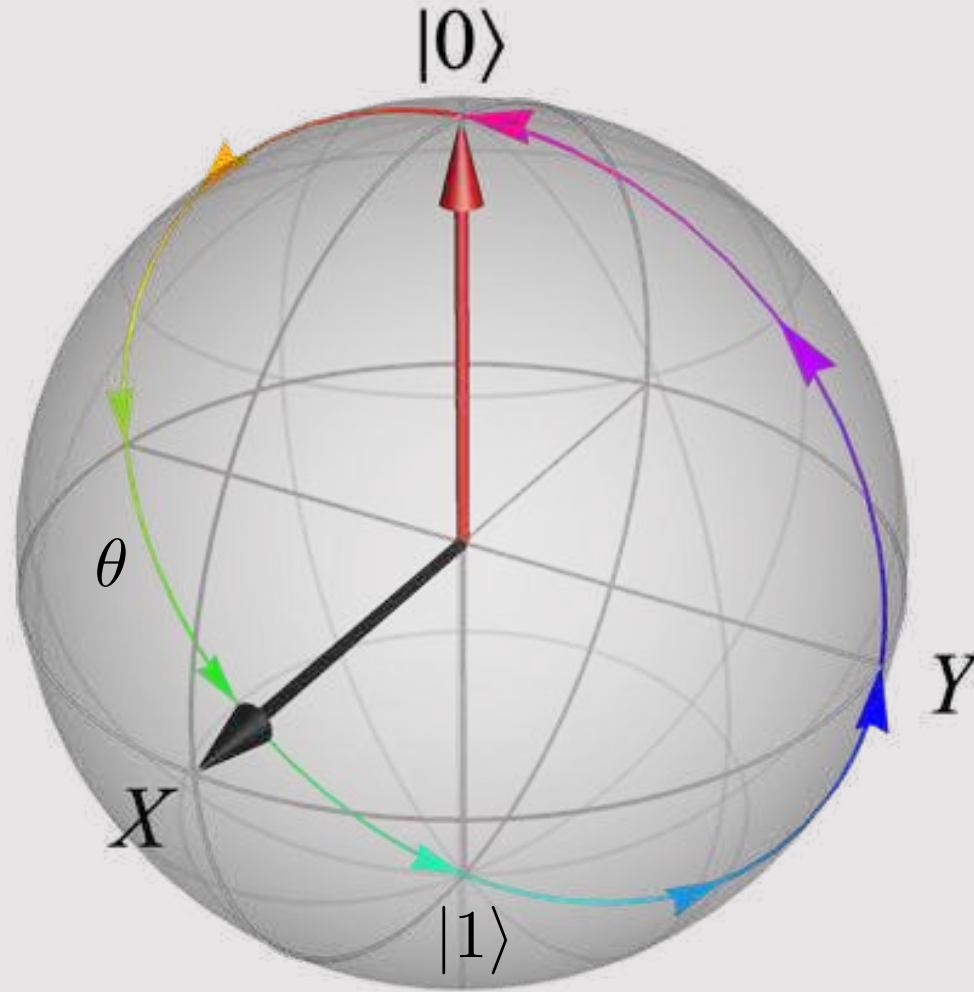


Visualize: Evolution on the Bloch sphere

$$R_X(\theta) = \exp\left(-\frac{i\theta}{2}X\right)$$

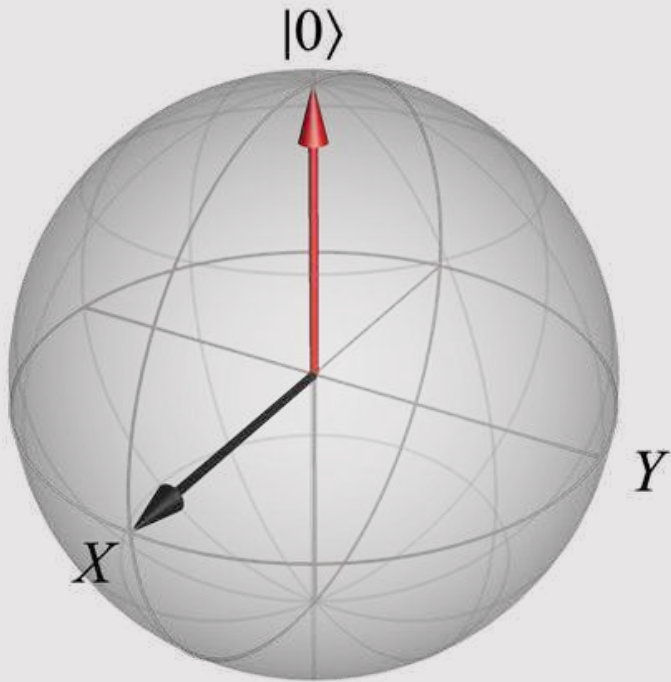
$$X|0\rangle = |1\rangle$$

$$X|1\rangle = |0\rangle$$

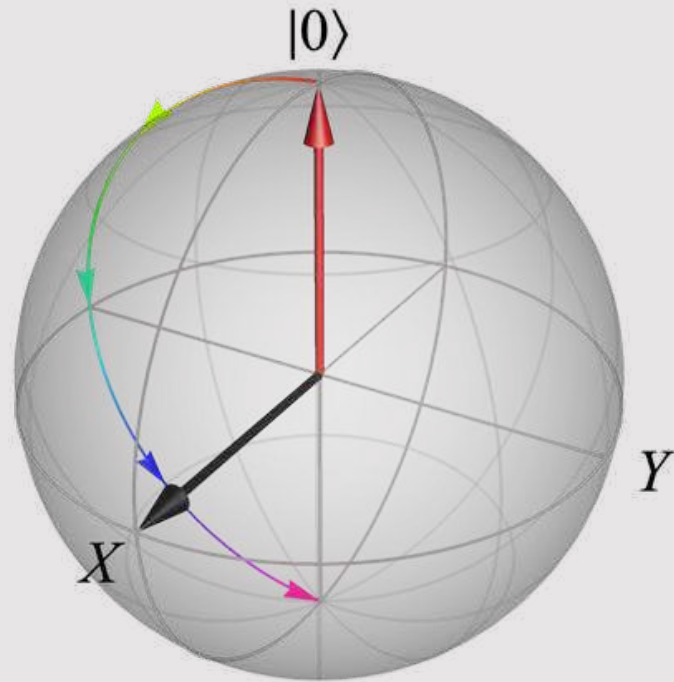


Evolution on the Bloch sphere

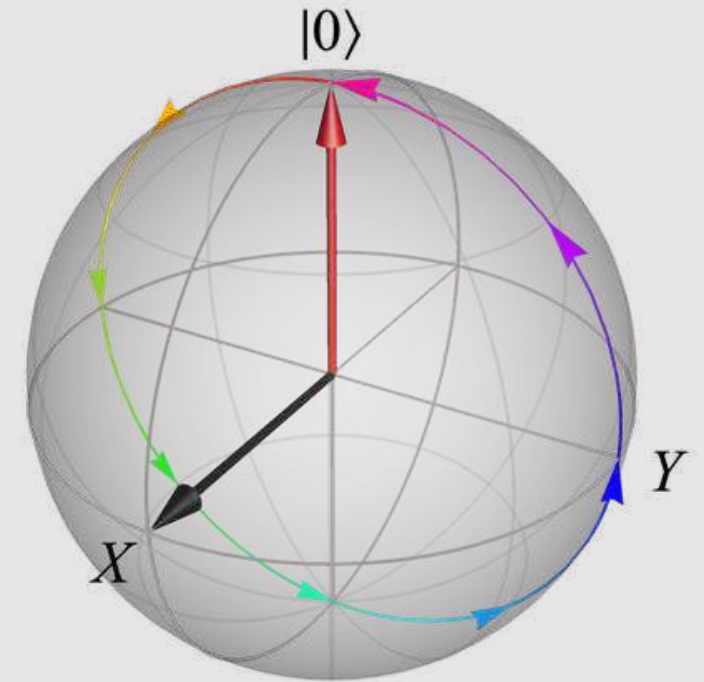
$|0\rangle$

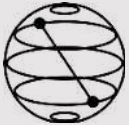


$|0\rangle$ — X —



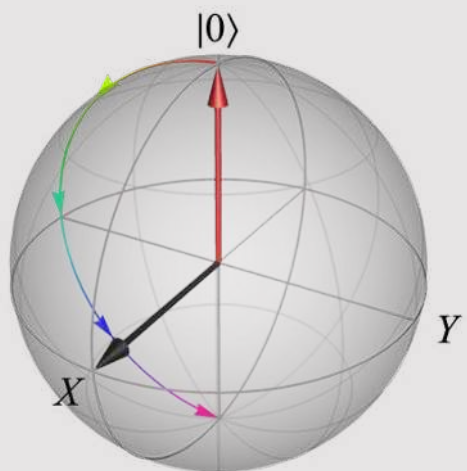
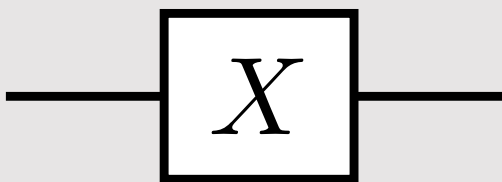
$|0\rangle$ — X — X —



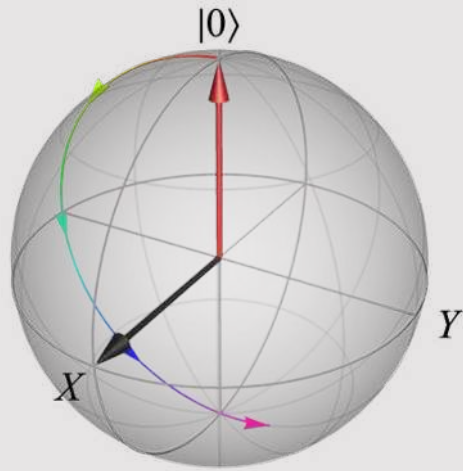
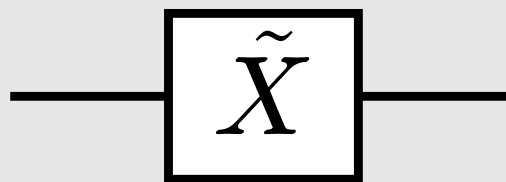


Miscalibrated gate

Ideal gate

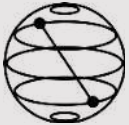


Noisy gate



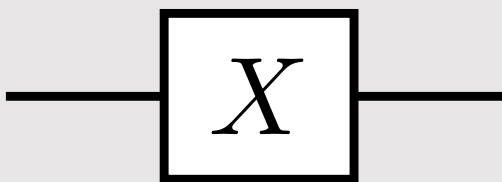
$$X = R_X(\pi)$$

$$\tilde{X} := R_X(\pi + \epsilon)$$

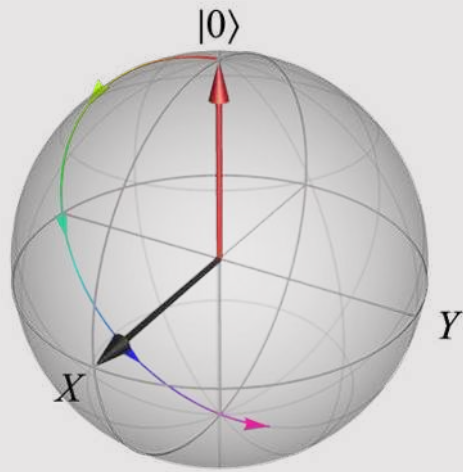
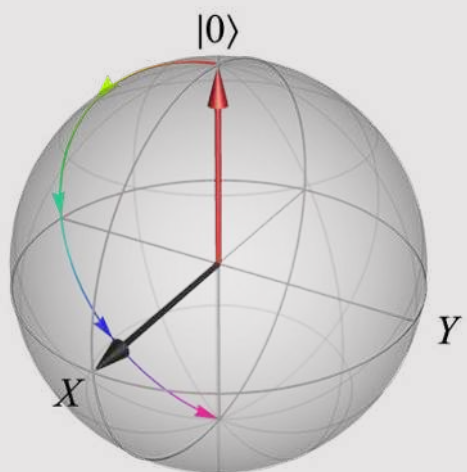
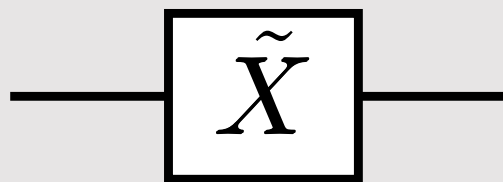


Miscalibrated gate

Ideal gate



Noisy gate



Shown

$$R_X (\theta + \phi) = R_X (\theta) R_X (\phi)$$

$$X = R_X (\pi)$$

$$\tilde{X} := R_X (\pi + \epsilon)$$

$$\begin{aligned} R_X (\theta) &= \exp \left(-\frac{i\theta}{2} X \right) \\ &= \cos (\theta/2) I - i \sin (\theta/2) X \end{aligned}$$

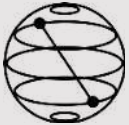
$$= \exp \left(-i \frac{\pi + \epsilon}{2} X \right)$$

$$= \exp \left(-i \frac{\pi}{2} X - i \frac{\epsilon}{2} X \right)$$

$$= \exp \left(-i \frac{\pi}{2} X \right) \exp \left(-i \frac{\epsilon}{2} X \right)$$

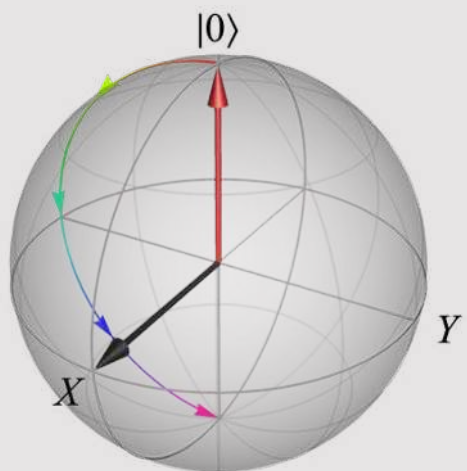
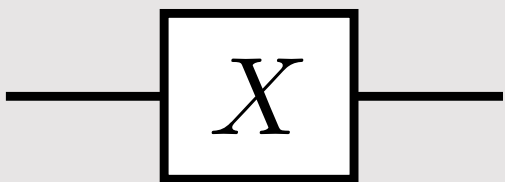
$$= R_X (\epsilon) R_X (\pi)$$

$$= X_\epsilon X$$

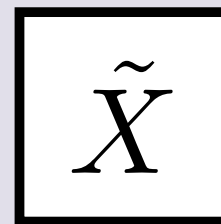
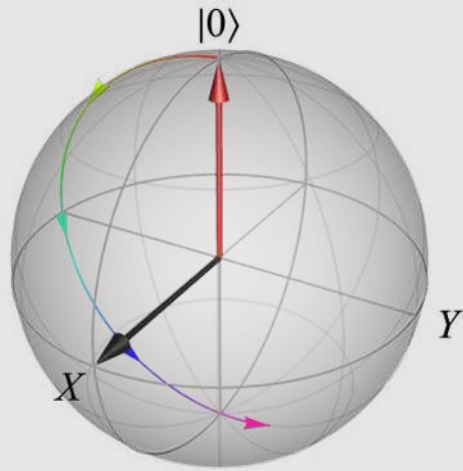
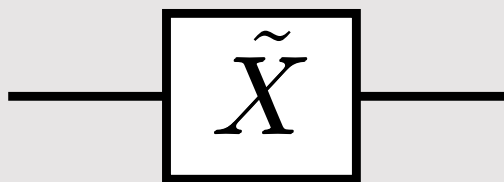


Noisy gate decomposition

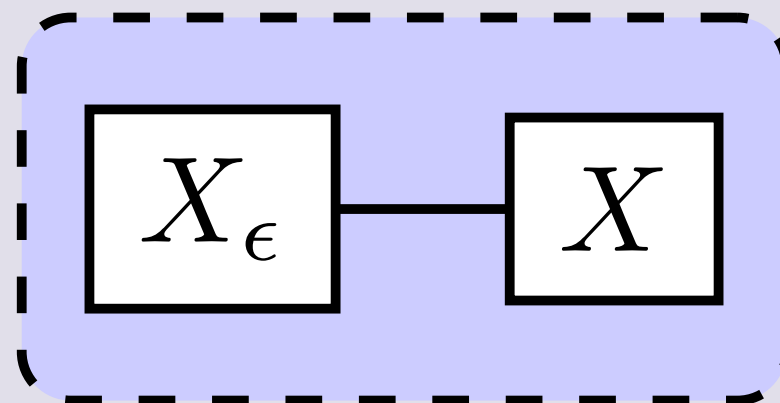
Ideal gate



Noisy gate



$\equiv$





Careful



Common pitfall

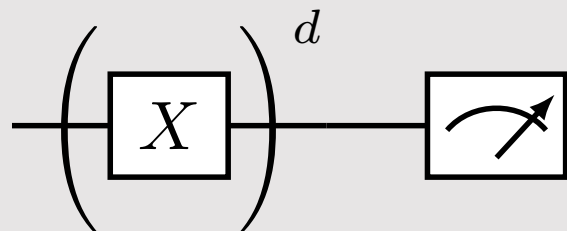
The order in which gates appear in a schematic is the reverse of how they appear in the algebra.

$$\text{---} \boxed{X} \text{---} \boxed{Y} \text{---} = \text{---} \boxed{YX} \text{---}$$



Using a noisy gate in a quantum circuit

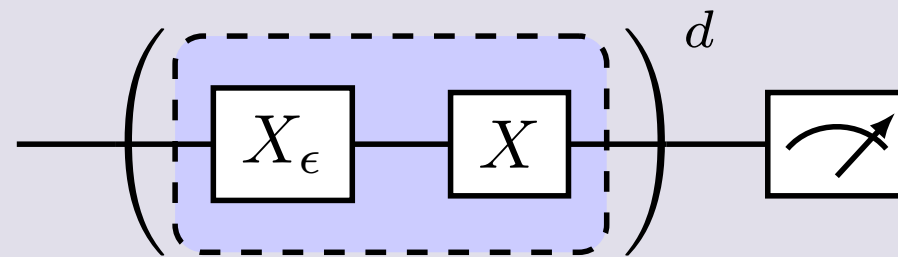
Ideal



$$X = R_X(\pi)$$

$$\begin{aligned} U_{\text{total}} &= X^d \\ &= [R_X(\pi)]^d \\ &= R_X(d\pi) \end{aligned}$$

Noisy



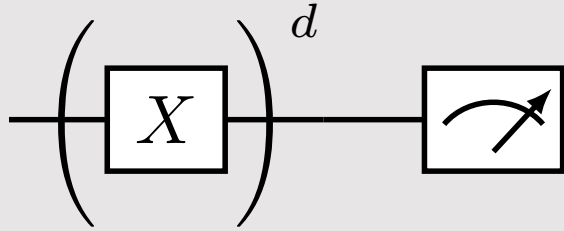
$$\tilde{X} := R_X(\pi + \epsilon) = X_\epsilon X$$

$$\begin{aligned} \tilde{U}_{\text{total}} &= \tilde{X}^d \\ &= [R_X(\epsilon) R_X(\pi)]^d \\ &= R_X(d\epsilon) R_X(d\pi) \\ \tilde{U}_{\text{total}} &= R_X(d\epsilon) U_{\text{total}} \end{aligned}$$



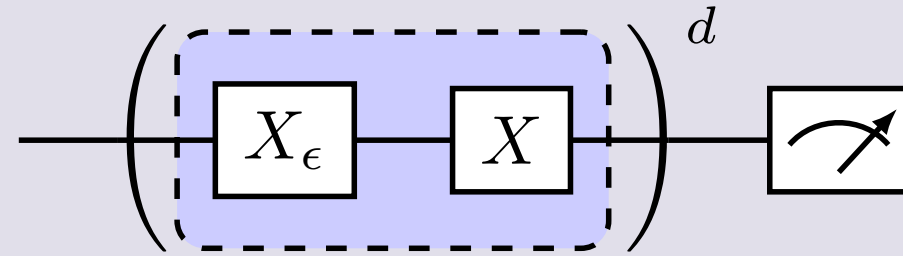
Using a noisy gate in a quantum circuit: final state

Ideal

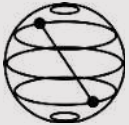


$$U_{\text{total}} = X^d = R_X(d\pi)$$

Noisy

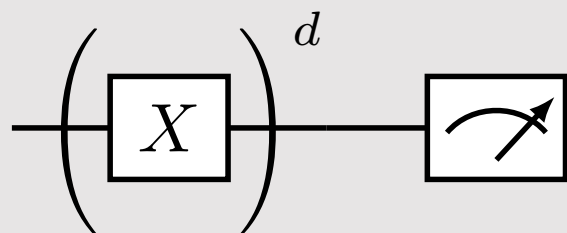


$$\tilde{U}_{\text{total}} = R_X(d\epsilon) U_{\text{total}}$$

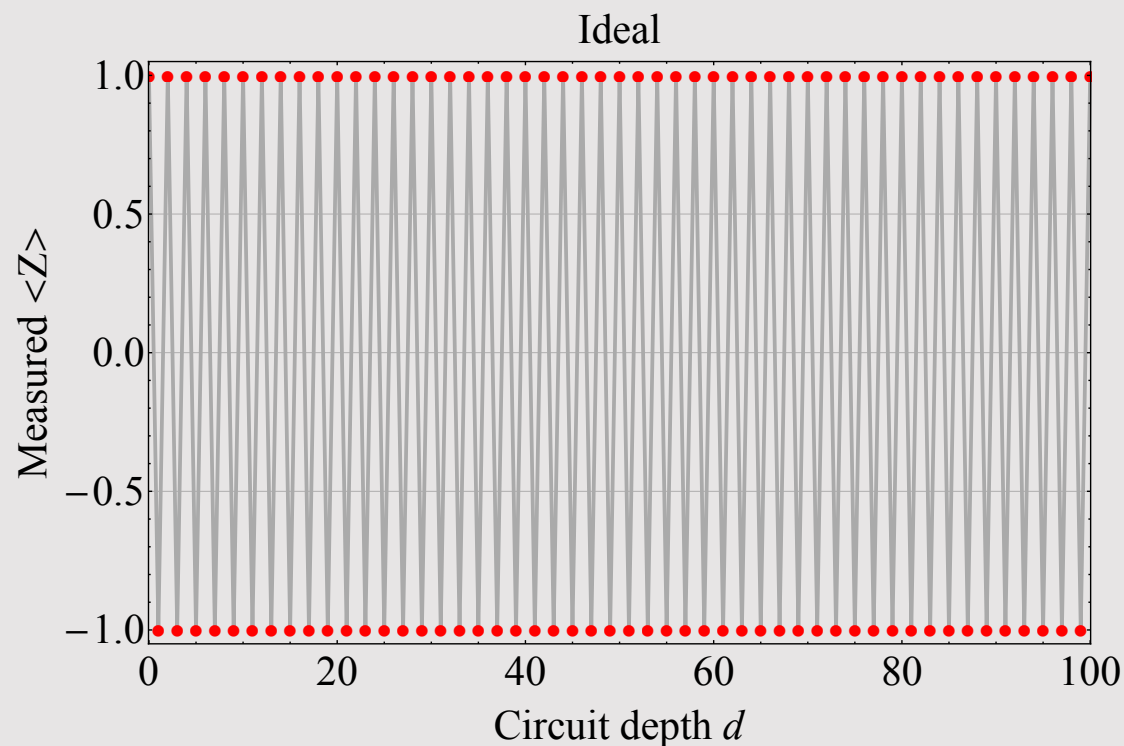


Ideal vs. noisy observable

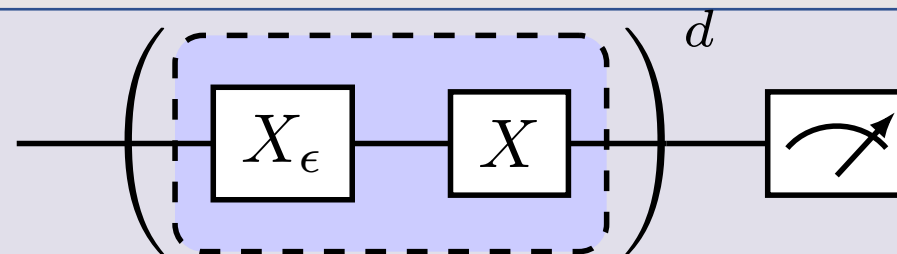
Ideal



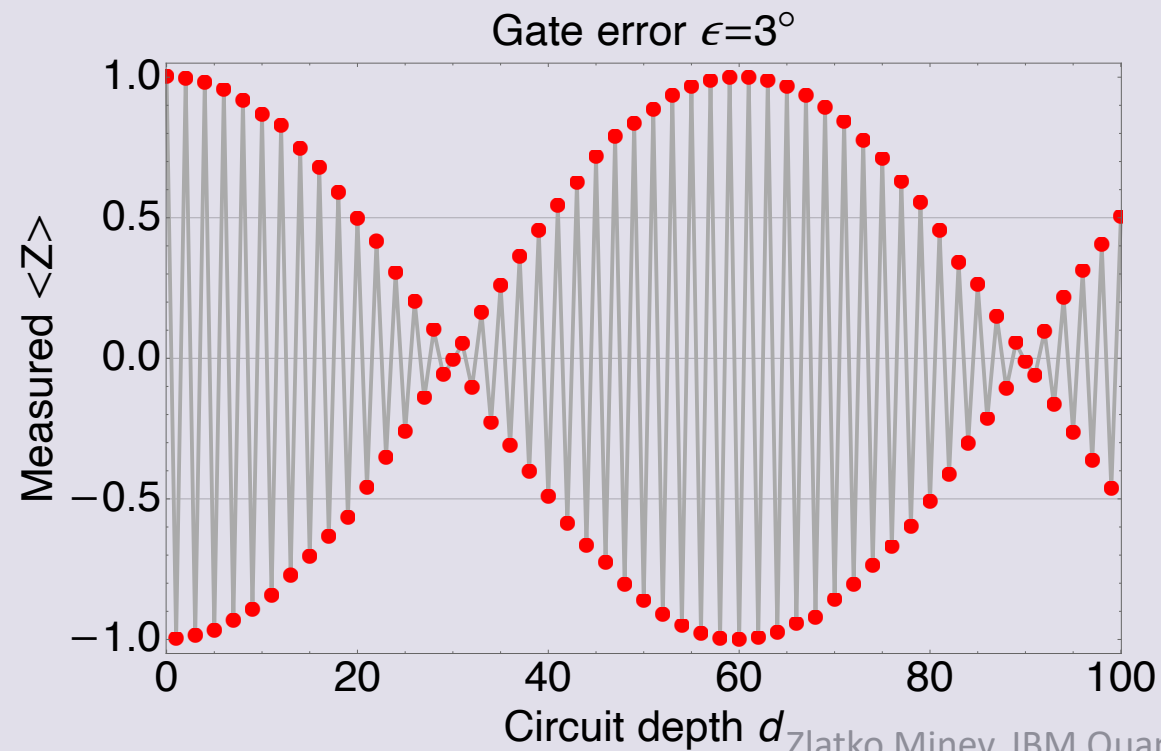
$$\langle \psi_f | Z | \psi_f \rangle = \cos(d\pi) = (-1)^d$$

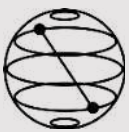


Noisy



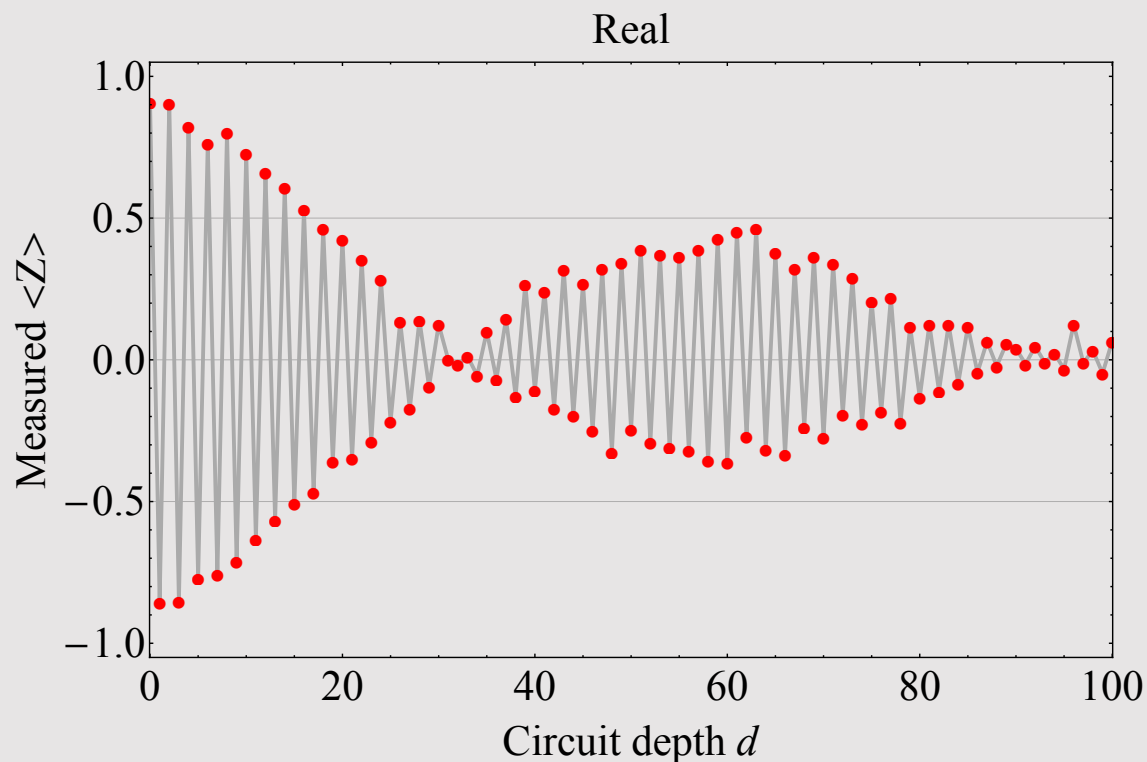
$$\langle \tilde{\psi}_f | Z | \tilde{\psi}_f \rangle = \cos(d\pi + d\epsilon)$$



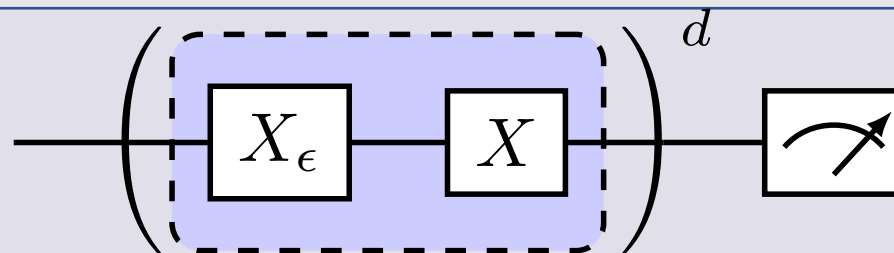


Compare to full experiment

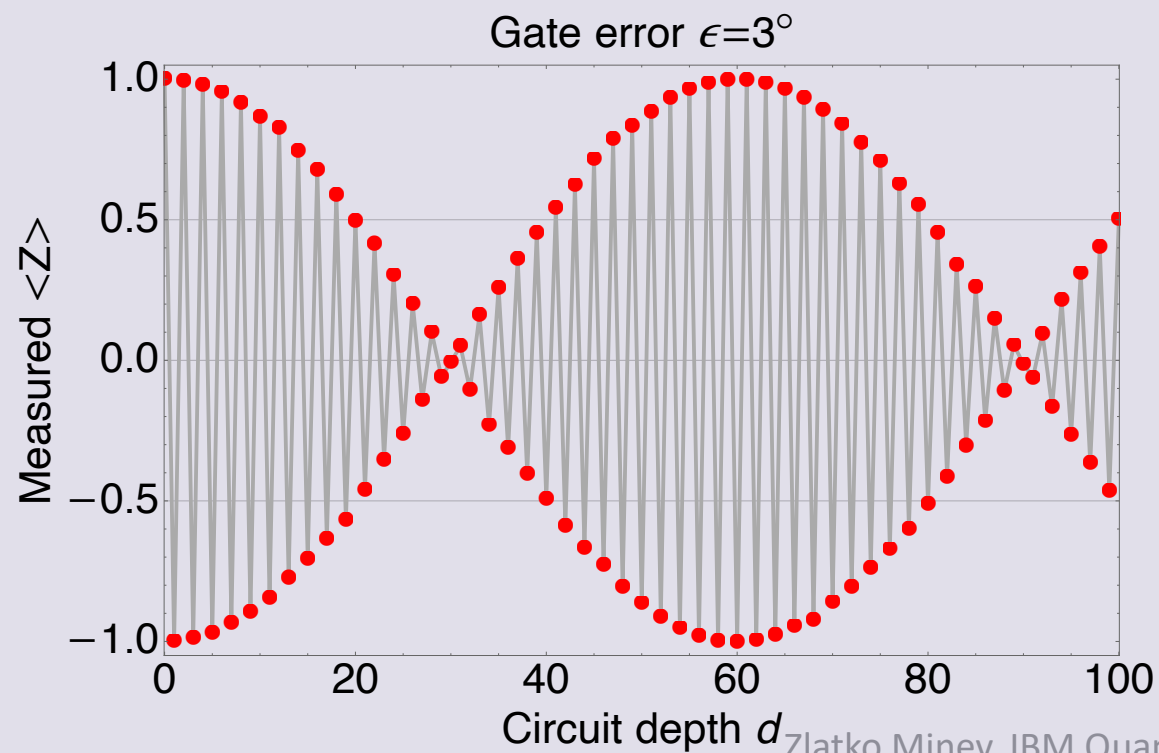
Full experiment



Noisy



$$\langle \tilde{\psi}_f | Z | \tilde{\psi}_f \rangle = \cos(d\pi + d\epsilon)$$





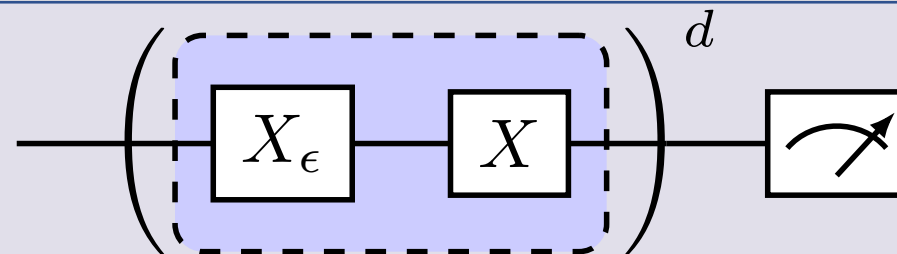
Coherent error is bad, quadratically so

Coherent errors have a *quadratic* impact on algorithmic accuracy (worst-case error)

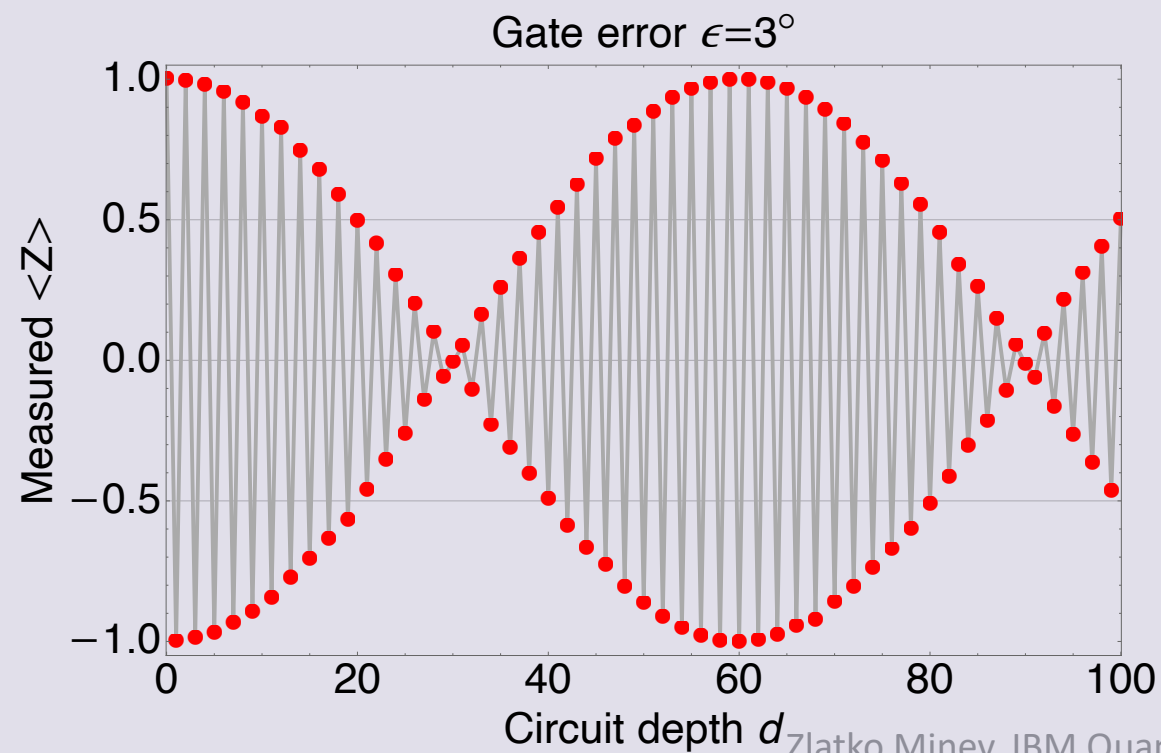
$$\langle \tilde{\psi}_f | Z | \tilde{\psi}_f \rangle - \langle \psi_f | Z | \psi_f \rangle$$

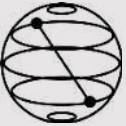
$$\cos(x) = 1 - \frac{1}{2}x^2 + \mathcal{O}(x^3)$$

Noisy



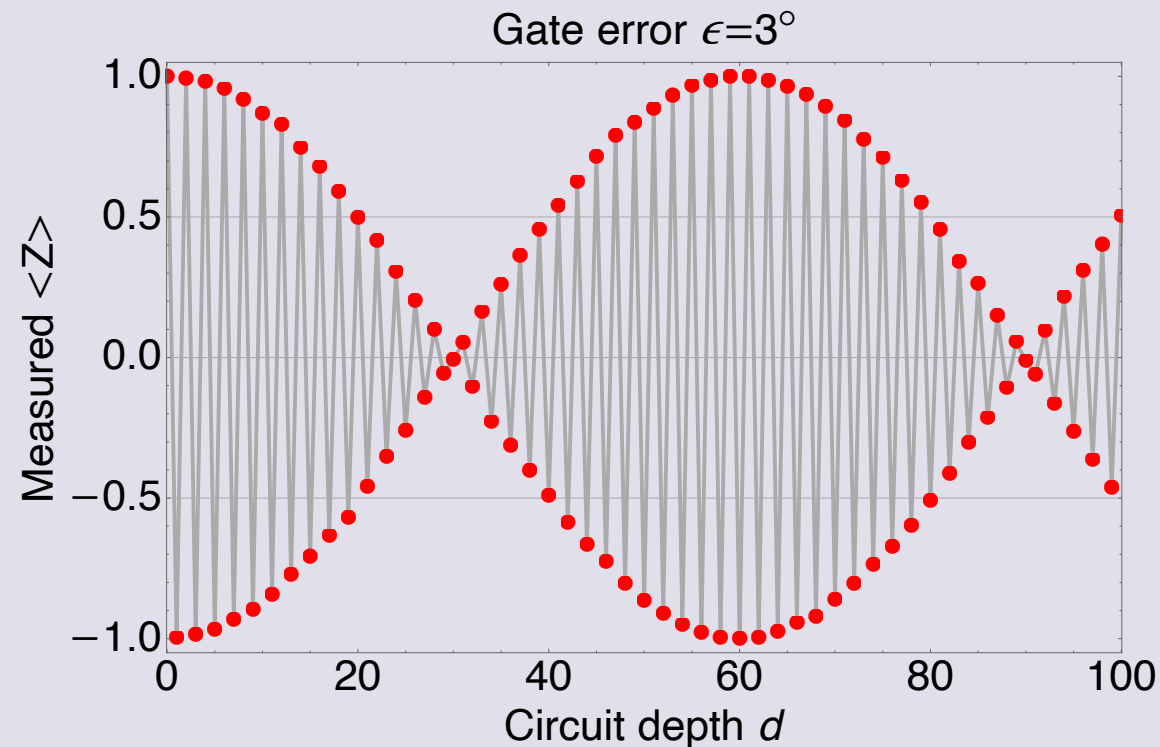
$$\langle \tilde{\psi}_f | Z | \tilde{\psi}_f \rangle = \cos(d\pi + d\epsilon)$$



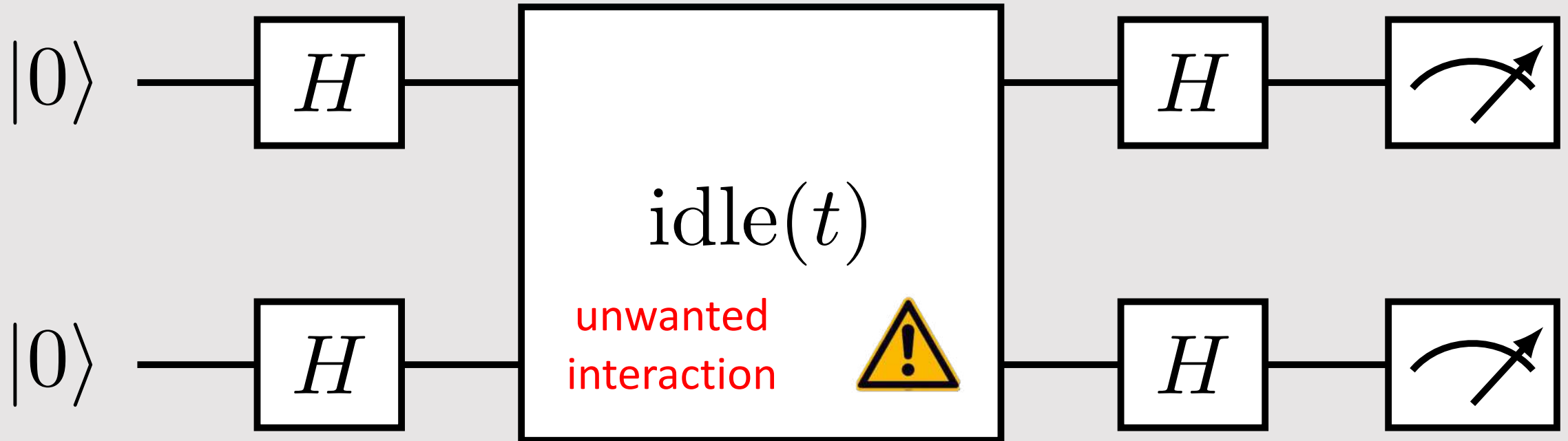


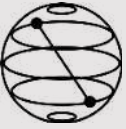
Coherent errors: brief summary

- are ubiquitous
- can be described by unitary operations
- do not lose quantum information
- data: can create oscillations in the data
- data: do not yield exponential decays
- have a *quadratic* impact on algorithmic accuracy (worst-case error)



Bonus content: two-qubit coherent ZZ error





Questions

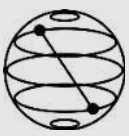
Answer these multiple-choice questions
in the chat; for example, type “1a 2b.”

1. Coherent noise can be caused by

- a) loss of energy of the qubit
- b) miscalibration, such as over-rotation
- c) unwanted coupling to neighboring qubit

2. Coherent noise can be really bad because

- a) it results in loss of information
- b) you cannot undo it
- c) the worst-case error often grows quadratically



Advanced questions to dive deeper

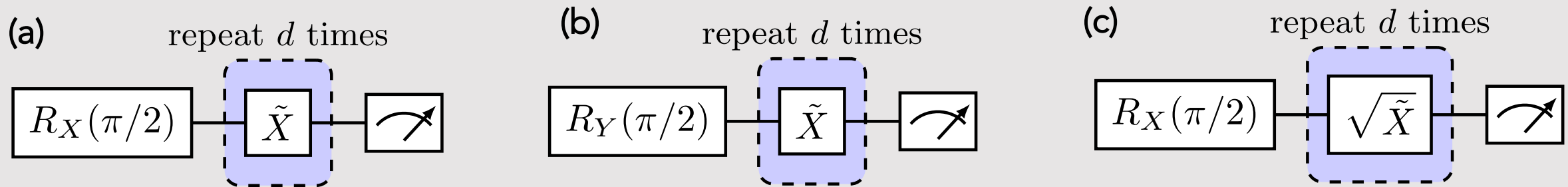


1. Amplitude-error amplification sequences.

(A) Calculate the expectation value $\langle Z(d) \rangle$ as a function of depth d of the sequence for the following sequences. Assume the noise model $\tilde{X} := R_X(\pi + \epsilon) = X_\epsilon X$

(B) How could you use this result to fine-tune your gates?

(C) Can you come up with an alternative or more clever error-amplification sequence?

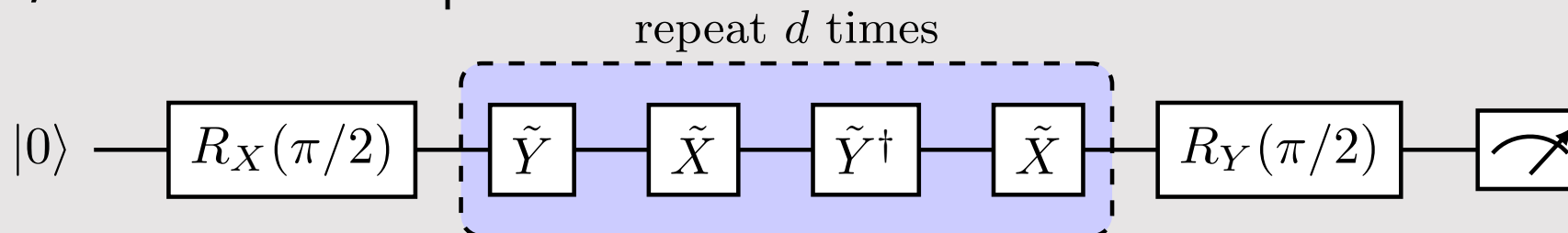


2. Phase-error amplification sequences. Use the noise model:

Repeat (A), (B), and (C) for Exercise (1) for the following sequences and assuming a phase error between the X and Y gates, rather than an amplitude error.

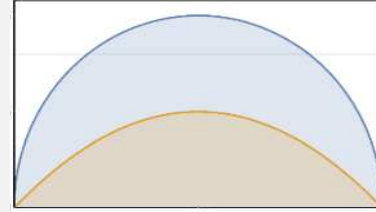
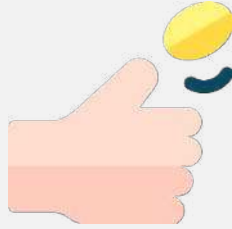
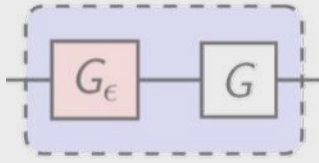
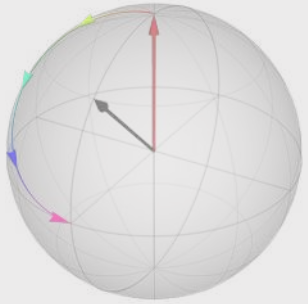
$$\tilde{X} = X,$$

$$\tilde{Y} = \exp \left[-i \frac{\pi}{2} (\cos \epsilon Y + \sin \epsilon X) \right].$$



Chapter 3: Measurement theory in a nutshell + projection noise

Coherent

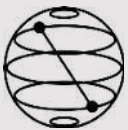


coin toss: flaticon; spam: make it move;
road based on: freepik

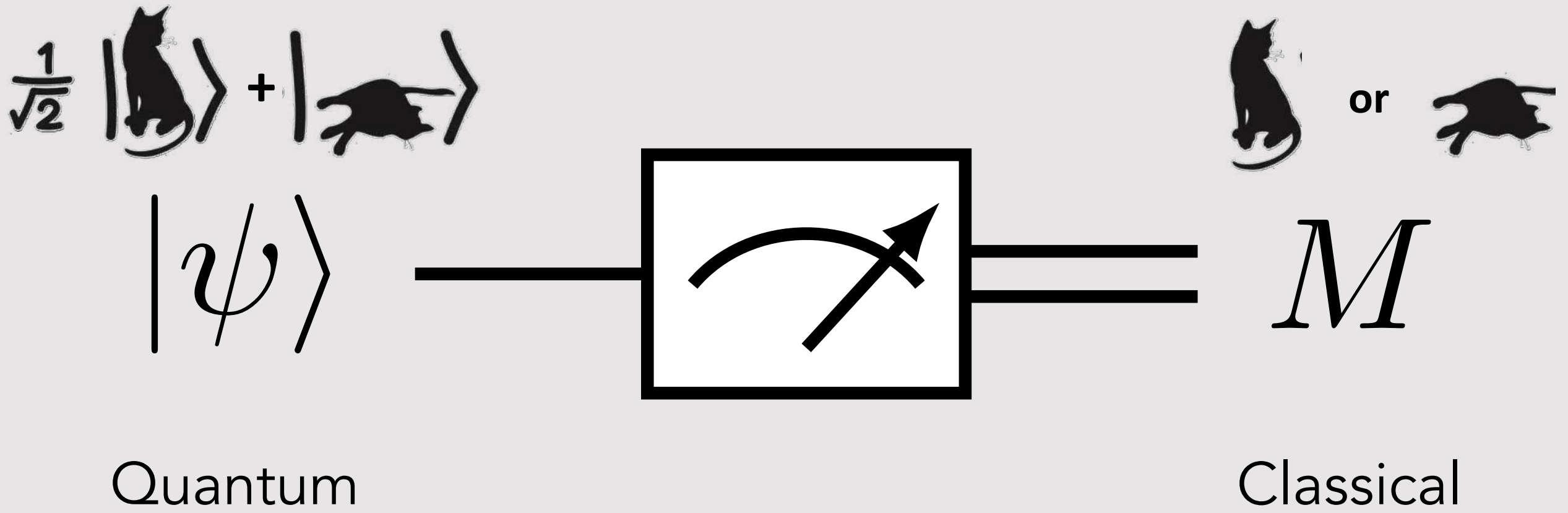
Quantum Measurement

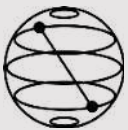


Bohr and Einstein (1925)
Source: Wikimedia

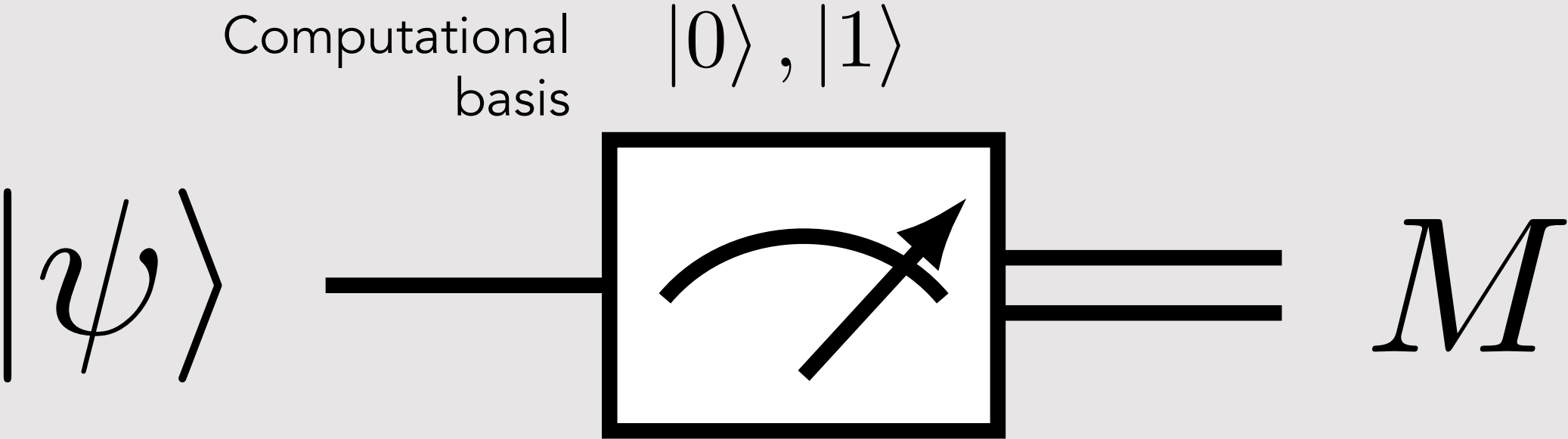


Measurement apparatus: general

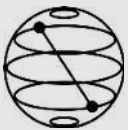




Measurement basis



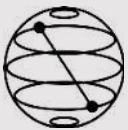
outcome	outcome probability
$M = 0$	$p(M = 0) = \langle 0 \psi \rangle ^2$
$M = 1$	$p(M = 1) = \langle 1 \psi \rangle ^2$



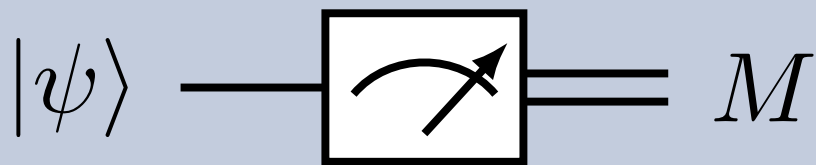
Resolution of the identity

$$|\psi\rangle \text{ --- } \boxed{\text{meter symbol}} = M$$

outcome	outcome probability
$M = 0$	$p(M = 0) = \langle 0 \psi \rangle ^2$
$M = 1$	$p(M = 1) = \langle 1 \psi \rangle ^2$

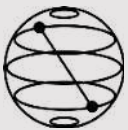


Resolution of the identity

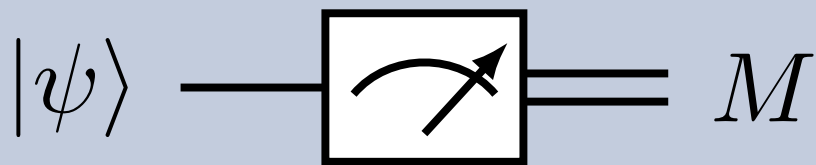


outcome	outcome probability
$M = 0$	$p(M = 0) = \langle 0 \psi \rangle ^2$
$M = 1$	$p(M = 1) = \langle 1 \psi \rangle ^2$

$$\begin{aligned} p(M = 0) + p(M = 1) &= |\langle 0 | \psi \rangle|^2 + |\langle 1 | \psi \rangle|^2 \\ &= \langle \psi | 0 \rangle \langle 0 | \psi \rangle + \langle \psi | 1 \rangle \langle 1 | \psi \rangle \\ &= \langle \psi | (|0\rangle \langle 0| + |1\rangle \langle 1|) | \psi \rangle \\ &= \langle \psi | \hat{I} | \psi \rangle \\ &= ||\psi\rangle|^2 \\ &= 1 \end{aligned}$$



Projects and overlaps



outcome	outcome probability
$M = 0$	$p(M = 0) = \langle 0 \psi \rangle ^2$
$M = 1$	$p(M = 1) = \langle 1 \psi \rangle ^2$

$$p(M = 0) + p(M = 1)$$

$$= \langle \psi | (|0\rangle \langle 0| + |1\rangle \langle 1|) | \psi \rangle$$

projectors



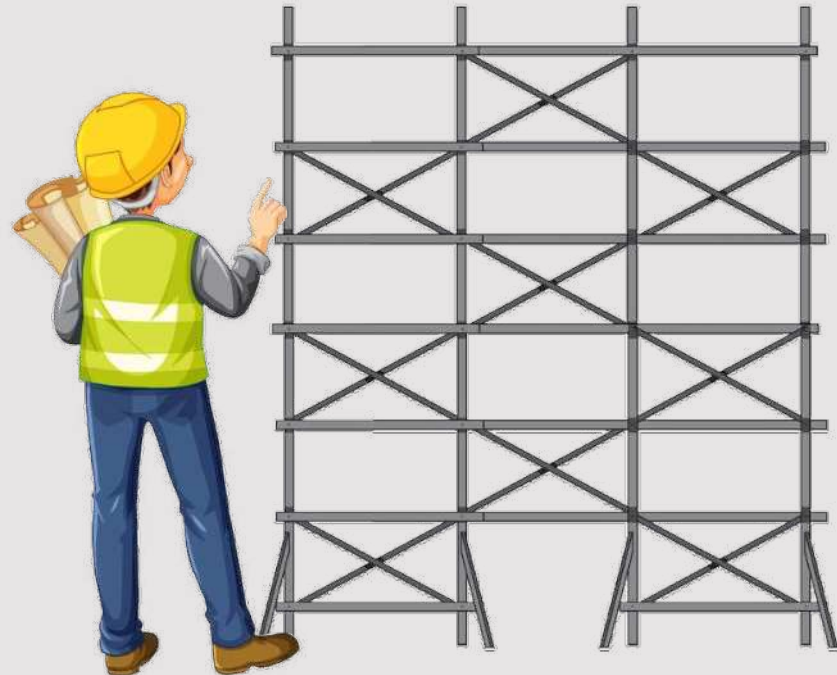
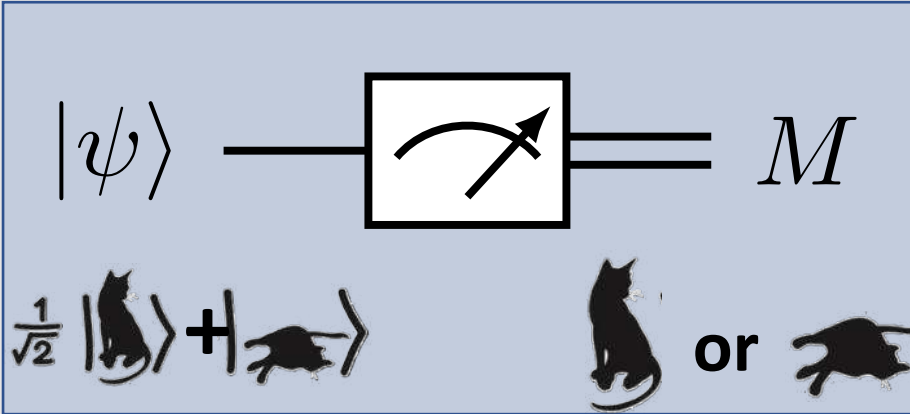
$$\hat{\Pi}_0 = |0\rangle \langle 0| \quad \hat{\Pi}_1 = |1\rangle \langle 1|$$

inner-product in operator space (overlap between states)

$$p(m) = \langle \hat{\Pi}_m, |\psi\rangle \rangle = \text{Tr} \left(\hat{\Pi}_m |\psi\rangle \langle \psi| \right)$$

Measurement theory 101 summary:

Scaffolding in terms of measurement operators



Measurement aspect

Measurement outcome M

Set of measurement outcomes Σ

Measurement operator Π_M

Probability of measuring $M = m$

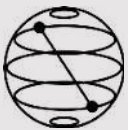
Example

$M = 0$

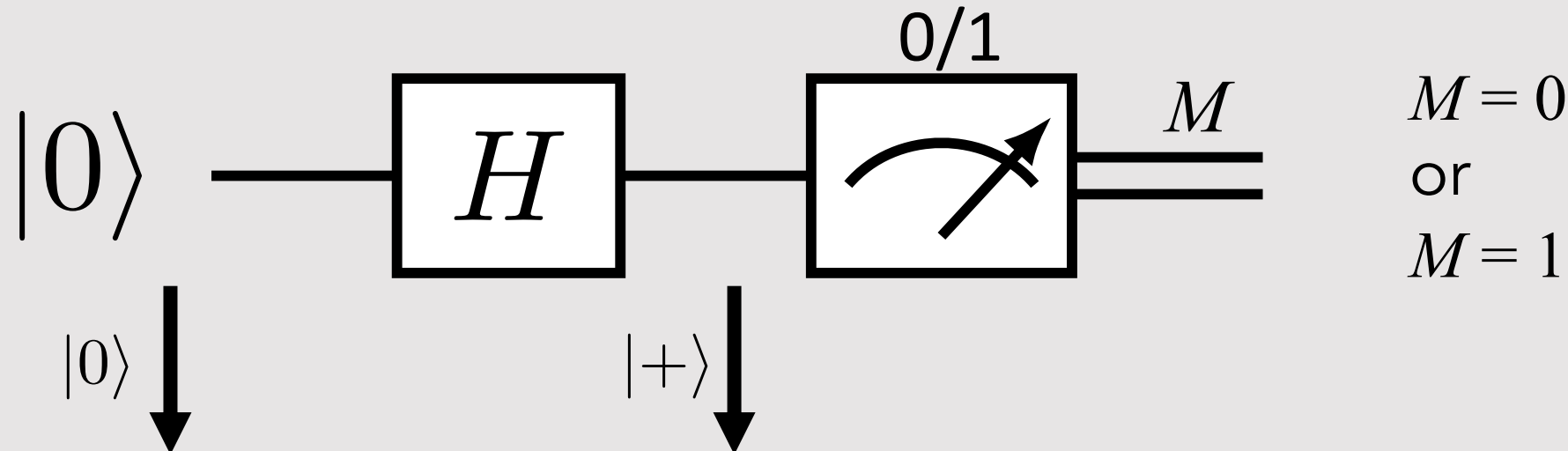
$\Sigma = \{0, 1\}$

$\Pi_M = |0\rangle\langle 0|$

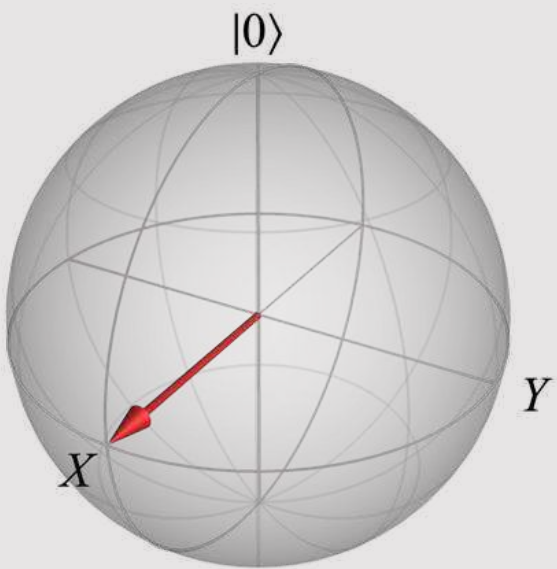
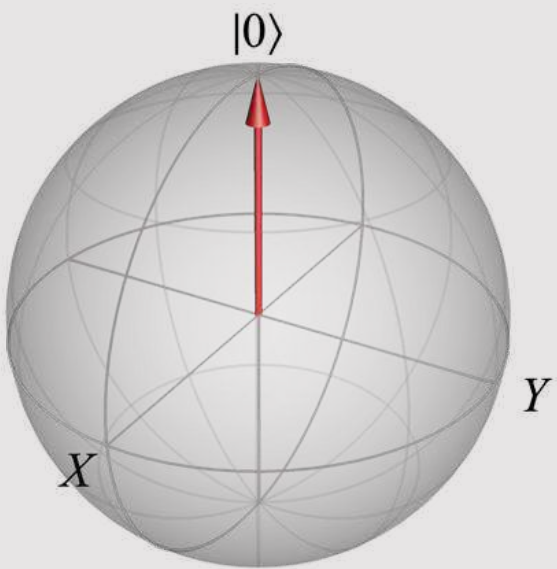
$$p(M=m) = \langle \Pi_m, \psi \rangle \\ = |\langle \Pi_m | \psi \rangle|^2$$



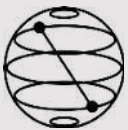
Putting it to use: example ideal circuit



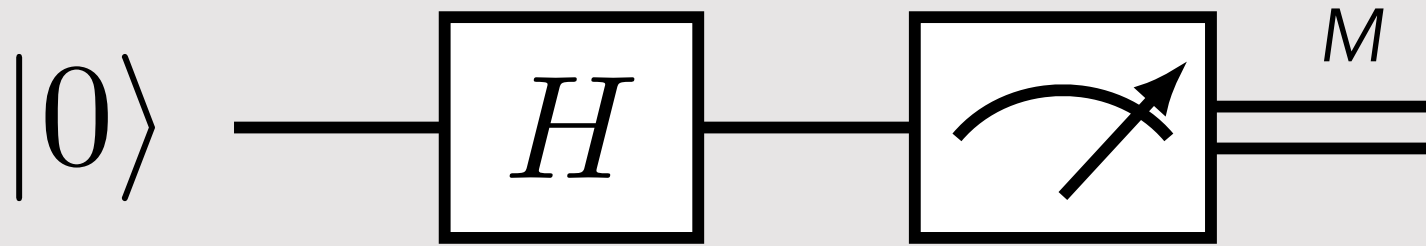
Quantum state



Classical random variable



Example ideal circuit



$M = 0$
or
 $M = 1$

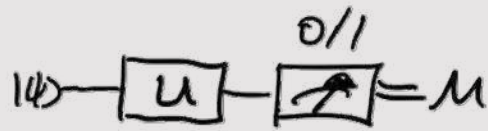
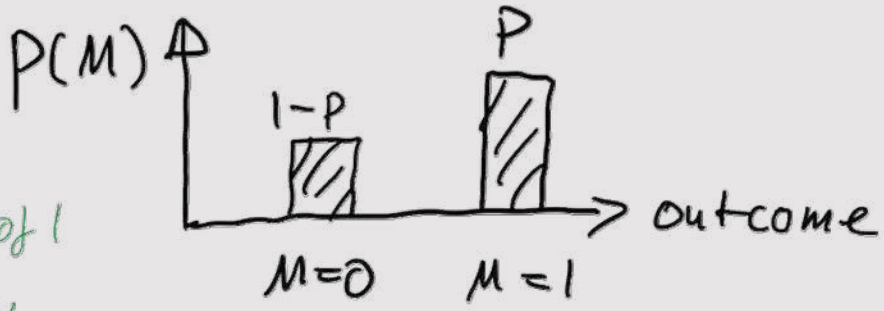
more general case

Prob to find $M=1$: $P(M=1) = p$ $p \in [0, 1]$

every possible qubit case

Bernoulli Distribution

$P(M) \sim B(p)$ Prob of 1
 $\uparrow$ classical rand variable
Probability



Mean

Statistics

$$\underset{\text{classical}}{\mathbb{E}}[\underset{\text{classical}}{M}] = \sum_m m P(m) = 0 P(m=0) + 1 P(m=1) \\ = P(m=1) \\ = p$$

$$= \sum_m m \langle \hat{\Pi}_m \rangle \\ = \sum_m m \langle |m\rangle \langle m| \rangle \\ = \langle \sum_m m |m\rangle \langle m| \rangle \\ = \langle \hat{M} \rangle$$

Quantum ↙ p^2

$$V[M] = \mathbb{E}[M^2] - \mathbb{E}[M]^2 \approx \langle \hat{M}^2 \rangle - \langle \hat{M} \rangle^2$$

$$\mathbb{E}[M^2] = \sum_m m^2 P(m)$$

$$= \sum_m m^2 \langle m \rangle \langle m |$$

$$= \langle \sum_m m^2 |m\rangle \langle m| \rangle$$

$$= \langle \hat{M}^2 \rangle$$

$$= \cancel{0^2 (1-p)} + 1^2 p$$

$$= p$$

$$V[M] = p - p^2$$

$$= p(1-p)$$

$$= \sigma_M^2$$

$= 0 \quad \text{if } p=0$
 $= 1 \quad \text{if } p=1$

Different observables

Different observables

For example $\hat{M} = 0|0\rangle\langle 0| + 1|1\rangle\langle 1| = \frac{1}{2}(\hat{I} + \hat{Z}) = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$

$\hat{M} = (+1)|0\rangle\langle 0| - 1|1\rangle\langle 1| = \hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

For $\hat{M} = \hat{Z}$

$\mathcal{B} = \{-1, +1\}$

Pauli Observables

$\hat{M} = \hat{X}$

$\mathcal{B} = \{+, -\}$

$\hat{\Pi}_+ = |+\rangle\langle +|$

$\hat{\Pi}_- = |- \rangle\langle -|$

$\hat{X} = \sum_{m \in \mathcal{B}} m |m\rangle\langle m|$

$= +1|+\rangle\langle +| - 1|-\rangle\langle -|$

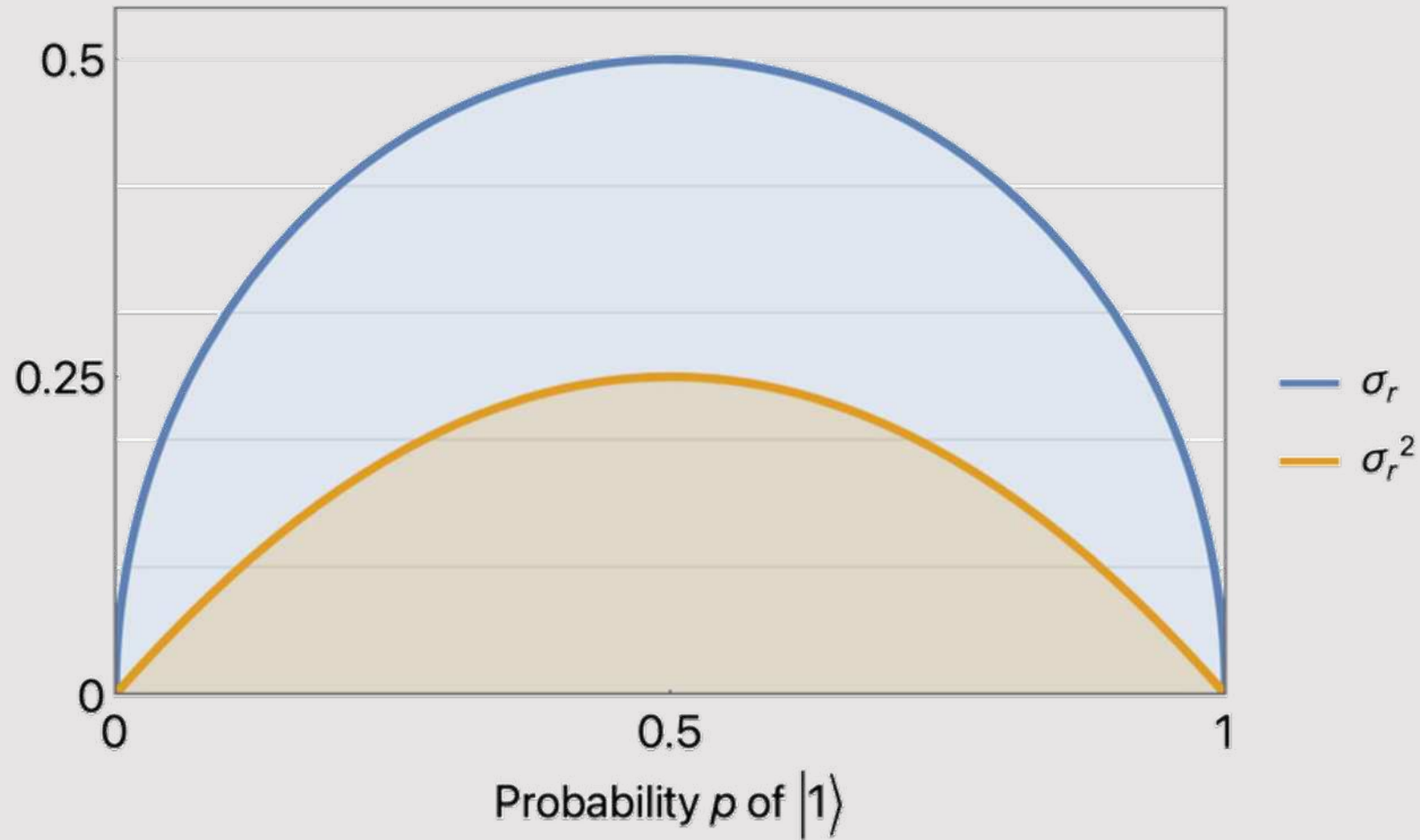
$\hat{X}|+\rangle = +1|x\rangle$

$\hat{X}|-\rangle = -1|x\rangle$

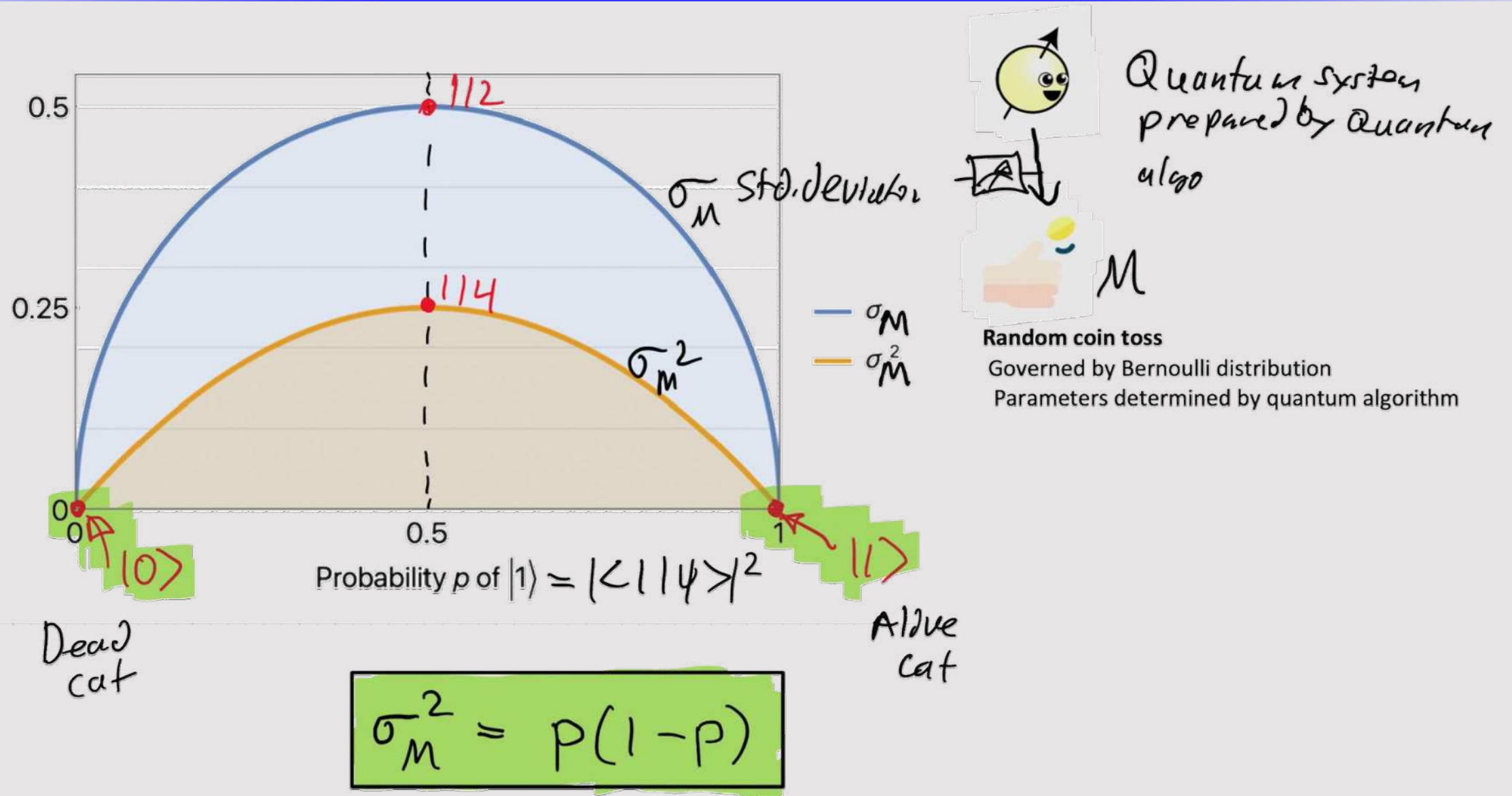
$|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$|-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

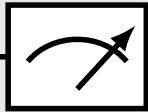
Variance of the random classical variable vs. probability to obtain 1

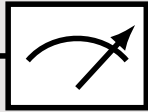


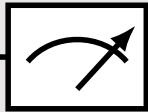
Variance of the random classical variable vs. probability to obtain 1

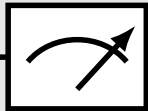


Shots, shots, shots

shot 1: $|0\rangle$ — H —  = $m_1 = 1$

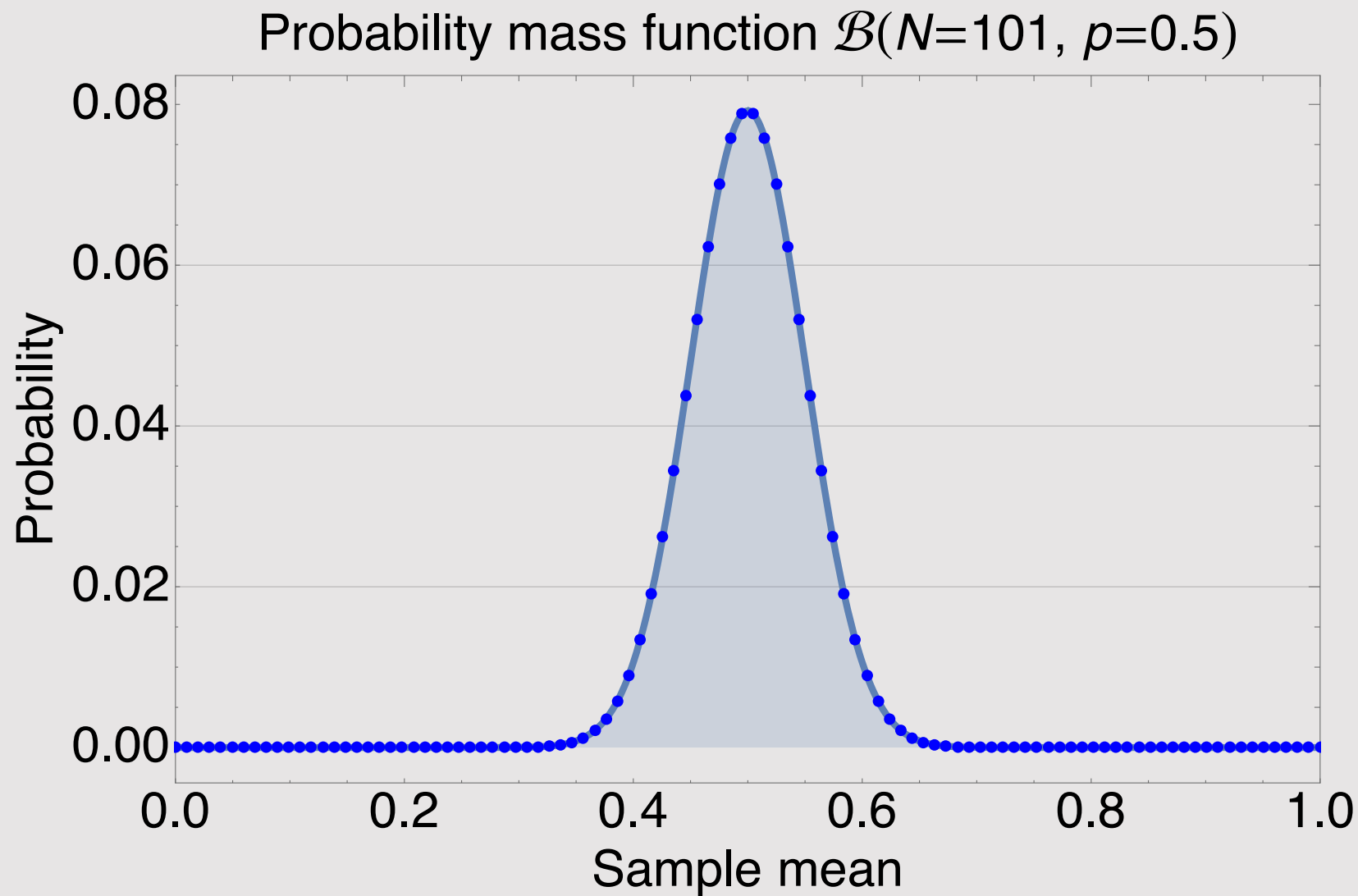
shot 2: $|0\rangle$ — H —  = $m_2 = 0$

shot 3: $|0\rangle$ — H —  = $m_4 = 1$

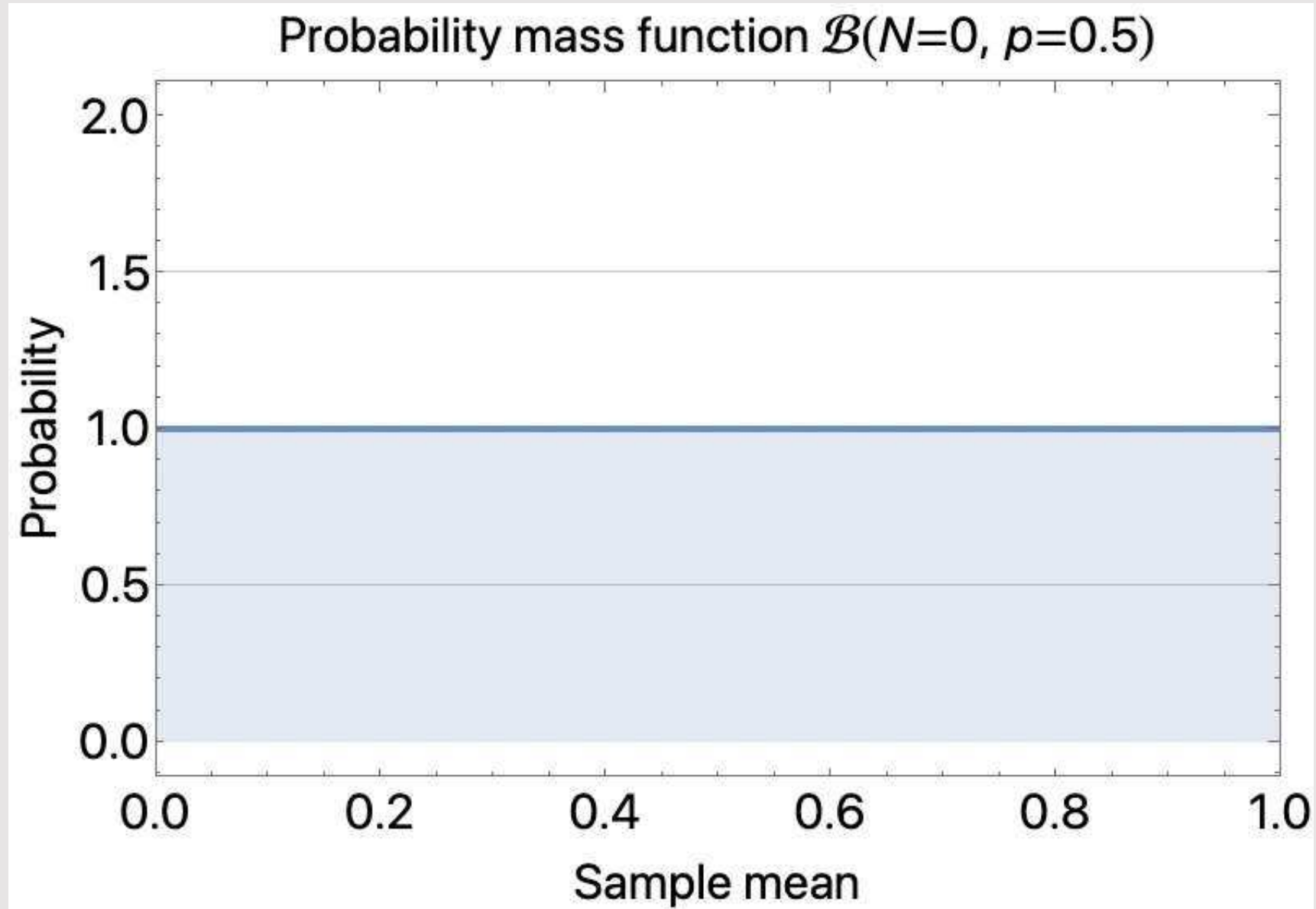
shot 4: $|0\rangle$ — H —  = $m_3 = 0$

⋮

Sampled output distribution



Animation of convergence of shots expectation value and mean



Concentration inequalities and tail bounds

Making a list,
checking it twice,
going to see
which inequality
is nice!

Markov? Hoeffding?
Jensen? Chebyshev?
Chernoff?

1. Probability (Technical note 11.9 v0.6)

1A. Concentration inequalities and tail bounds

Unless otherwise specified, all variables are real $\mathbb{R}$. Inequalities come as one-sided $\Pr(\dots \leq \dots)$ and two-sided $\Pr(|\dots| \leq \dots)$. Notation: X is a random variable, $\mu := \mathbb{E}[X]$, $\sigma^2 := \text{Var}[X]$, $S_n := X_1 + \dots + X_n$.

Inequality	Conditions	Common form	Notes / Alternate form	
			Single random variable	
Markov ¹	Non-negative	$X \geq 0$	$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$	$\forall a > 0$ $\Pr[X \geq k\mathbb{E}[X]] \leq \frac{1}{k}$ $k > 1$ [3, Sec. 5.1][6, Thm 1.13]
extension	+ non-negative, strictly increasing func Φ	$X \geq 0$ $\Phi(X) \geq 0$ increasing	$\Pr[X \geq a] = \Pr[\Phi(X) \geq \Phi(a)] \leq \frac{\mathbb{E}[\Phi(X)]}{\Phi(a)}$	$\forall a > 0$ Wiki
Reverse Markov	upper-bounded by U (can be positive)	$\max X = U$	$\Pr[X \leq a] \leq \frac{U - \mathbb{E}[X]}{U - a}$	$\forall a > 0$ [1, Sec. 3.1]
Chebyshev ²	Finite mean and variance	$\mathbb{E}[X], \text{Var}[X]$ finite	$\Pr[X - \mathbb{E}[X] \geq a] \leq \frac{\sigma^2}{a^2}$	$\Pr[X - \mathbb{E}[X] \geq a \cdot \sigma] \leq \frac{1}{a^2}$ [1, Sec. 3.2] $\forall a > 0, \sigma^2 = \text{Var}[X]$ [3, Sec. 5.1][2, Thm 18.11]
Cantelli	Improved Chebyshev	(same; but one-sided)	$\Pr[X - \mathbb{E}[X] \geq a] \leq \frac{\sigma^2}{\sigma^2 + a^2}$	$\forall a > 0, \sigma^2 = \text{Var}[X]$ Wiki
Chernoff ³	Generic		$\Pr[X \geq a] = \Pr[e^{tX} \geq e^{ta}]$	$\forall t > 0, a \in \mathbb{R}$ [1, Sec. 3.3]
Jensen		$f: \mathbb{R} \rightarrow \mathbb{R}; f$ convex	$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$	[3, Prob. 5.3][6, Thm 1.14]
Hoeffding's lemma		$\mathbb{E}[X] = \mu$ $a \leq X \leq b$	$\mathbb{E}[e^{\lambda X}] \leq e^{\lambda \mu} e^{\frac{\lambda^2 (b-a)^2}{8}}$	$\lambda \in \mathbb{R}$ [1, Sec. 3.4]
Sum of random variables				
Chernoff-Hoeffding (one-sided)	n independent random vars	$X_1, \dots, X_n$ indep $S_n = X_1 + \dots + X_n$ $X_i \in [a_i, b_i] \quad \forall i$	$\Pr[S_n - \mathbb{E}[S_n] \geq t] \leq \exp\left(\frac{-t^2 n^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$	[1, Sec. 3.5]
(two-sided) ⁴	(same as above)		$\Pr[S_n - \mathbb{E}[S_n] \geq t] \leq 2 \exp\left(\frac{-t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$	$\forall t \in (0, \frac{1}{2})$ [5, Thm.1.1]
(two-sided iid)	same plus iid, range, mean μ for each	$X_1, \dots, X_n \in [0, 1]$ $\mathbb{E}[X_i] = \mu$ iid	$\Pr\left[\left \frac{S_n}{n} - \mu\right \geq \epsilon\right] \leq 2 \exp(-2n\epsilon^2)$	$\forall \epsilon > 0$ [6, Thm 1.16]
Thm 1.3	n independent random vars	$X_1, \dots, X_n$ indep $S_n = X_1 + \dots + X_n$	$\Pr[S_n - \mathbb{E}[S_n] > \epsilon] \leq 2 \exp\left(\frac{-\epsilon^2}{4 \sum_{i=1}^n \text{Var}[X_i]}\right)$	$\epsilon \in (0, 2 \text{Var}[S_n] / (\max_i X_i - \mathbb{E}[X_i]))$ [5, Thm. 1.3]
Azuma				
Weak law of large numbers	n independent iid random vars	$X_1, \dots, X_n$ indep $\mathbb{E}[X_i] = \mu$ iid	$\lim_{n \rightarrow \infty} \Pr\left[\left \frac{1}{n} S_n - \mu\right \geq \epsilon\right] = 0$	$\forall \epsilon > 0$ [3, Sec. 5.2][6, Thm 1.15]
Strong law of large numbers	(same)	(same)	$\Pr\left[\lim_{n \rightarrow \infty} \frac{1}{n} S_n = \mu\right] = 1$	[3, Sec. 5.5]
Advanced				
Bennett	n independent zero-mean	$X_1, \dots, X_n$ indep $\mathbb{E}[X_i] = 0$ iid	$\Pr[S_n > \epsilon] \leq \exp\left(-n\sigma^2 h\left(\frac{\epsilon}{n\sigma^2}\right)\right)$	$\sigma^2 := \frac{1}{n} \sum_{i=1}^n \text{Var}[X_i], \forall \epsilon > 0,$ $h(a) := (1+a) \log(1+a) - a$ for $a \geq 0$ [1, 4.1]
Bernstein	(same)	(same)	$\Pr[S_n > \epsilon] \leq \exp\left(\frac{-n\epsilon^2}{2(\sigma^2 + \epsilon/3)}\right)$	(same) [1, 4.2]
Efron-Stein	scalar func of vars $f: \chi^n \rightarrow \mathbb{R}$	$X_1, \dots, X_n$ indep w/ values in set χ	$\text{Var}[Z] \leq \sum_{i=1}^n \mathbb{E}\left[(Z - \mathbb{E}[Z])^2\right]$	$Z := g(X_1, \dots, X_n)$ $\mathbb{E}_i[Z] := \mathbb{E}[Z X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n]$ [1, 4.3]
McDiarmid's	scalar func of vars $f: \chi^n \rightarrow \mathbb{R}$	$X_1, \dots, X_n$ indep w/ values in set χ	$\Pr[f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)] \geq \epsilon] \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right)$	condition: c -bounded difference property $\forall \epsilon > 0$ $ f(X_1, \dots, X_i, \dots, X_n) - f(X_1, \dots, X'_i, \dots, X_n) \leq c_i$ [1, 4.4]

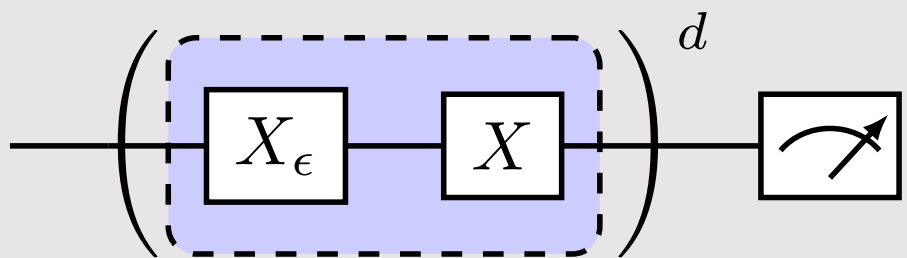
¹Markov's inequality bounds the first moment of random variable. Use it when a constant probability bound is sufficient [1, Sec. 3.3].

²Chebyshev is derived from Markov. It bounds the second moment. It is the appropriate one when the variance σ is known. If σ is unknown, we can use the bounds of $X \in [a, b]$.

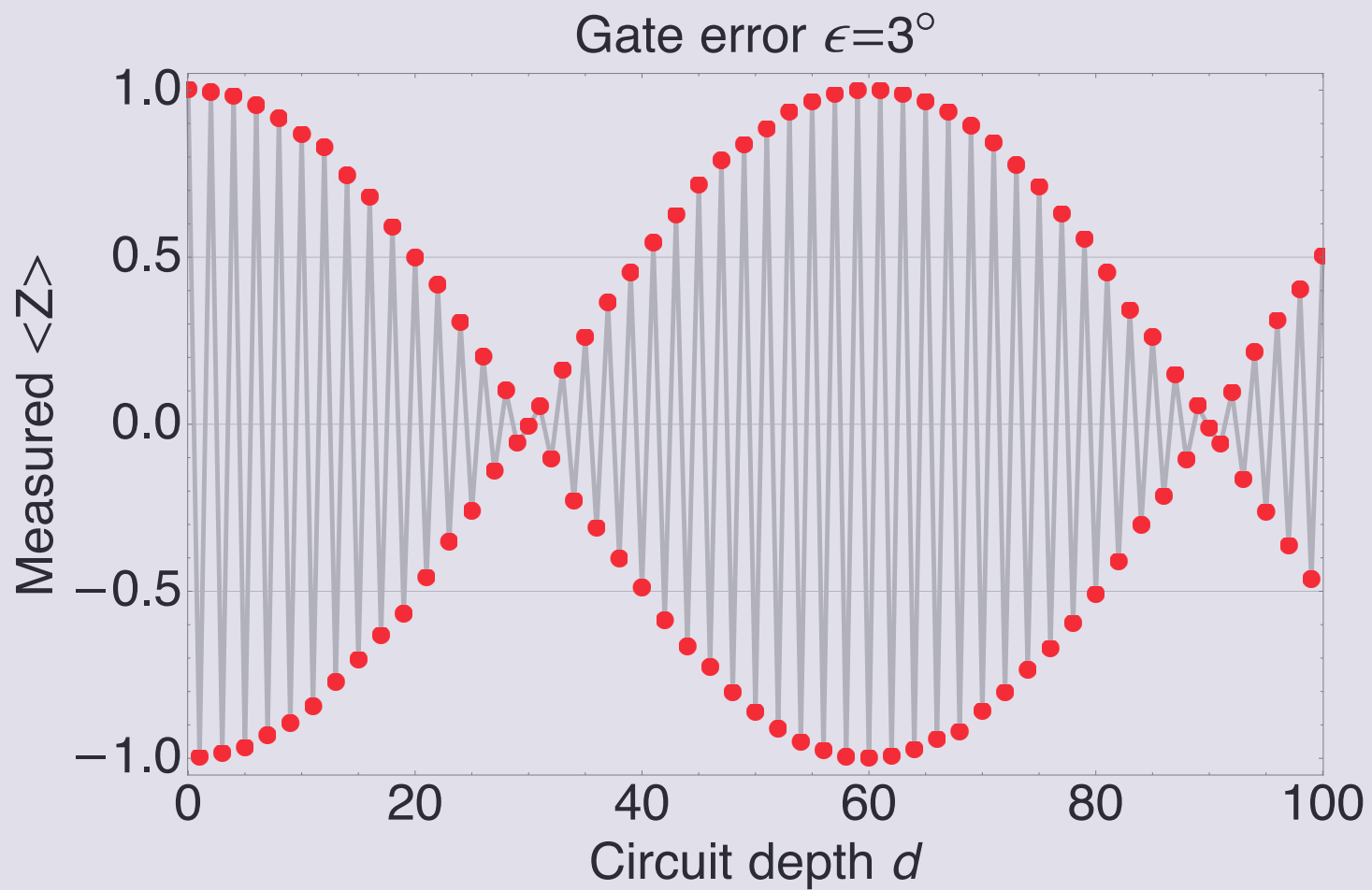
³Chernoff bound is used to bound the tails of the distribution for a sum of independent random variables. By far the most useful tool in randomized algorithms [1, Sec. 3.3].

⁴This probability can be interpreted as the level of significance ϵ (probability of making an error) for a confidence interval around the mean of size 2ϵ . Therefore, we require at least $\log(2\epsilon)/2t^2$ samples to acquire $1 - \alpha$ confidence interval $\mathbb{E}[X] \pm t$.

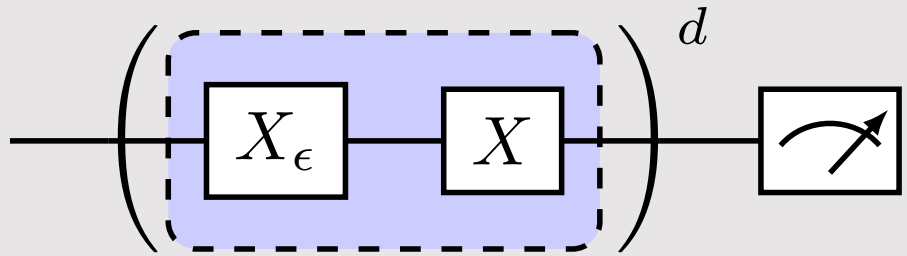
Recall gate error result



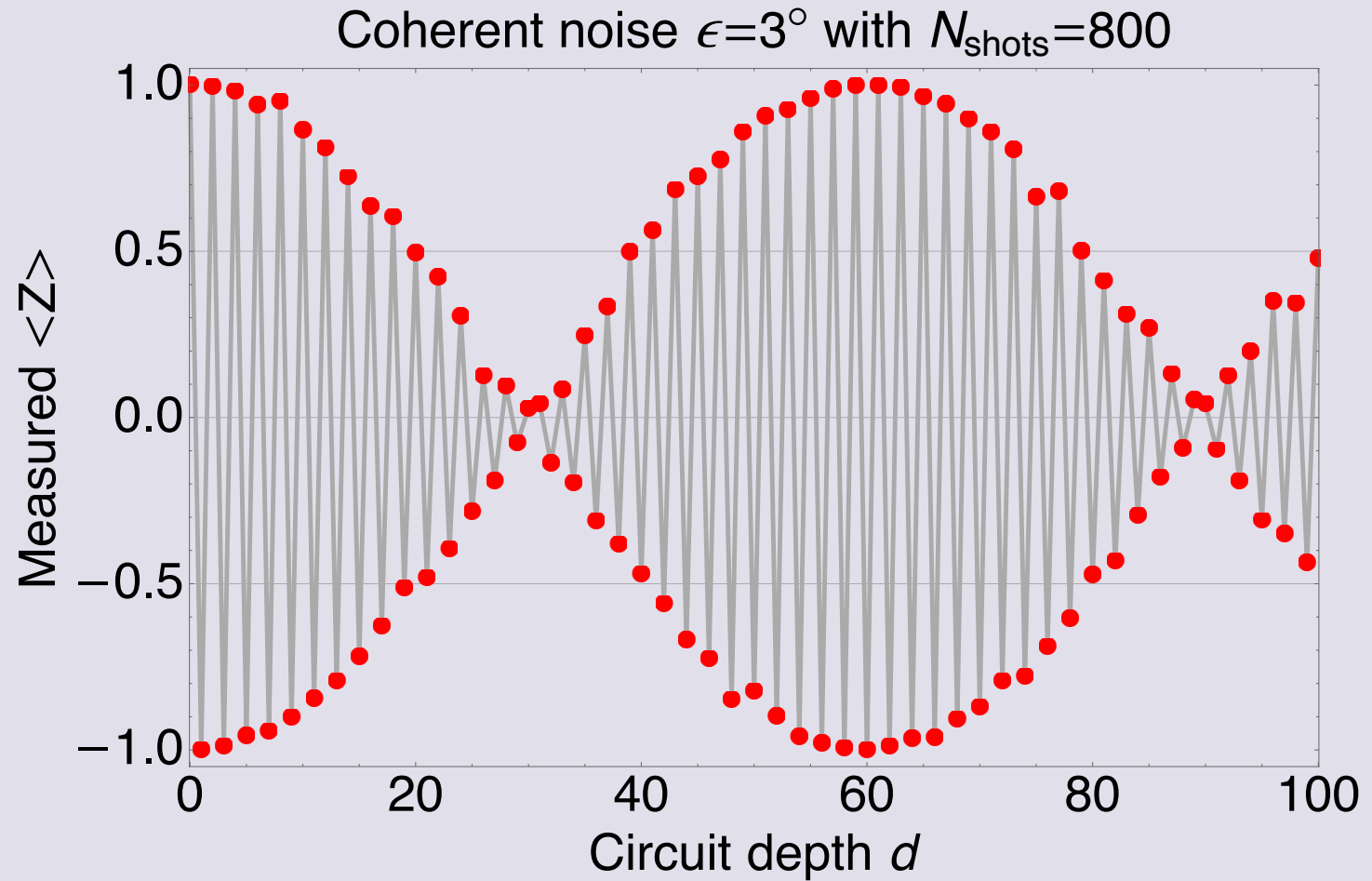
$$\langle \tilde{\psi}_f | Z | \tilde{\psi}_f \rangle = \cos(d\pi + d\epsilon)$$



Projection & sampling noise



$$\langle \tilde{\psi}_f | Z | \tilde{\psi}_f \rangle = \cos(d\pi + d\epsilon)$$





Questions

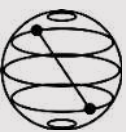
Answer these multiple-choice questions
in the chat; for example, type “1a 2b.”

1. Projection noise is due to

- a) measurement apparatus that could be made more efficient
- b) classical limitations
- c) core nature of quantum physics

2. To reduce projection noise

- a) increase the number of sample
- b) you cannot undo it
- c) apply readout error mitigation



Dive deeper? Try the following



1. Calculate the following for a qubit

1. The expectation value of the sample variance for N shots of the observable $|1\rangle\langle 1|$.

where the sample mean is defined as

$$S = \frac{1}{N} \sum_{n=1}^N M_n$$

and the sample variance is defined as

$$V = \frac{1}{N} \sum_{n=1}^N (M_n - S)^2$$

2. Is the estimate biased?

3. The variance of V .

4. Can you find an expression for an unbiased estimate of the sample variance?

2. What about two qubits?

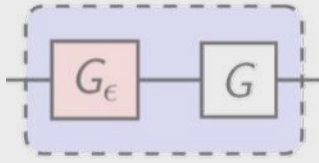
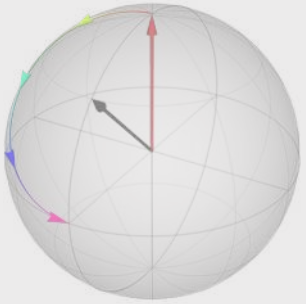
1. Can you find the projection operators for the observable ZZ ?

2. Find the probability distribution for the observables ZI , IZ , and ZZ for a general state.

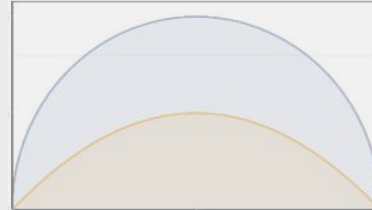
3. If you take 10 shots and find all 10 outcomes to be 1, what is the probability the qubit is in the $|0\rangle$ state? (hint: it's not zero!)

State preparation & measurement

Coherent



Projection & measurement theory



State preparation & measurement

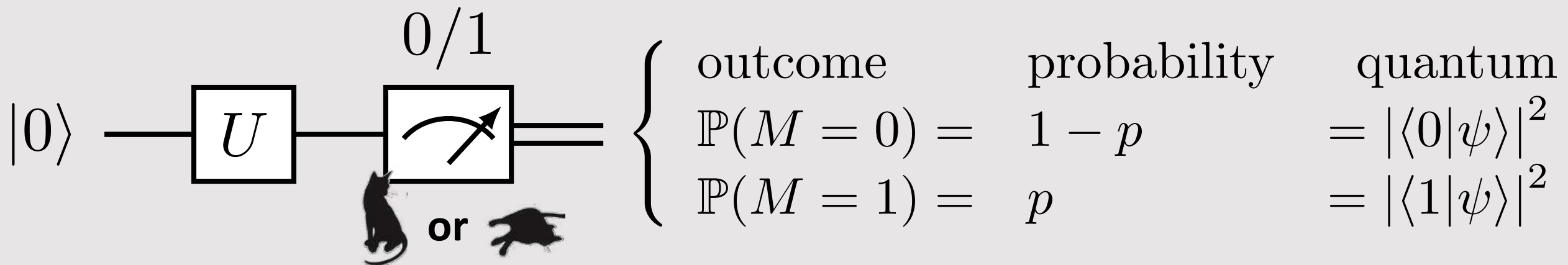


$|0\rangle$ —



coin toss: flaticon; spam: make it move;
road based on: freepik

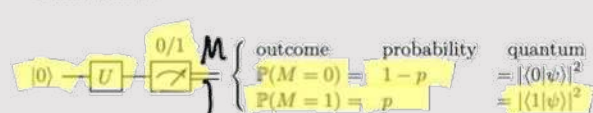
Qubit example



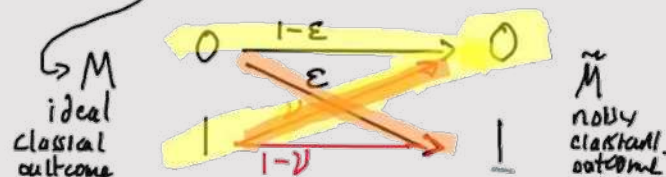
Measurement error

Qiskit Global Summer School on Quantum Machine Learning

Zlatko K. Minev



$$\hat{M} = |1\rangle\langle 1| = \frac{1}{2}(\hat{I} - \hat{Z})$$



$$P_M = \begin{pmatrix} P(M=0) \\ P(M=1) \end{pmatrix} = \begin{pmatrix} 1-p \\ p \end{pmatrix}$$

$$P_{\tilde{M}} = \begin{pmatrix} P(\tilde{M}=0) \\ P(\tilde{M}=1) \end{pmatrix} = \begin{pmatrix} 1-\tilde{p} \\ \tilde{p} \end{pmatrix}$$

$$\begin{aligned} P(\tilde{M}=0) &= P(\tilde{M}=0|M=0)P(M=0) + P(\tilde{M}=0|M=1)P(M=1) \\ P(\tilde{M}=1) &= P(\tilde{M}=1|M=0)P(M=0) + P(\tilde{M}=1|M=1)P(M=1) = \epsilon(1-p) + (1-\nu)p = \tilde{p} \end{aligned}$$

$$P_{\tilde{M}} = A P_M$$

ϵ noisy $\leftarrow$ readout error matrix $\leftarrow$ ideal

$$A = \begin{matrix} & M=0 & M=1 \\ \begin{matrix} \tilde{M}=0 \\ \tilde{M}=1 \end{matrix} & \begin{pmatrix} P(\tilde{M}=0|M=0) & P(\tilde{M}=0|M=1) \\ P(\tilde{M}=1|M=0) & P(\tilde{M}=1|M=1) \end{pmatrix} \end{matrix}$$

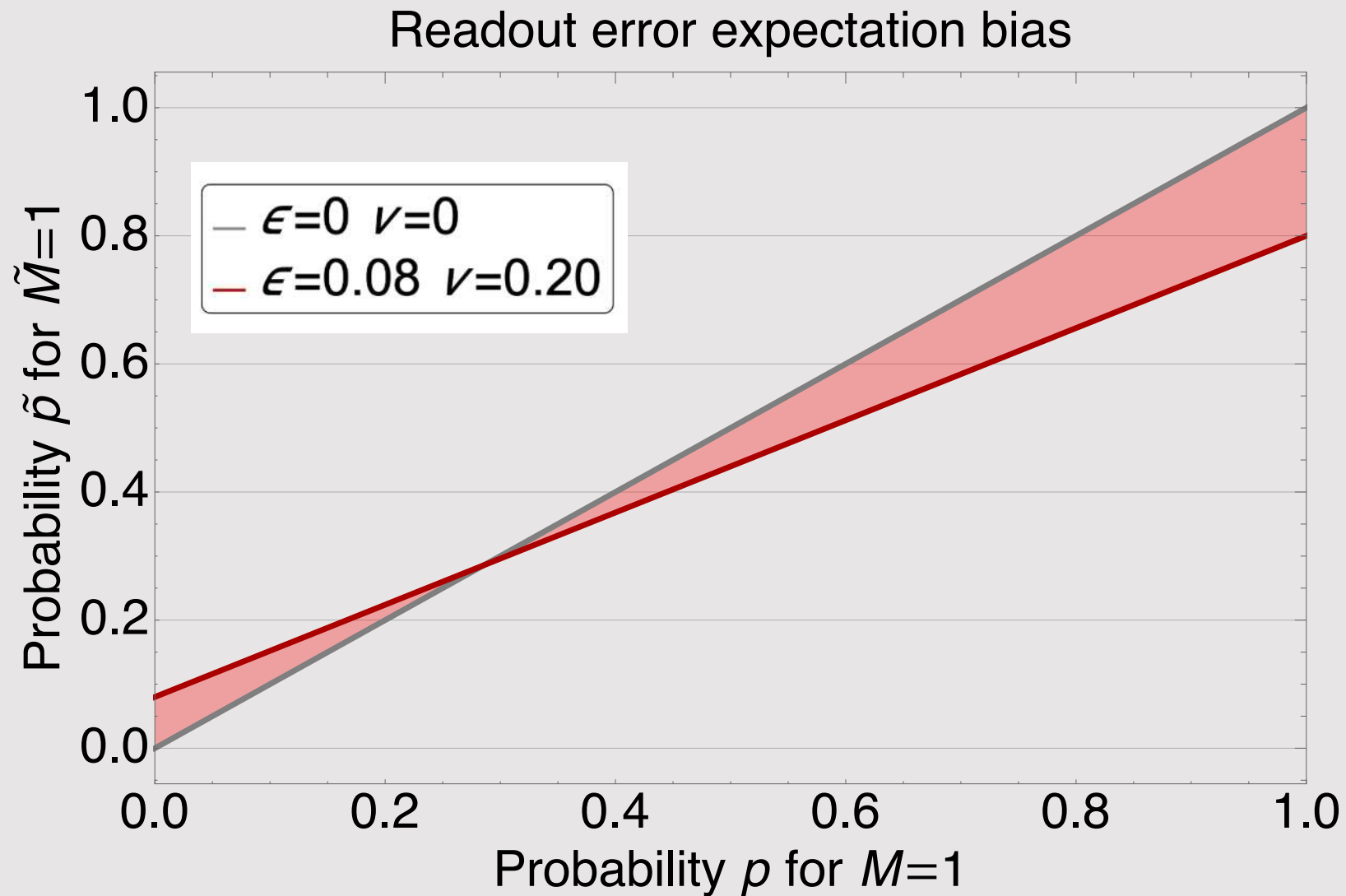
$$= \begin{pmatrix} 1-\epsilon & \nu \\ \epsilon & 1-\nu \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ for an ideal measurement}$$

$$\sum_m A_{mn} = 1 \text{ for any } n$$

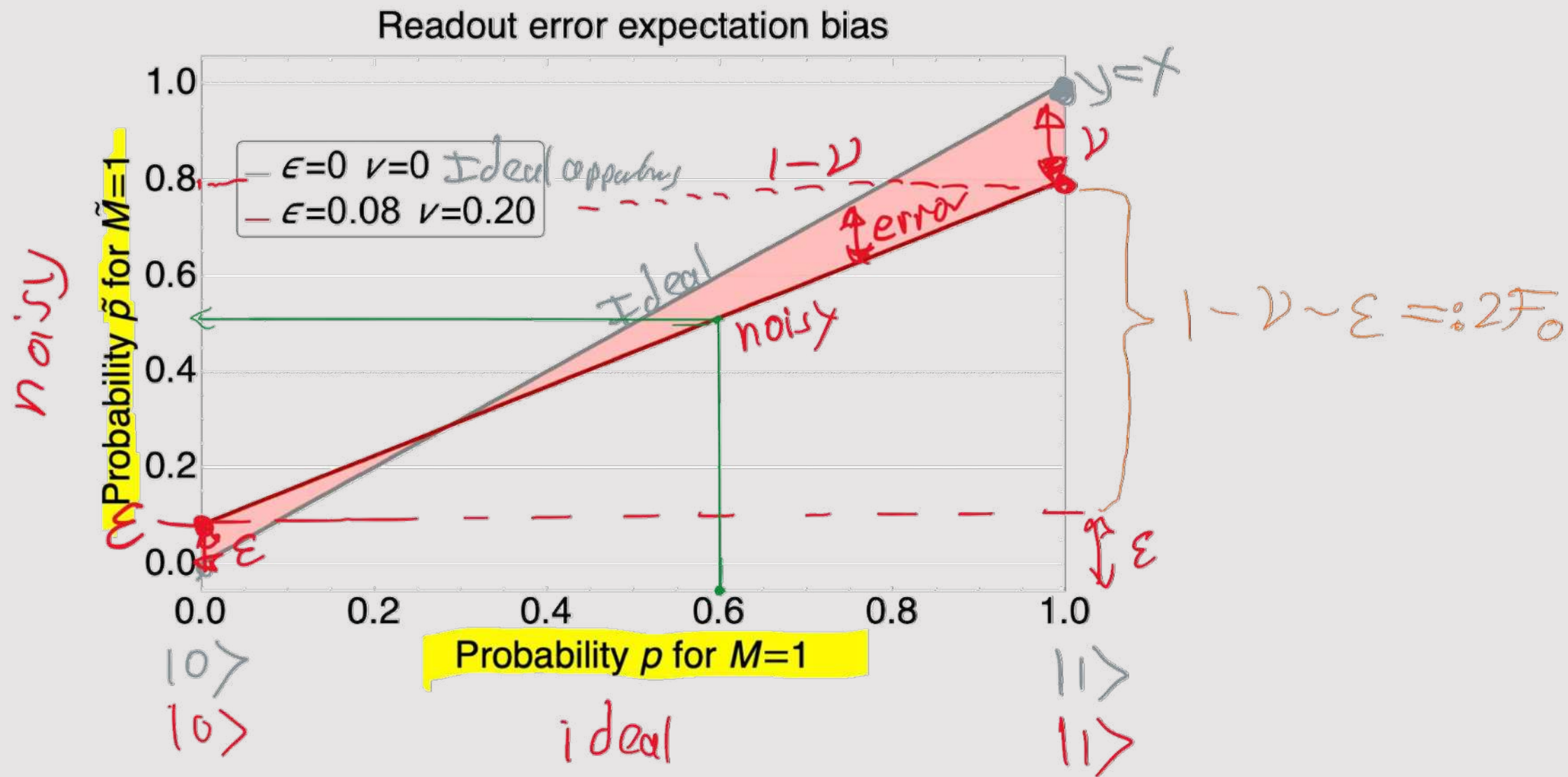
Stochastic matrix

$$\begin{aligned} \tilde{p} &= \epsilon(1-p) + (1-\nu)p \\ &= \epsilon - p\epsilon + p - \nu p \\ &= p + \epsilon - (\nu + \epsilon)p \end{aligned}$$

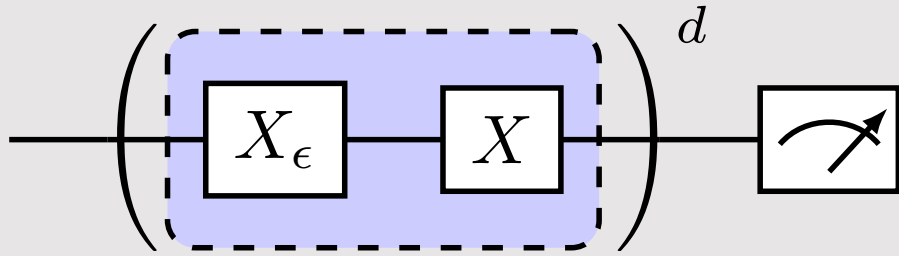
Readout error



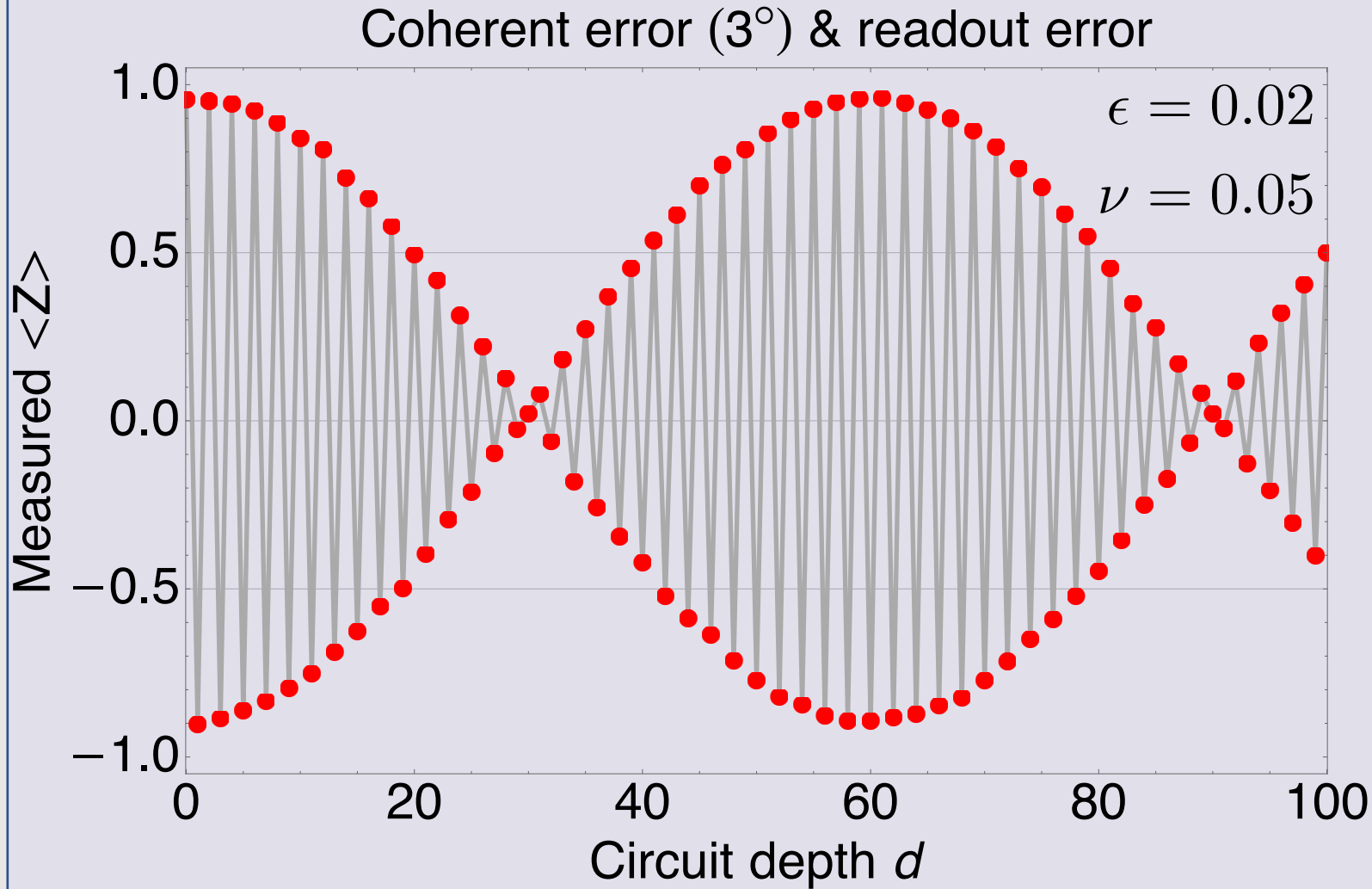
Assignment fidelity



Projection & sampling noise



$$A = \begin{matrix} & \tilde{M} = 0 & \tilde{M} = 1 \\ \begin{matrix} M = 0 \\ M = 1 \end{matrix} & \begin{pmatrix} 1 - \epsilon & \nu \\ \epsilon & 1 - \nu \end{pmatrix} \end{matrix}$$





Questions

Answer these multiple-choice questions
in the chat; for example, type “1a 2b.”

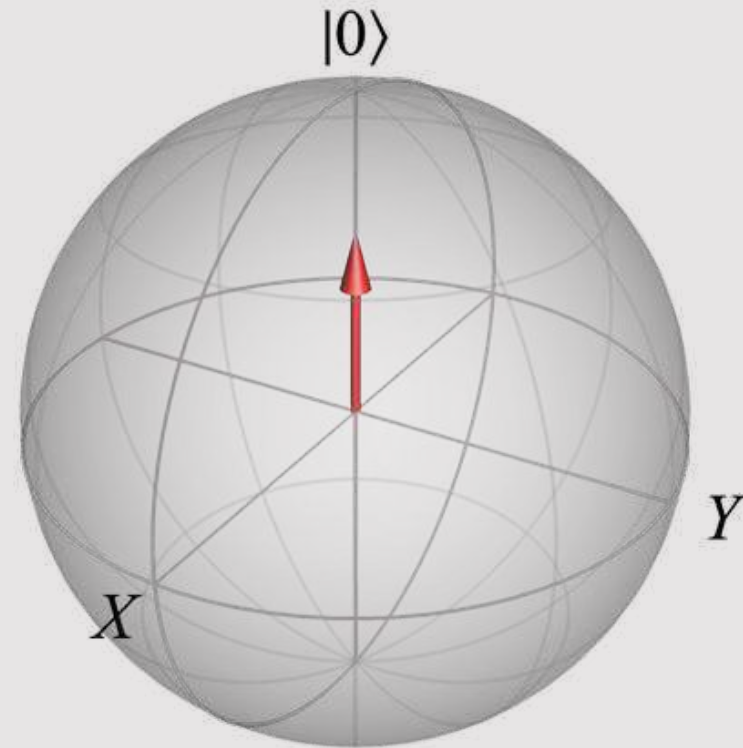
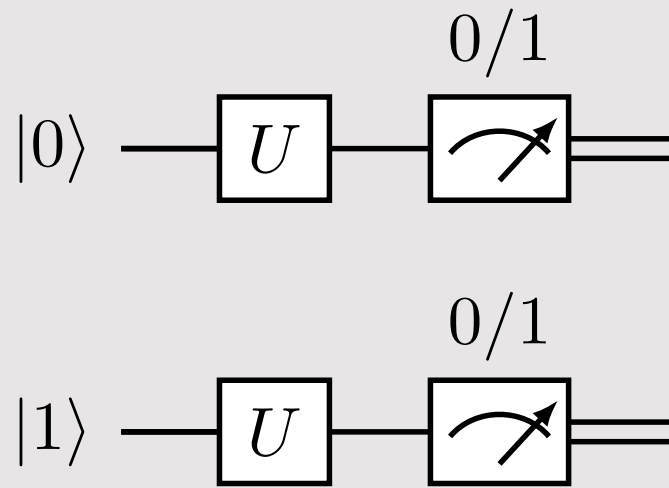
1. Readout error is due to

- a) measurement apparatus that could be made more efficient
- b) classical limitations
- c) core nature of quantum physics

2. To reduce readout error bias

- a) increase the number of sample
- b) you cannot undo it
- c) apply readout error mitigation

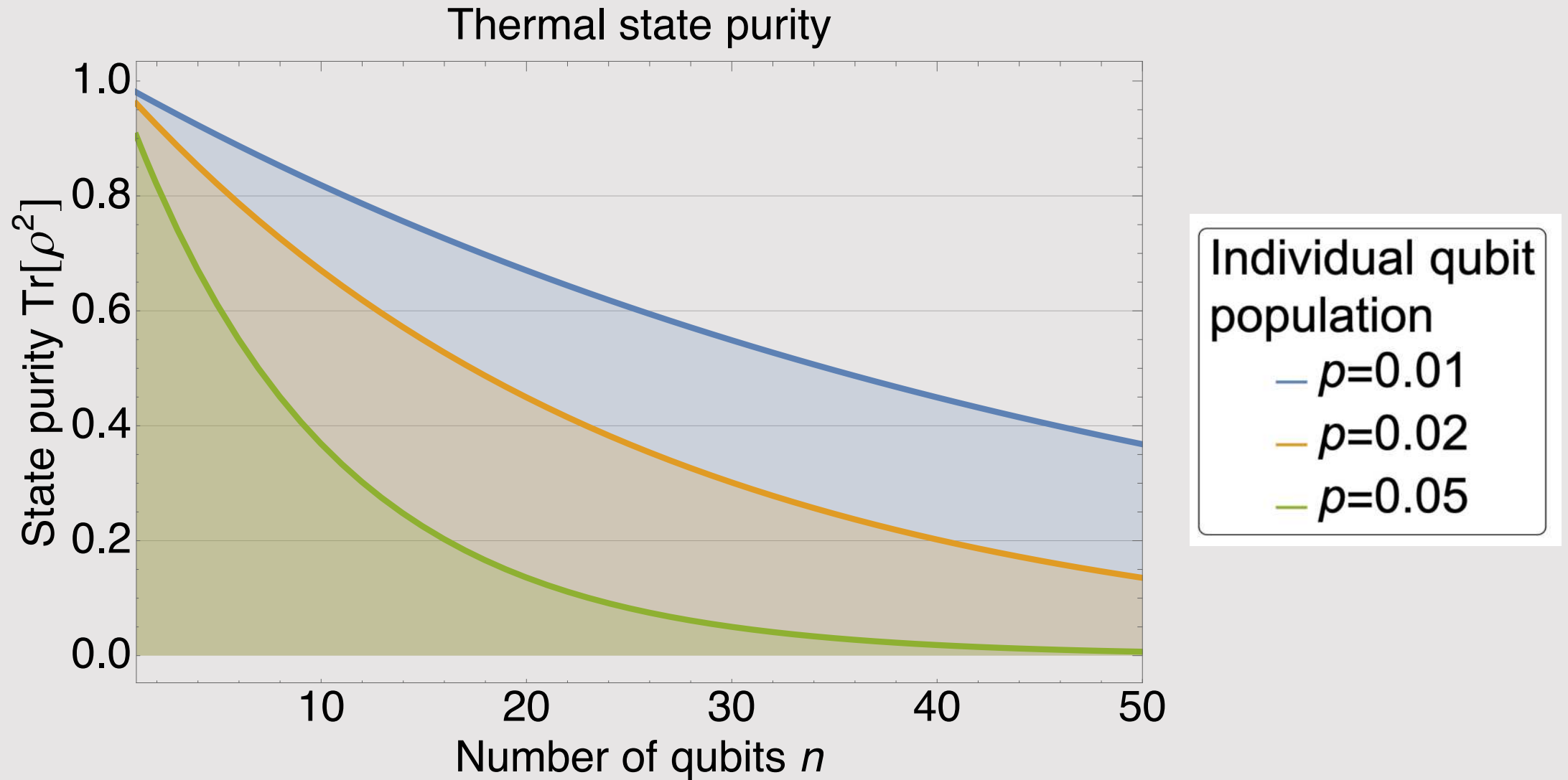
State prep



Multiple qubits

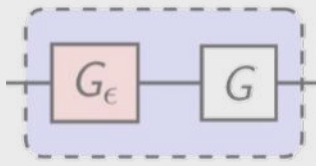
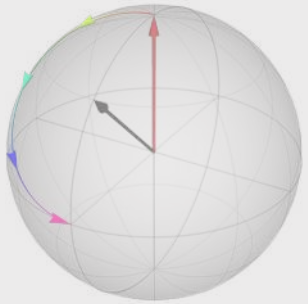
$$[(1 - p) |0\rangle\langle 0| + p |1\rangle\langle 1|]^{\otimes n} \equiv U \equiv \text{Measurement}$$

Thermal state

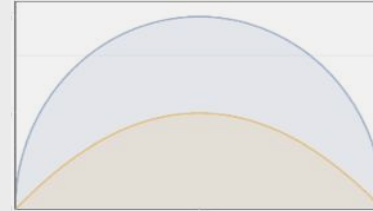


Incoherent noise

Coherent



Projection & measurement theory



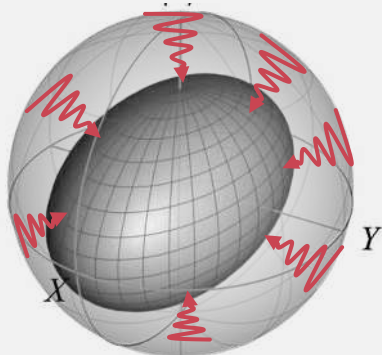
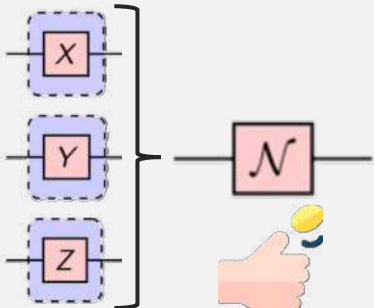
State preparation & measurement



$|0\rangle$ —



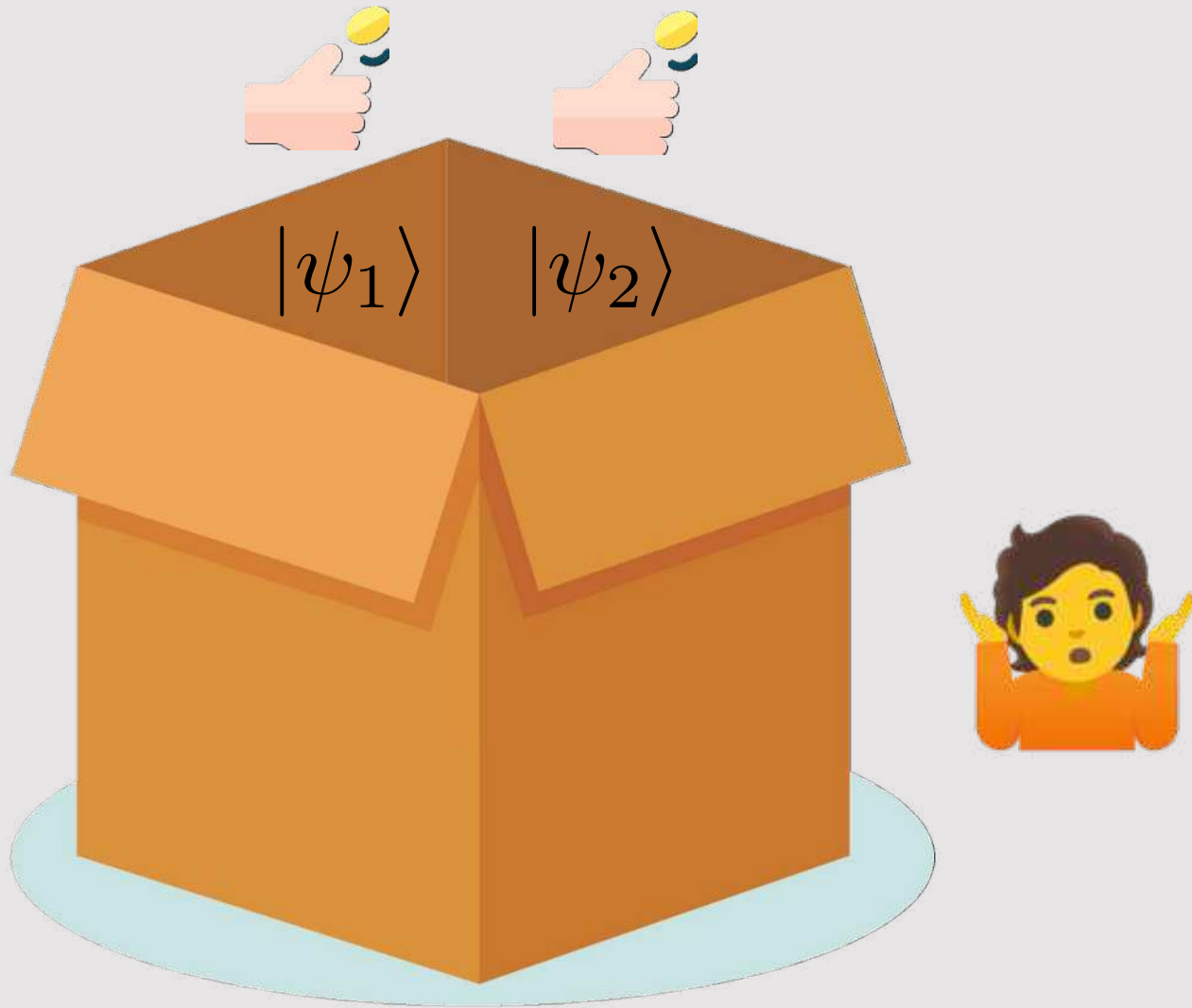
Incoherent noise



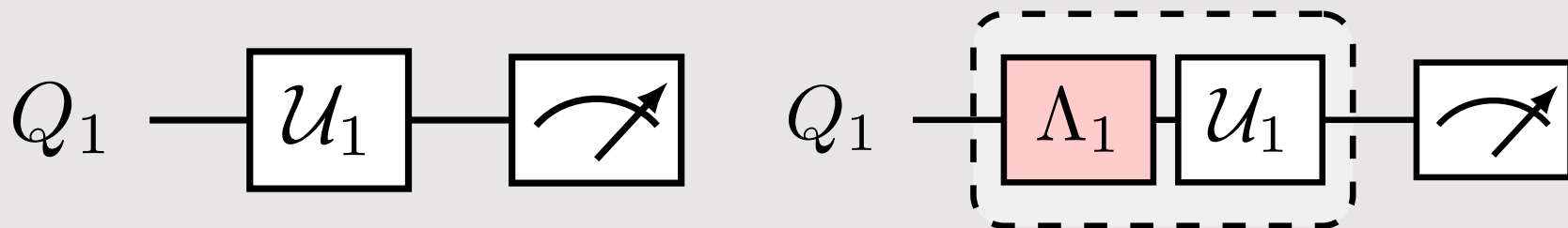
coin toss: flaticon; spam: make it move;
road based on: freepik

Zlatko Minev, IBM Quantum (73)

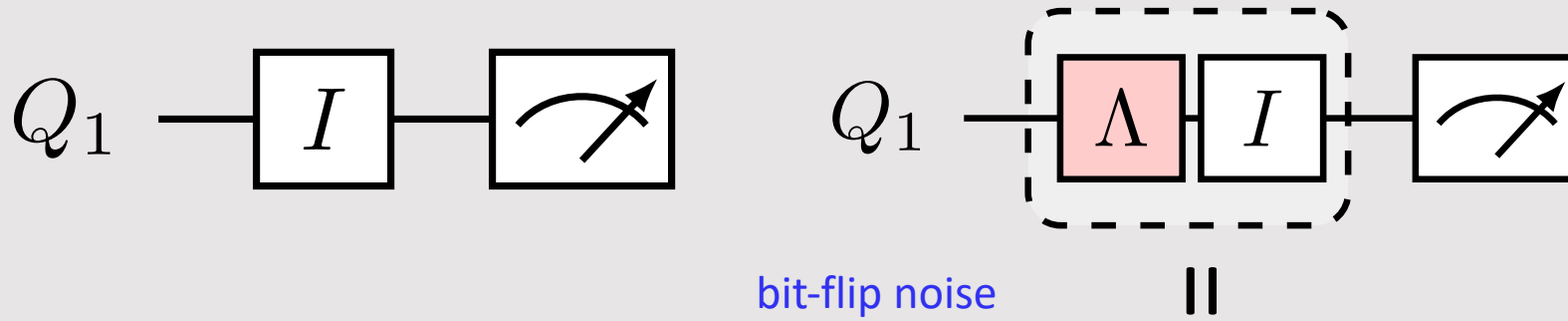
Review: mixed state (density matrix)



Toy model

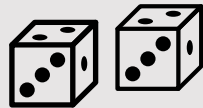


Toy model: noise unraveling into quantum trajectories

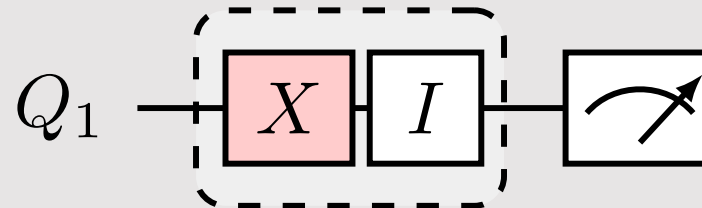
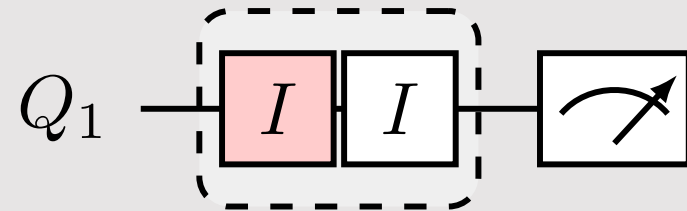


unraveling
(quantum trajectories)

probability $1-p$

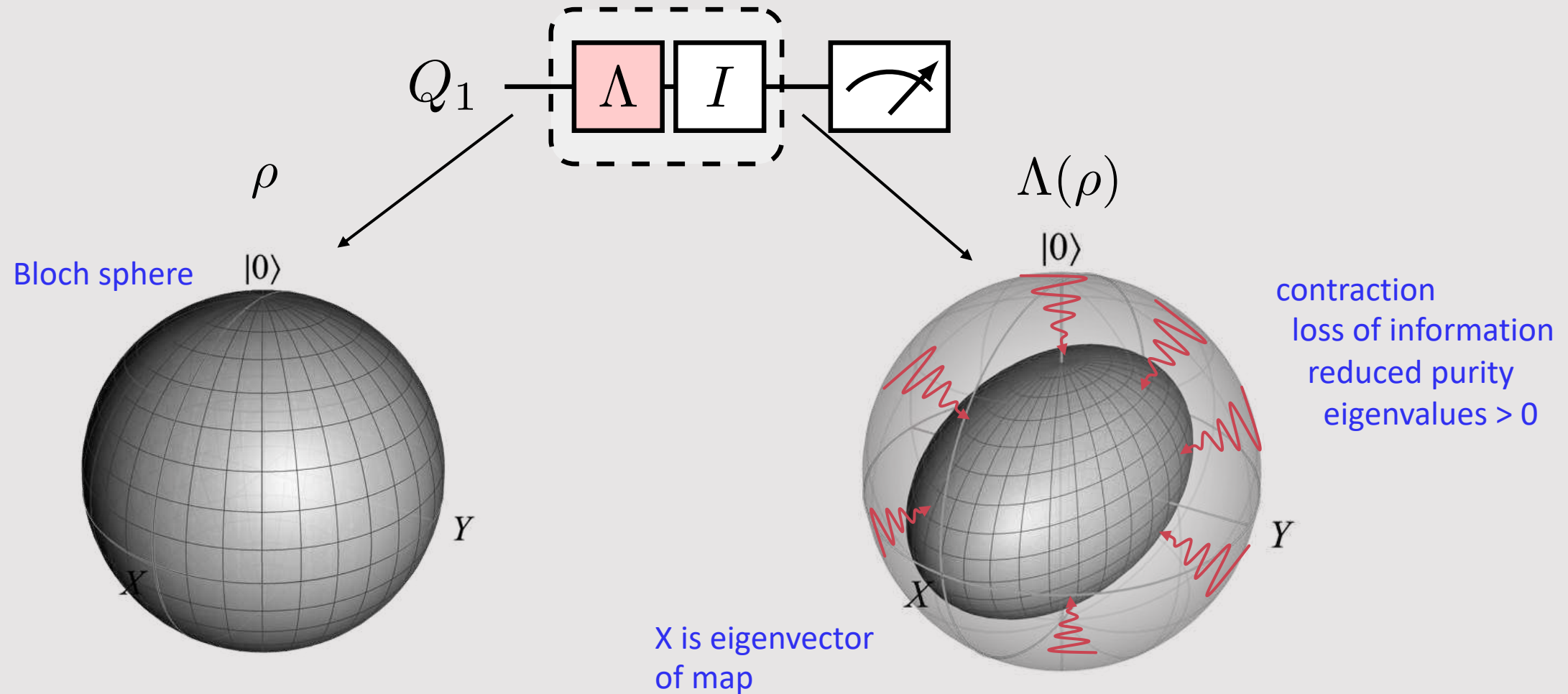


probability p

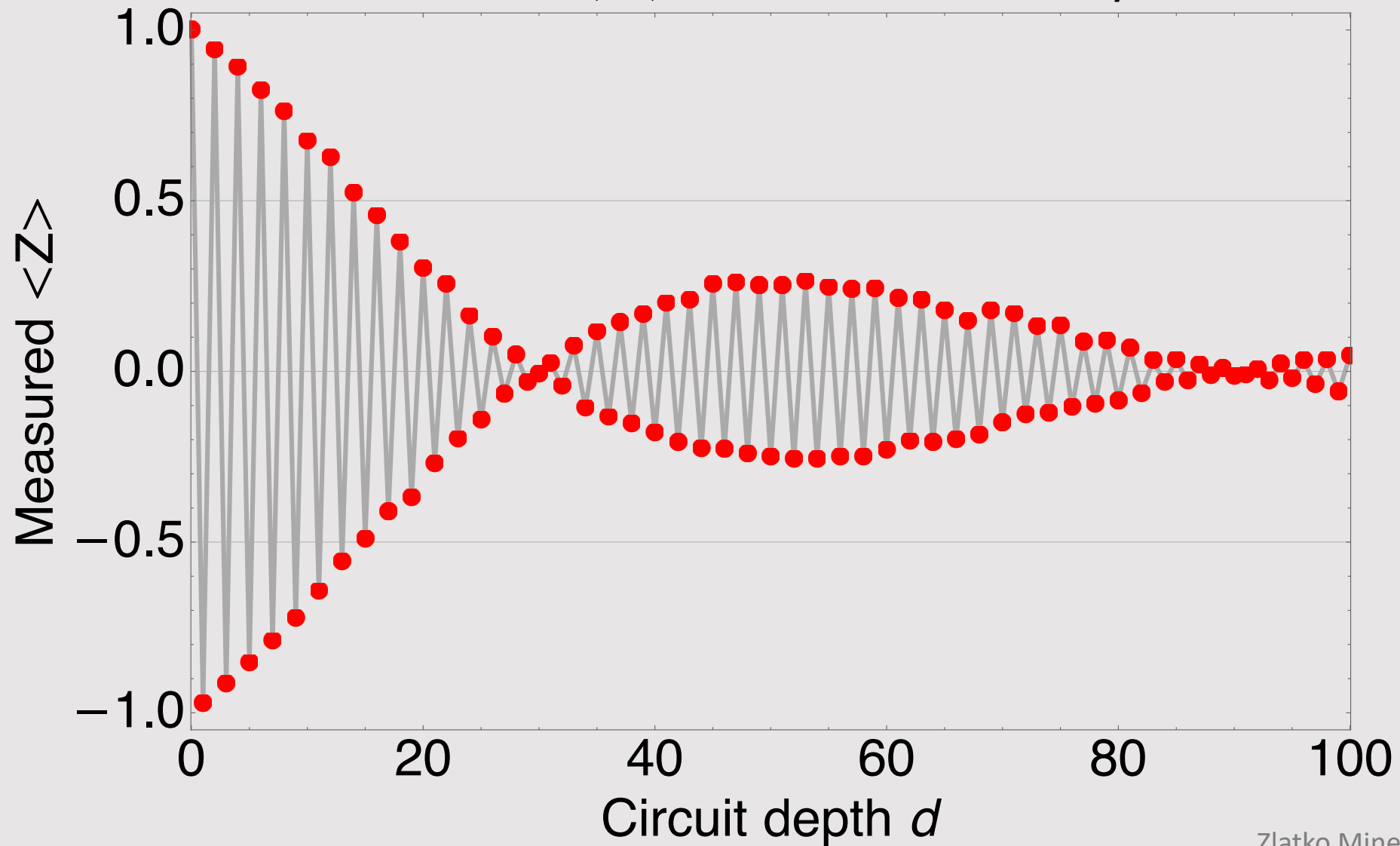


$$\Lambda(\rho) = (1-p)I\rho I + pX\rho X$$

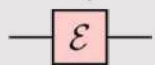
Toy model: noise unraveling into quantum trajectories



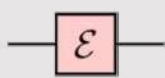
Coherent error (3°) & incoherent error $p=0.012$



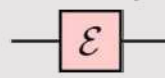
X bit-flip noise



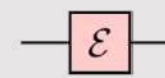
Phase noise



Depolarizing noise

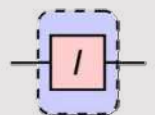


Stochastic Pauli noise

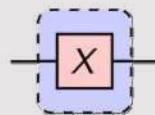


probability circuit instance

$1 - p$

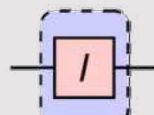


p

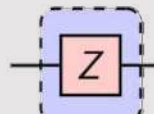


probability circuit instance

$1 - p$

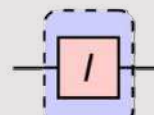


p

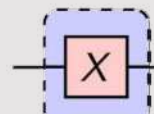


probability circuit instance

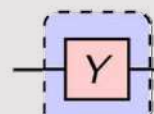
$1 - 3p/4$



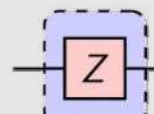
$p/4$



$p/4$

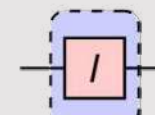


$p/4$

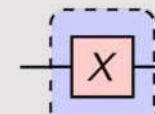


probability circuit instance

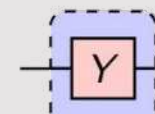
$1 - p_X - p_Y - p_Z$



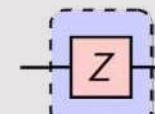
p_X



p_Y



p_Z



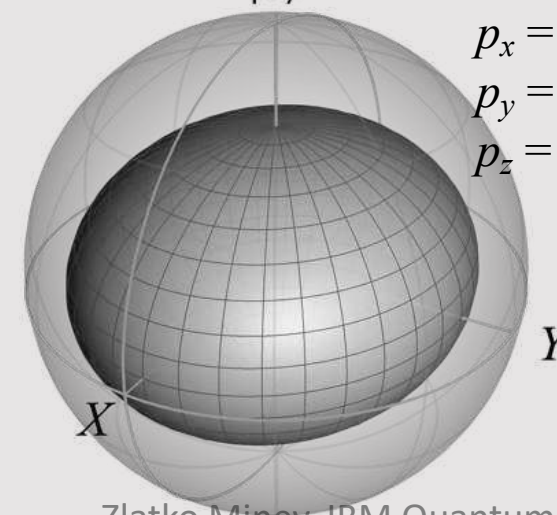
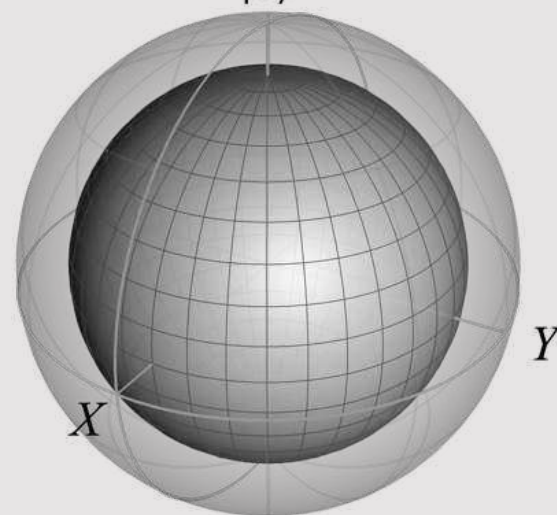
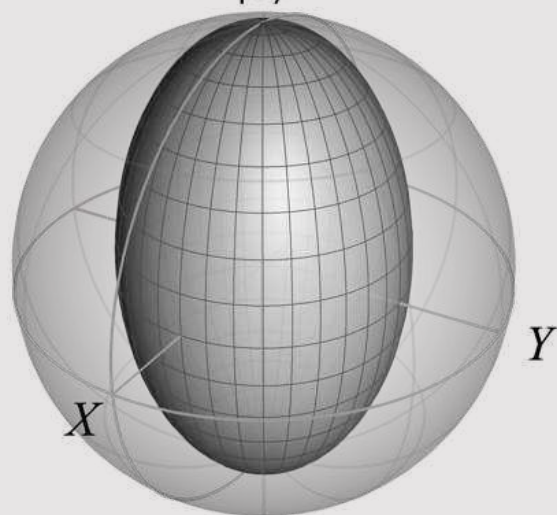
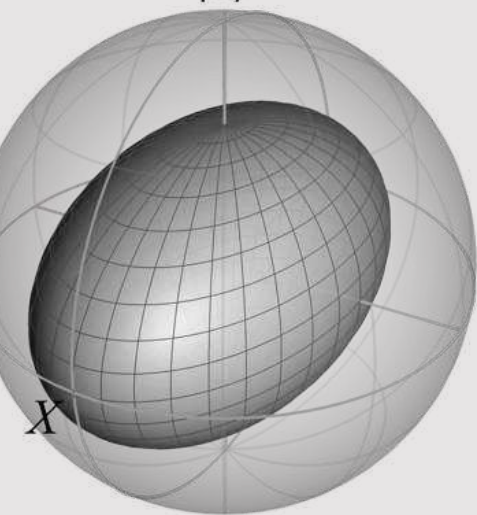
$|0\rangle$

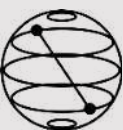
$|0\rangle$

$|0\rangle$

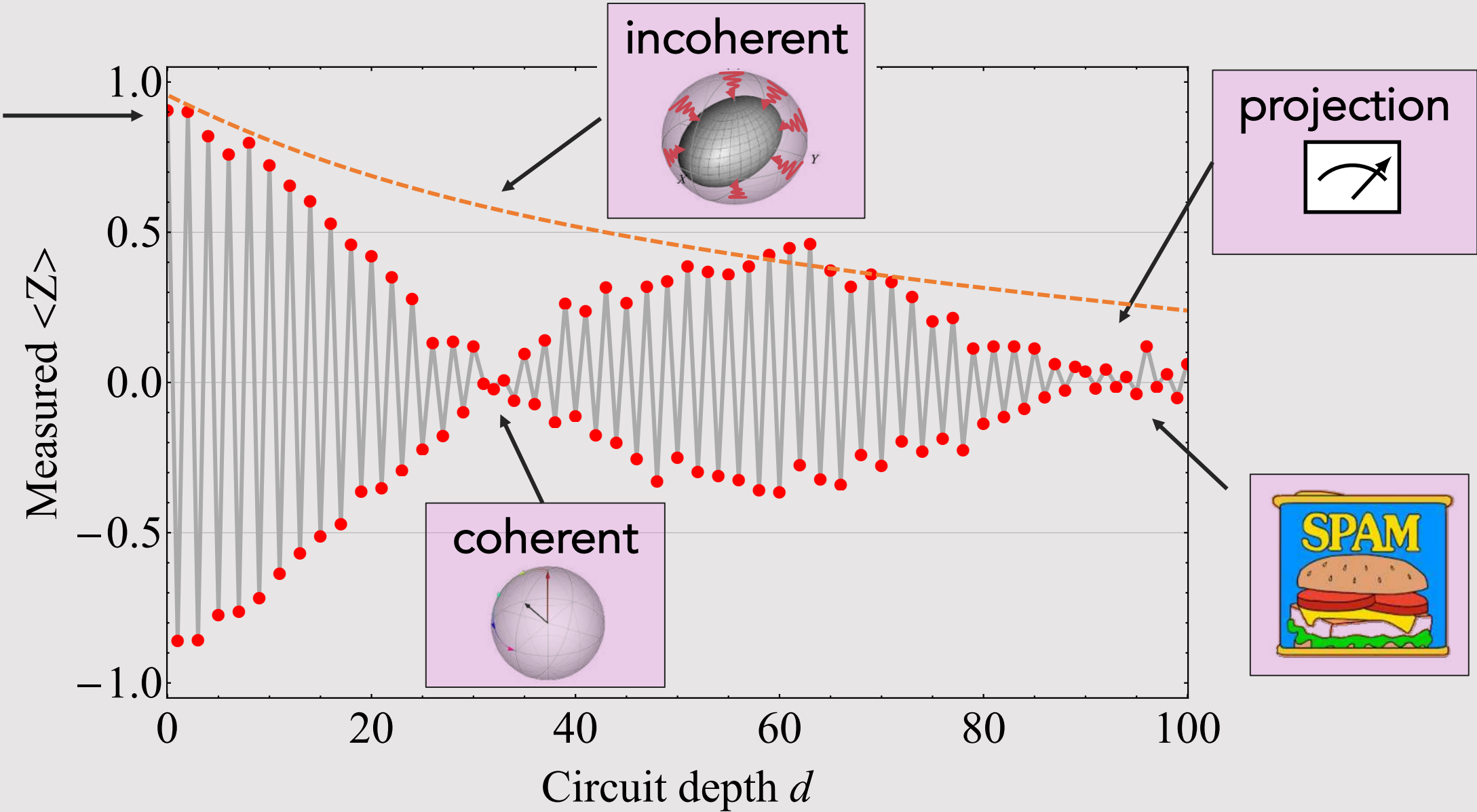
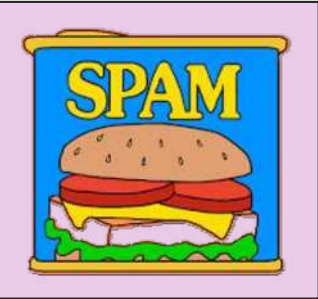
$|0\rangle$

$p_X = 0.1$
 $p_Y = 0.2$
 $p_Z = 0.4$



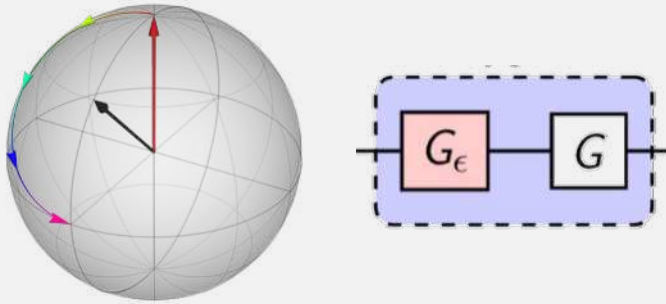


Elements of noise

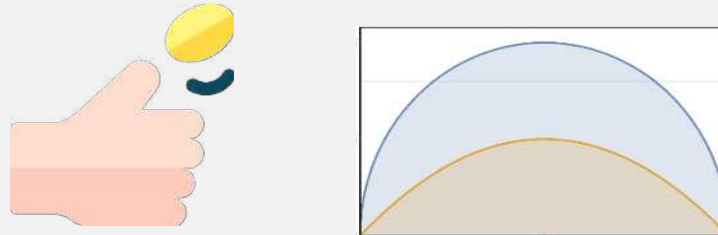


Our journey so far

Coherent noise



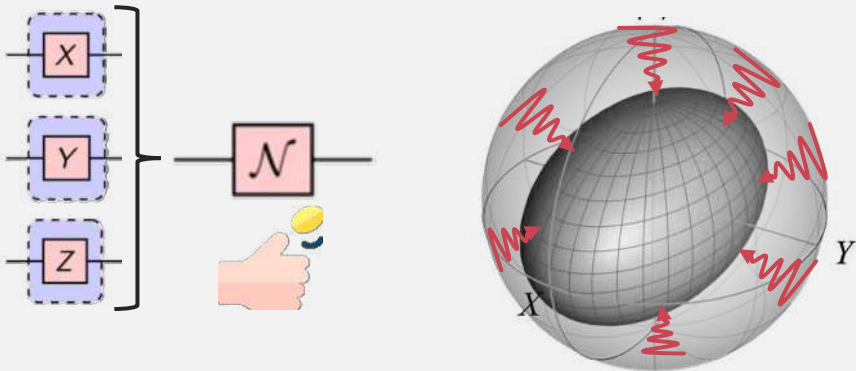
Measurement in a nutshell Projection noise



SPAM: Noisy meters



Incoherent noise

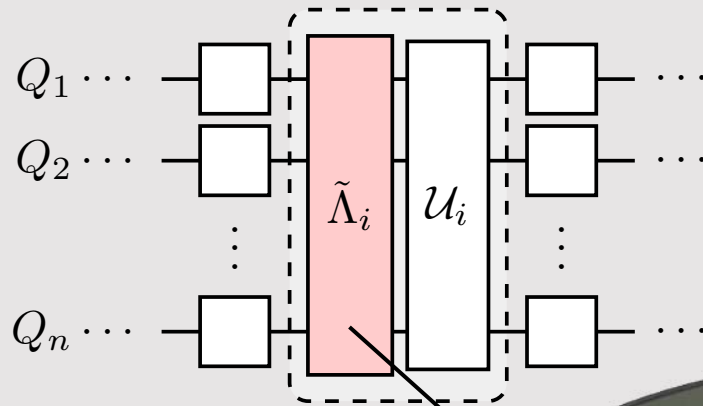


Bonus content Coherent ZZ State preparation

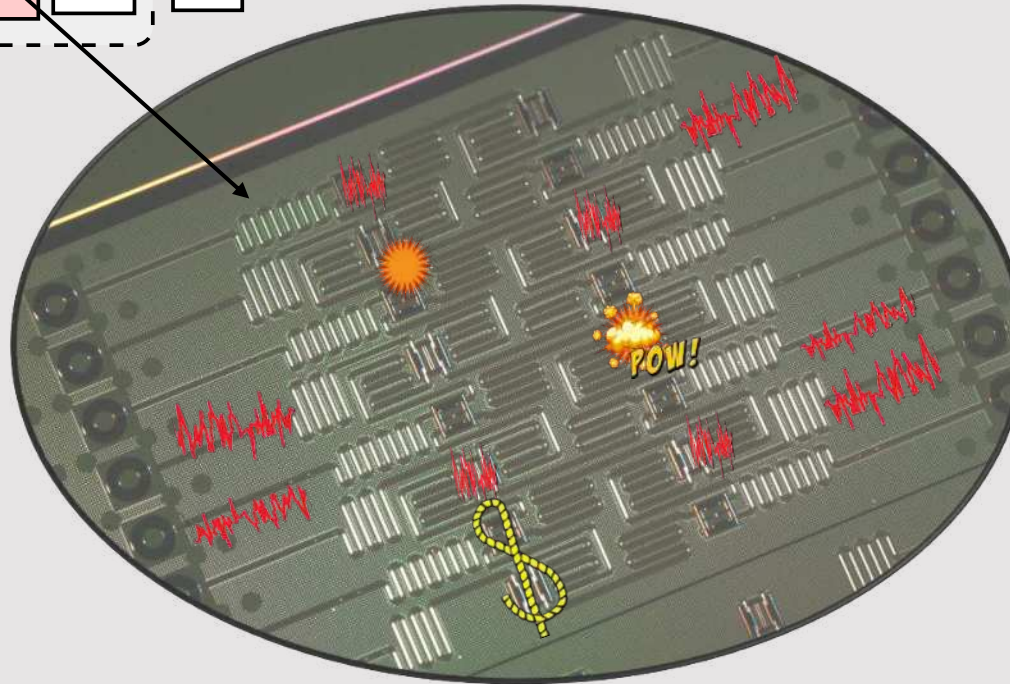


coin toss: flaticon; spam: make it move;
road based on: freepik

Outlook: Is it possible to learn & correct the noise with accuracy, efficiency, and scalability?

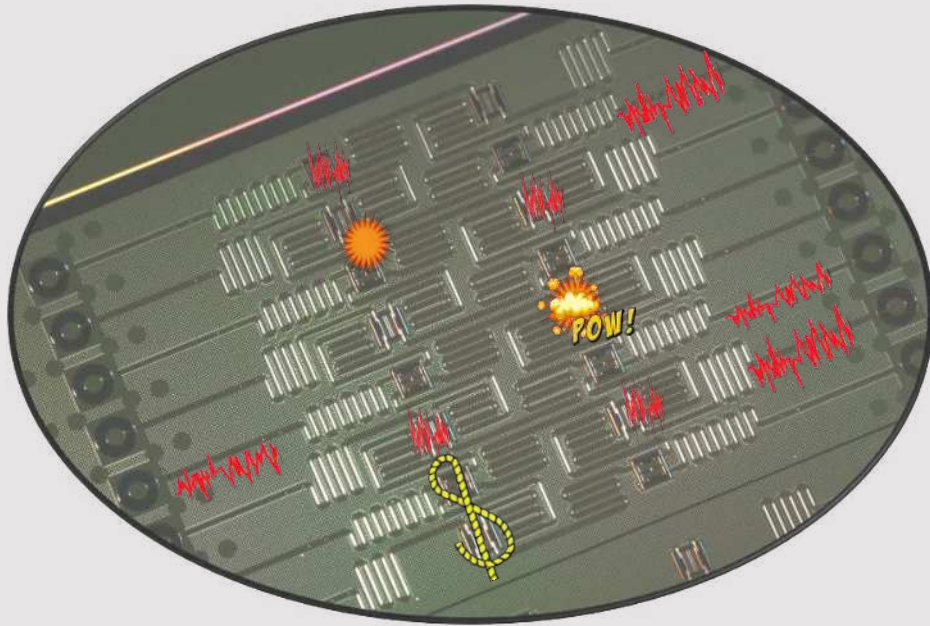


Energy relaxation T_1
Dephasing T_2
Coherent errors ZZ
Classical crosstalk
Quantum crosstalk
State preparation error
Measurement correlated errors
...



Control errors
Photon shot noise
1/f charge noise
1/f flux noise
Nonequilibrium quasiparticles
Leakage
Cosmic rays
...

Error mitigation and error correction



Error mitigation: working with what you have

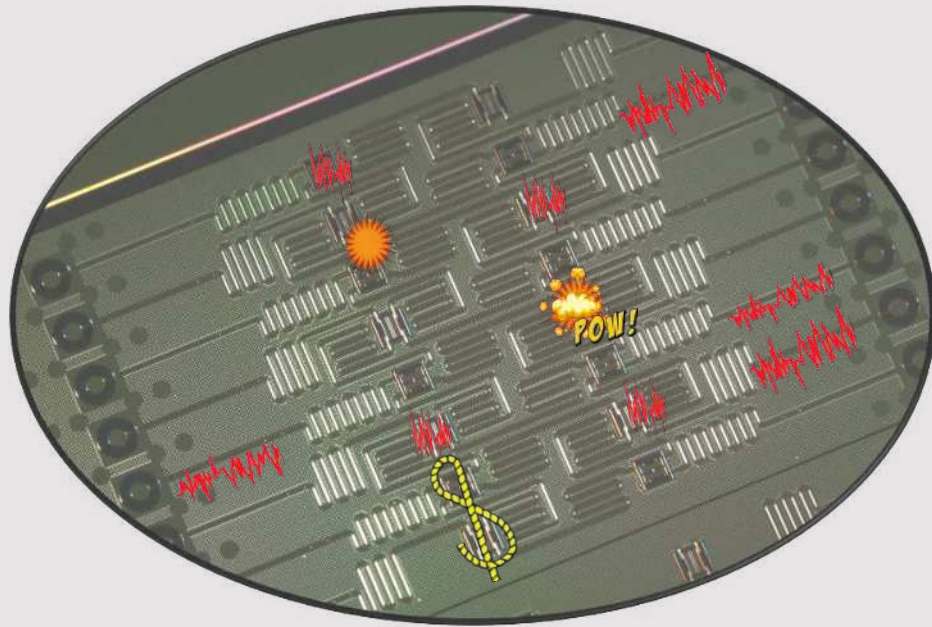
- **benefit** suppress errors on classical results (expectation values)
- **q-cost** no extra qubits or hardware resources needed
- **c-cost** trades classical resources (post-processing) for lower error
- **limitation** bad asymptotic scaling: high number of samples & circuits

Error correction: protecting quantum information

- **benefit** suppress & correct errors to arbitrarily small level
- **q-cost** very large qubit and hardware overhead
- **c-cost** decoding and encoding can be classically costly
- **challenge** requires fault-tolerant operations and readout

Latest: Qiskit Quantum Seminar YouTube Series

Ways to learn more
TODO



Introduction to Quantum Noise

Lab work with Qiskit

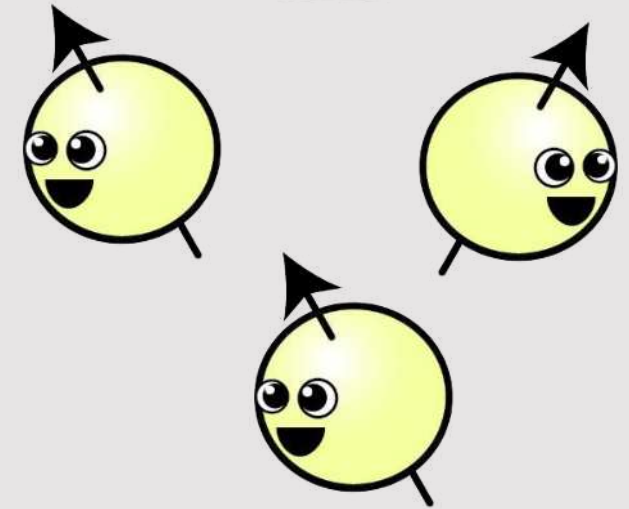
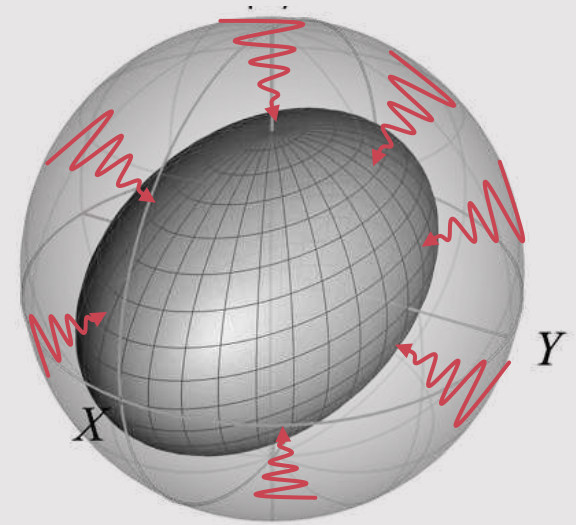
Run experiments on real devices

Check out references, problems given in the lecture, dangerous bends

Stay in touch

Thank you!

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IBM Quantum

The important thing is not to stop questioning.
Curiosity has its own reason for existence.

One cannot help but be in awe when he
contemplates the mysteries of eternity, of life, of the
marvelous structure of reality.

It is enough if one tries merely to comprehend a
little of this mystery each day.

Albert Einstein

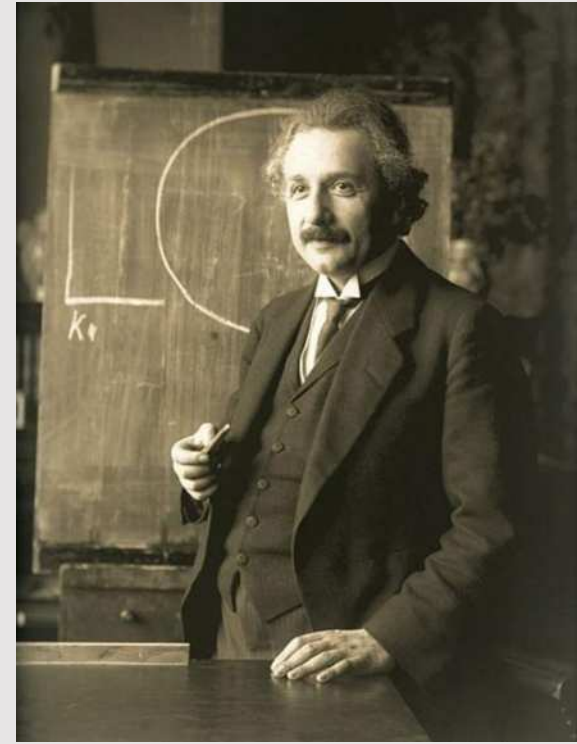


Photo: F. Schmutzer



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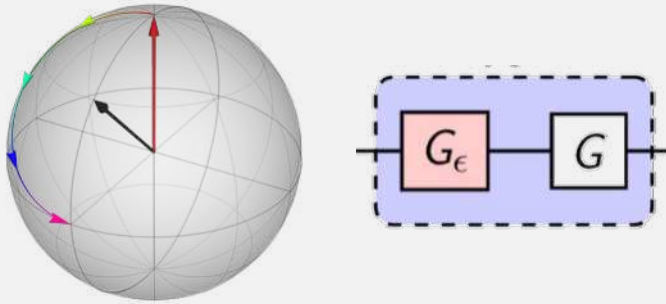
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IBM Quantum

Bonus content

Bonus content

Coherent noise



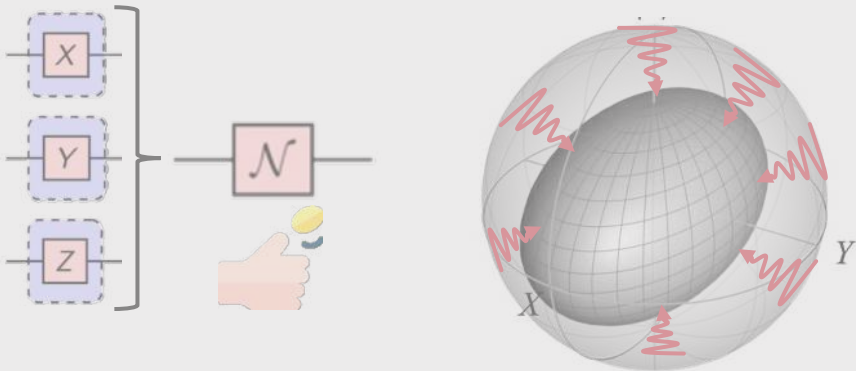
Measurement in a nutshell Projection noise



SPAM: Noisy meters



Incoherent noise

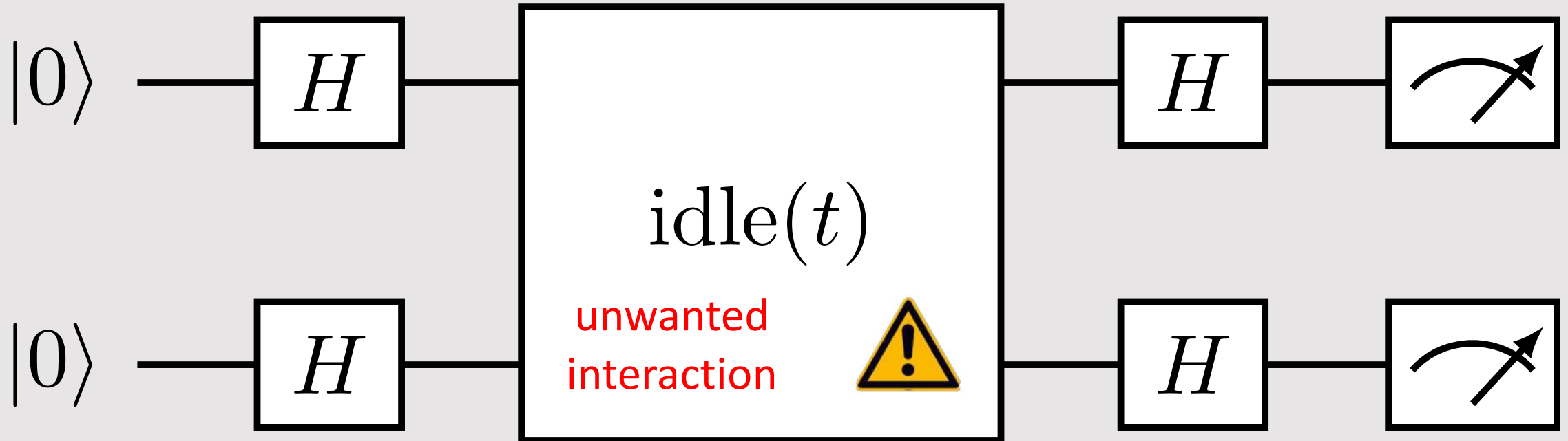


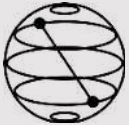
Bonus content



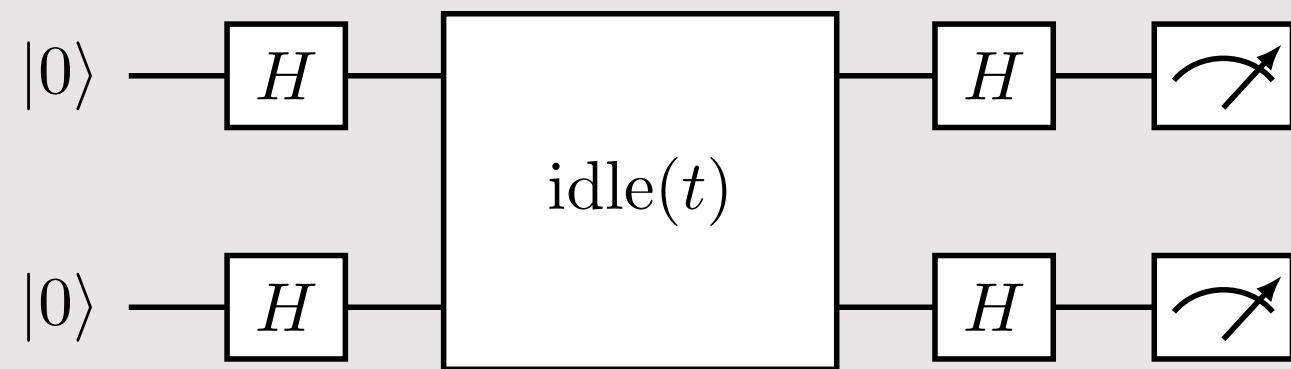
coin toss: flaticon; spam: make it move;
road based on: freepik

Bonus content: two-qubit coherent ZZ error





Bonus content: two-qubit ZZ error



Hadamard gate



$$H = \begin{matrix} |0\rangle \\ |1\rangle \end{matrix} \begin{pmatrix} \langle 0| & \langle 1| \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{pmatrix}$$

$$H |0\rangle = |+x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$H |1\rangle = |-x\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

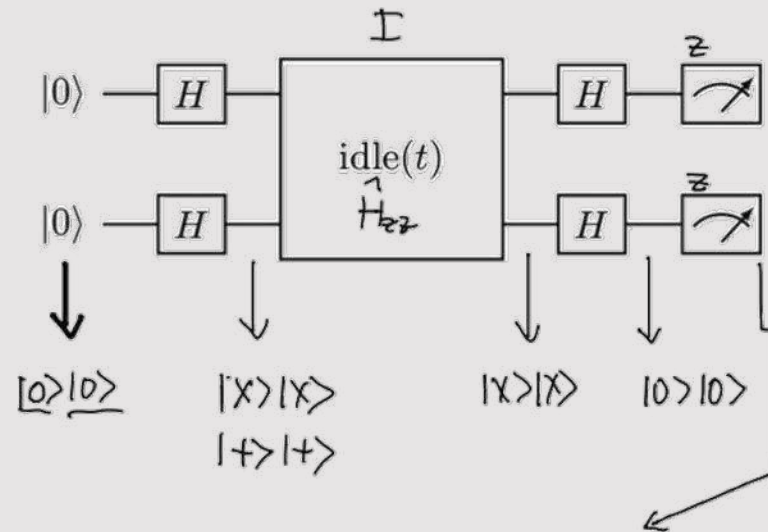
Breakout to notebook

Introduction to quantum noise

Coherent errors

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$$\begin{aligned}
 \langle ZI \rangle &= +1 \\
 \langle IZ \rangle &= +1 \\
 \langle ZZ \rangle &= \langle 00 | Z | 00 \rangle + \langle 01 | Z | 01 \rangle + \langle 10 | Z | 10 \rangle + \langle 11 | Z | 11 \rangle \\
 &= \langle 00 | Z | 00 \rangle + \langle 01 | Z | 01 \rangle + \langle 10 | Z | 10 \rangle + \langle 11 | Z | 11 \rangle \\
 &= (+1) + (+1) \\
 &= +1
 \end{aligned}$$

Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} \langle 0 | & \langle 1 | \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \end{pmatrix}$$

$$\begin{cases} H |0\rangle = |+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \\ H |1\rangle = |-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \end{cases}$$

$$\begin{aligned}
 X |+\rangle &= + |+\rangle \\
 X |-\rangle &= - |-\rangle
 \end{aligned}$$

$$\begin{aligned}
 |+\rangle &::= |+\rangle \\
 |-\rangle &::= |-\rangle
 \end{aligned}$$

$$\begin{cases} Z |0\rangle = +1 |0\rangle \\ Z |1\rangle = -1 |1\rangle \end{cases}$$

Noisy

ZZ Interaction

$$\hat{H} = \frac{1}{2} \hbar \omega \hat{Z} \hat{Z}$$

$$\begin{aligned} \hat{U}(t) &= \exp(-i \hbar^{-1} \hat{H} t) \\ &= \exp(-i \frac{\omega t}{2} \hat{Z} \hat{Z}) \end{aligned}$$

$$= \cos(\frac{\omega t}{2}) \hat{I} - i \sin(\frac{\omega t}{2}) \hat{Z} \hat{Z}$$

$$= \hat{R}_{ZZ}(\theta = \omega t)$$

$$|0\rangle|0\rangle \xrightarrow{HH} |+\rangle|+\rangle \xrightarrow{\text{idle}} R_{ZZ}(\theta) |+\rangle|+\rangle = \cos \frac{\theta}{2} |+\rangle|+\rangle - i \sin \frac{\theta}{2} |-\rangle|-\rangle \xrightarrow{HH} \underline{\cos \frac{\theta}{2} |0\rangle|0\rangle - i \sin \frac{\theta}{2} |1\rangle|1\rangle}$$

$$R_{ZZ}(\theta) |+\rangle|+\rangle = \cos \frac{\theta}{2} |+\rangle|+\rangle - i \sin \frac{\theta}{2} (Z|+\rangle)_1 \otimes (Z|+\rangle)_2$$

$$Z|+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle$$

$$Z|+\rangle = |-\rangle$$

$$Z|-\rangle = |+\rangle$$

$$\langle ZI \rangle = \cos \omega t$$

$$\langle IZ \rangle = \cos \omega t$$

$$\langle ZZ \rangle = 1$$

$$\hat{R}_X(\theta) = \exp(-i \frac{\theta}{2} \hat{X}) \quad X^2 = I$$

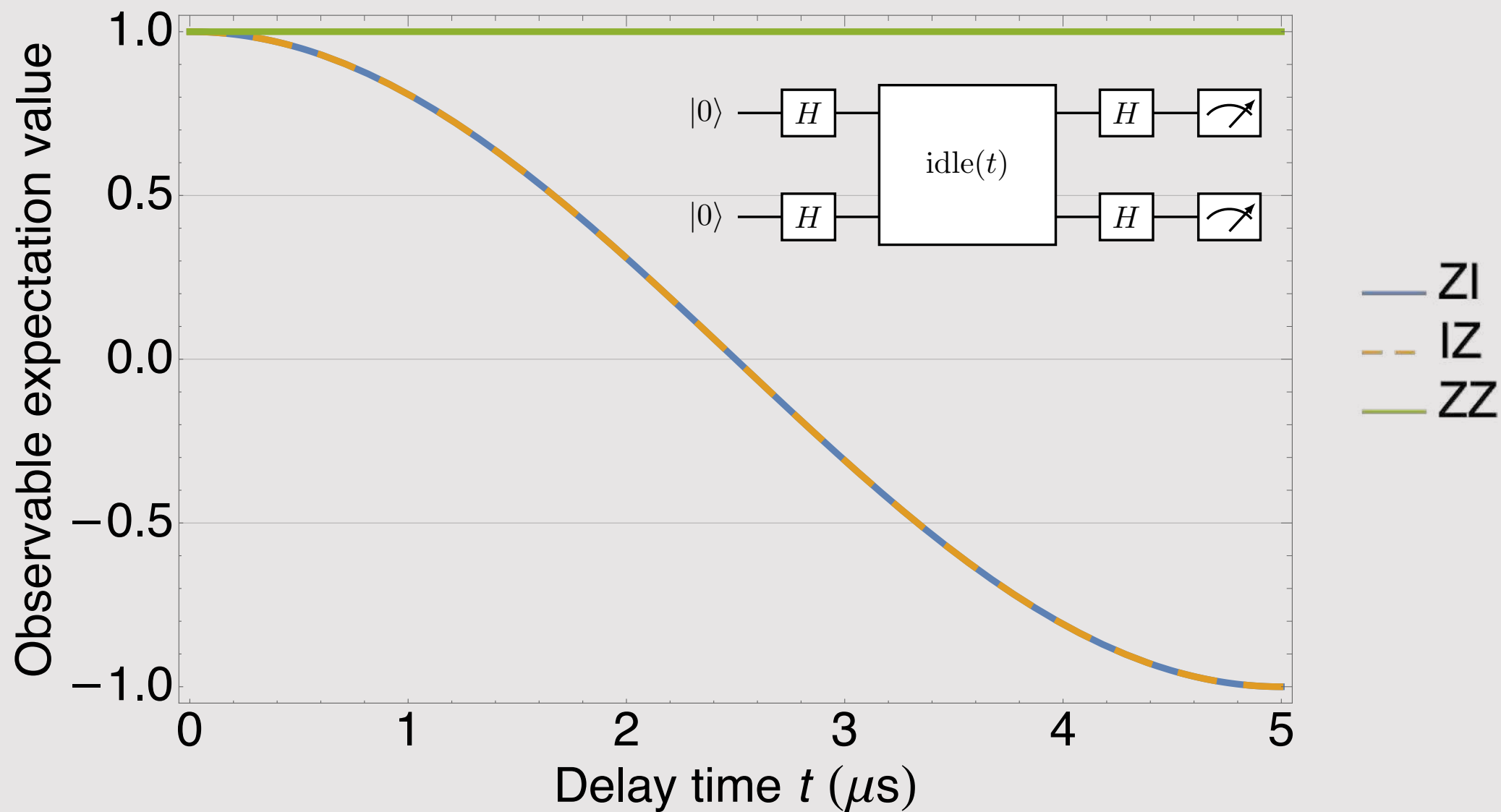
$$= \cos(\frac{\theta}{2}) I - i \sin(\frac{\theta}{2}) X$$

$$(\hat{Z} \hat{Z})^2 = Z^2 Z^2 = I \otimes I = I_4$$



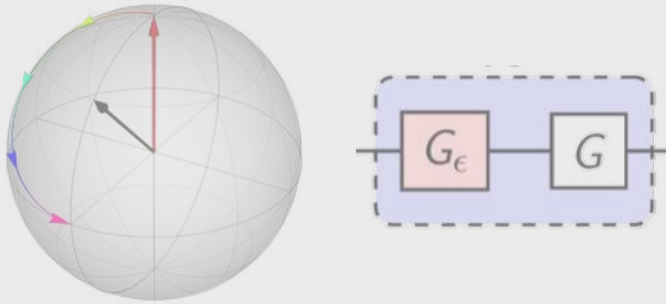
Bonus content: two-qubit ZZ error

Gate error $\omega = 10$ kHz

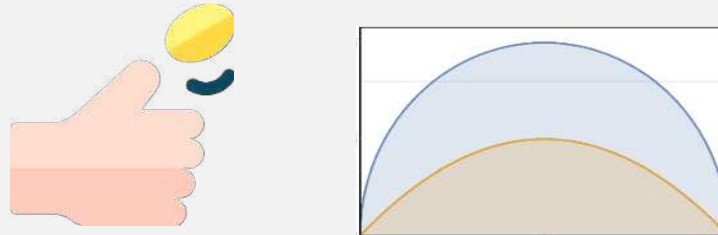


Bonus content

Coherent noise



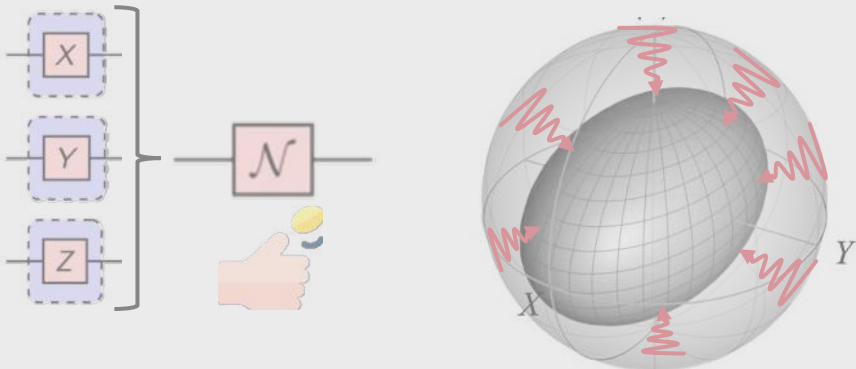
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Incoherent noise



Bonus content



coin toss: flaticon; spam: make it move;
road based on: freepik

A matrix

Bonus section content:

Reconstruct A matrix

$$|0\rangle \xrightarrow{\text{A}} (M=0) \xrightarrow{p=0} \tilde{M} \xrightarrow{\tilde{p}=\epsilon}$$

$$|0\rangle \xrightarrow{\text{X}} |1\rangle \xrightarrow{\text{A}} (M=1) \xrightarrow{p=1} \tilde{M} \xrightarrow{\tilde{p}=1-\nu} \leftarrow \text{we have data}$$

Noise mitigation

We know A

measure $\tilde{p}$, $\tilde{P}_M$ noisy

find p , P_M ideal

$$\dim A = 2^n \times 2^n \quad n = \text{qubits}$$

$$\tilde{P}_M = A P_M$$

$$P_M = A^{-1} \tilde{P}_M$$

$$p =$$

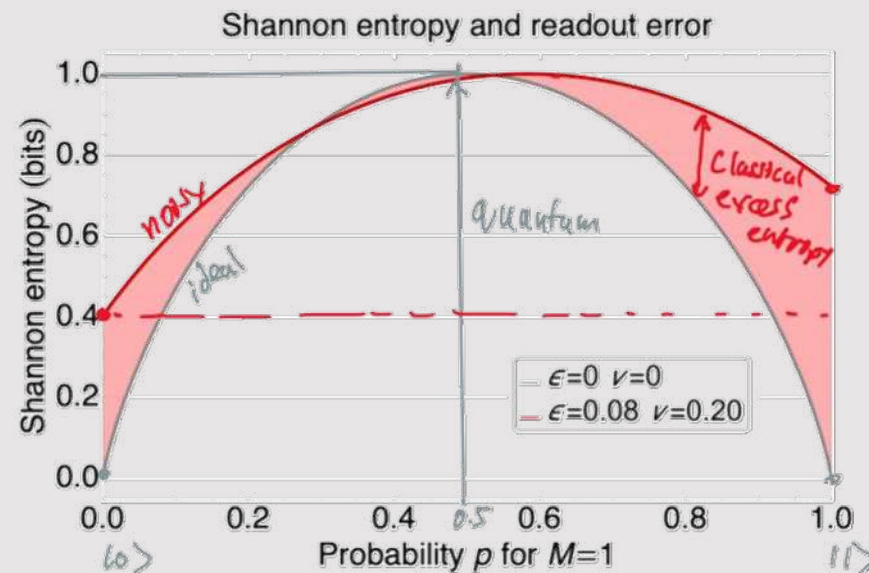
Assignment Fidelity

$$\begin{aligned} \mathcal{F}_0 &= 1 - \frac{1}{2} [p(\tilde{M}=1|M=0) + p(\tilde{M}=0|M=1)] \\ &= \frac{1}{d} \text{Tr}(A) \\ &= 1 - \frac{1}{2} (M + D) \end{aligned} \quad d = 2^n, n = \text{qubits}$$

Shannon Entropy

$$\mathcal{H}(A) = \mathcal{H}(P_M) = - \sum_m P_m \log_2 P_m = - (1-p) \log_2 (1-p) - p \log_2 p$$

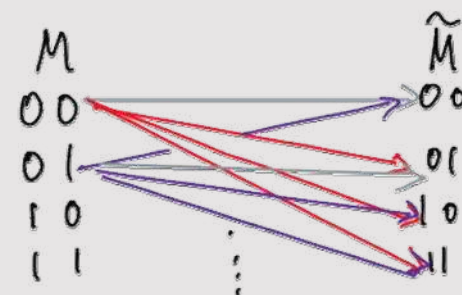
Binary entropy



Larger System

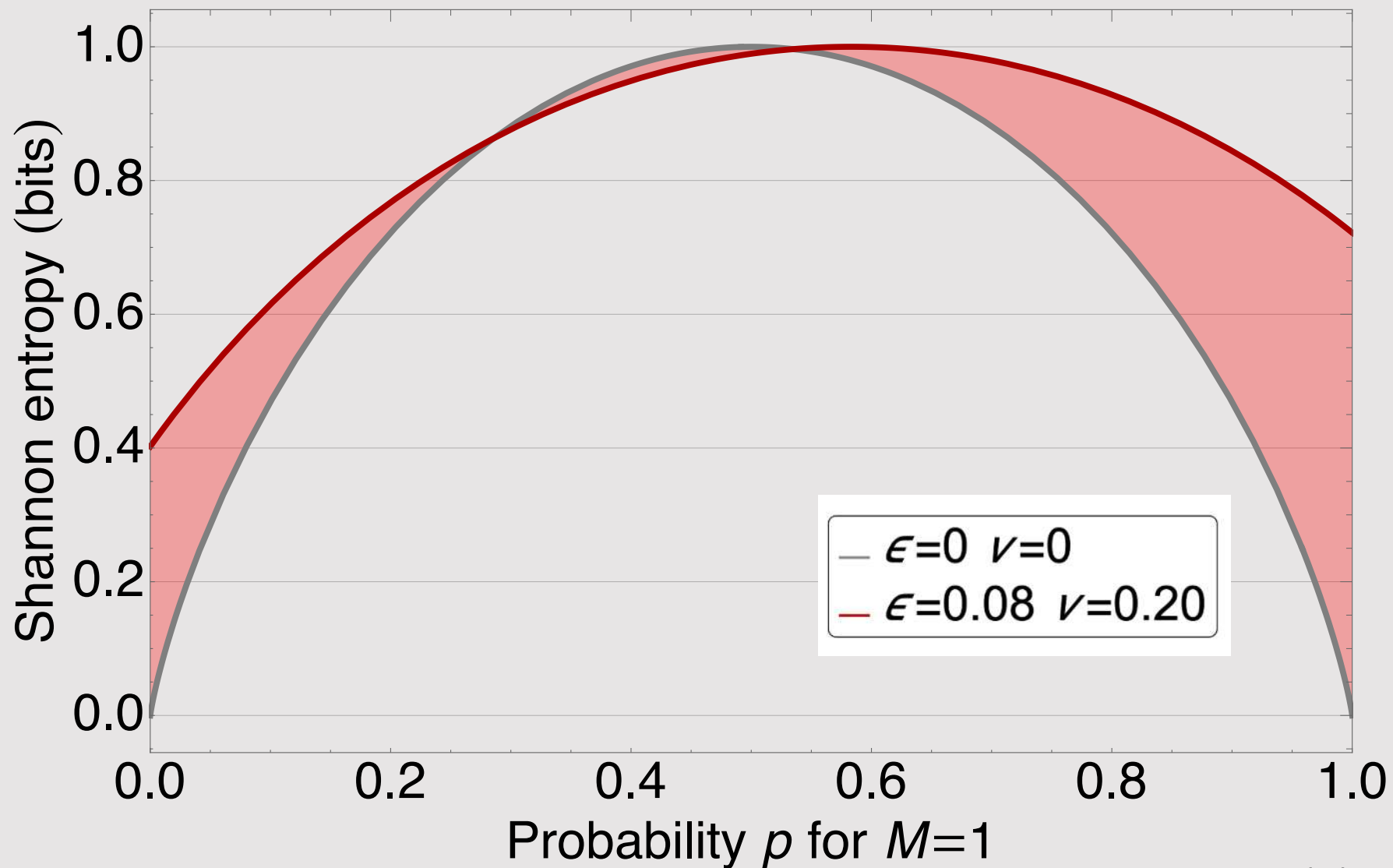
n qubits

$$\dim A = 2^n \times 2^n$$



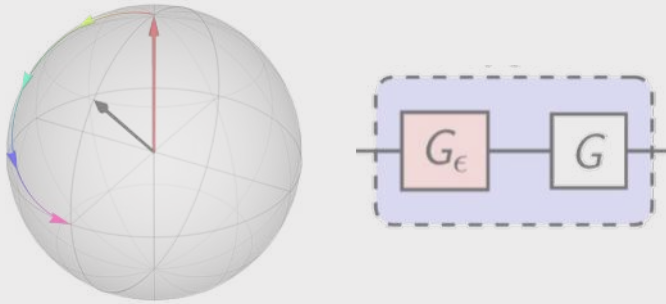
Entropy

Shannon entropy and readout error

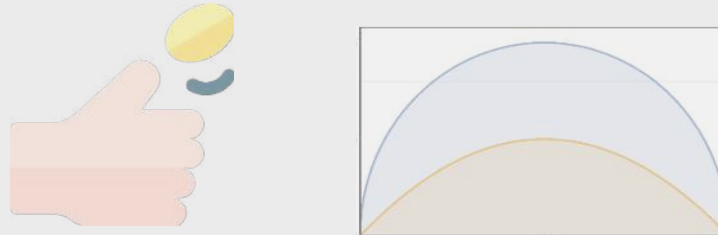


Bonus content

Coherent noise



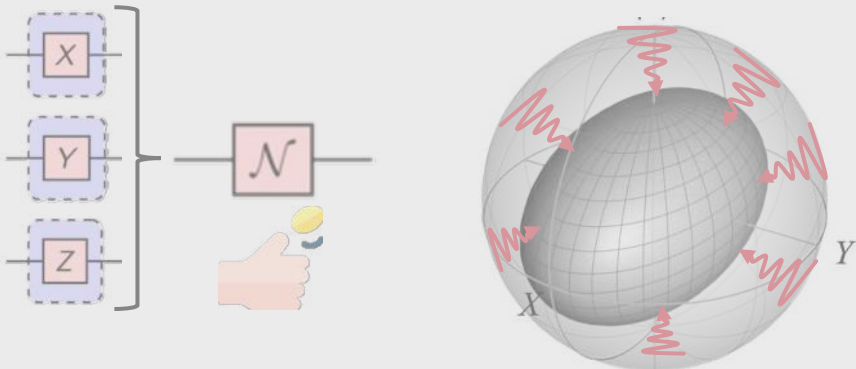
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Bonus content



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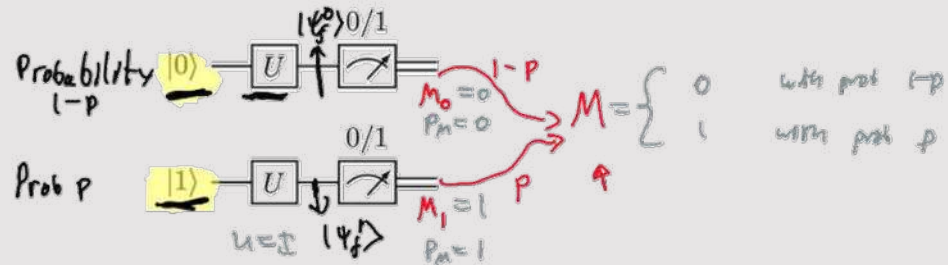
State prep noise

Introduction to quantum noise

State preparation noise

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Density operator

$$\rho = (1-p) |0\rangle\langle 0| + p |1\rangle\langle 1|$$

$$= \rho_{\text{no error}} \rho_{\text{ideal}} + \rho_{\text{error}} \rho_{\text{error}}$$

$$\begin{aligned} \rho_f &= \rho_{\text{no error}} \rho_{\text{ideal}}^{\text{final}} + \rho_{\text{error}} \rho_{\text{error}}^{\text{final}} \\ &= (1-p) |\psi_f^0\rangle\langle\psi_f^0| + p |\psi_f^1\rangle\langle\psi_f^1| \\ &= (1-p) U |0\rangle\langle 0| U^\dagger + p U |1\rangle\langle 1| U^\dagger \\ &= U [(1-p) |0\rangle\langle 0| + p |1\rangle\langle 1|] U^\dagger \\ &= U \rho U^\dagger \end{aligned}$$

$$\rho_v = U \rho U^\dagger$$

$$(U|\psi\rangle)^\dagger = (|\psi\rangle)^\dagger U^\dagger$$

$$\langle\psi_f| = \langle\psi| U^\dagger$$

$$\rho_v = U \rho U^\dagger$$

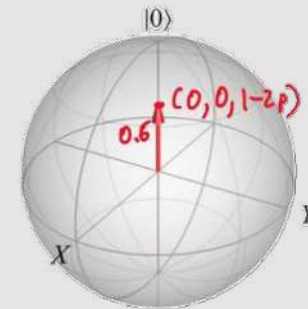
$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$|1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\rho = \begin{pmatrix} 1-p & 0 \\ 0 & p \end{pmatrix} = \frac{1}{2} (\hat{I} + (1-2p) \hat{Z})$$

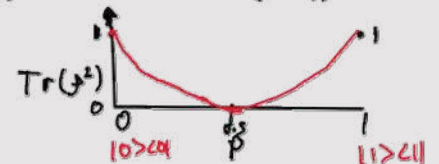
$$\begin{cases} \langle X \rangle = \text{Tr}[\rho X] = 0 \\ \langle Y \rangle = \text{Tr}[\rho Y] = 0 \\ \langle Z \rangle = \text{Tr}[\rho Z] = 1-2p \end{cases}$$

$$\text{Tr}[\rho_\alpha \rho_\beta] = 2\delta_{\alpha\beta} \quad \text{for } \alpha, \beta \in \{X, Y, Z\}$$



$$\rho = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \times (1-p) + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \times p$$

$$\text{Tr}(\rho^2) = \langle X \rangle^2 + \langle Y \rangle^2 + \langle Z \rangle^2 = 0^2 + 0^2 + (1-2p)^2$$



State prep noise

Scaling to larger number of qubits

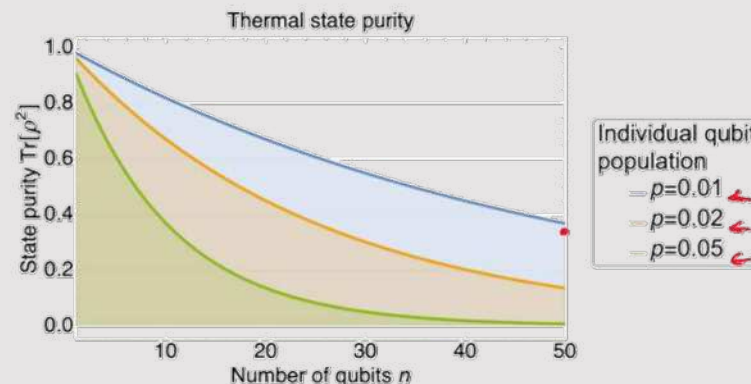
$$[(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|]^{\otimes n} \equiv U \equiv \text{[Circuit Diagram]}$$

$$\begin{aligned} \rho_0 &= [(1-p)|0\rangle\langle 0| + p|1\rangle\langle 1|]^{\otimes n} \\ &= (1-p)^n |000\dots 0\rangle\langle 000\dots 0| + \dots + p^n |111\dots 1\rangle\langle 111\dots 1| \end{aligned}$$

$$P(M=0000\dots 0) = \begin{cases} 1 & \text{if } p=0 \text{ (ideal case)} \\ (1-p)^n \approx 1-np + O(p^2) & \text{for } p>0 \end{cases}$$

$$\langle Z_k \rangle = \langle ||| Z ||| \rangle = 1-2p \quad \text{single qubit}$$

$$\begin{aligned} \langle Z Z Z \dots Z \rangle &= \text{Tr} [Z^{\otimes n} \rho] = (1-2p)^n \\ \rho &= \prod_{k=1}^n \frac{1}{2} (\hat{I}_k + (1-2p) \hat{Z}_k) \\ &= \frac{1}{2^n} (1+\underline{Z}) \otimes (1+\underline{Z}) \otimes \dots \otimes (1+\underline{Z}) \\ &= \frac{1}{2^n} [(1-2p)^n \underline{Z}^{\otimes n} + \dots] \\ \text{Tr} [\rho^2] &= \prod_{k=1}^n \text{Tr} (\rho_k^2) = (1-2p)^{2n} \end{aligned}$$



The End!